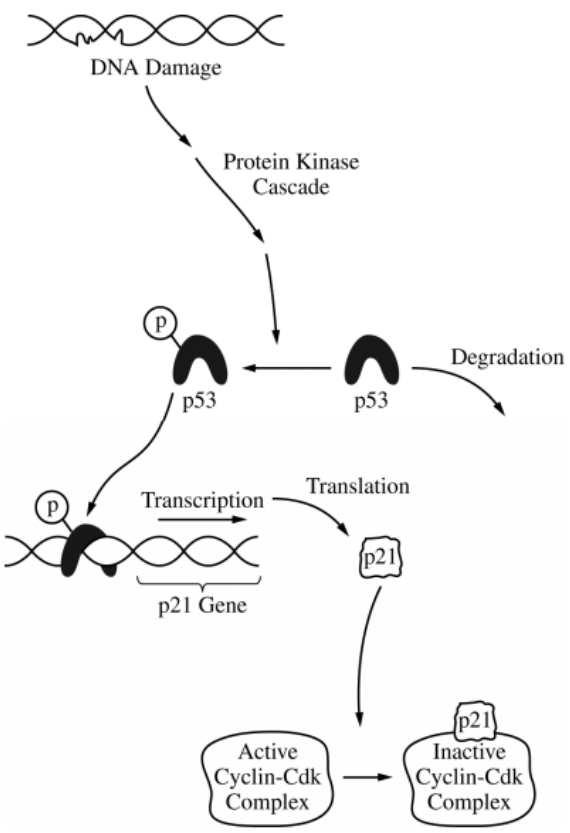
6 Gene Expression and Regulation MCQ

1.



The p53 protein regulates a cellular response to DNA damage. Based on the diagram above, which of the following best describes the role of p53 in the response to DNA damage?

(A) Phosphorylated p53 binds to DNA and repairs the damage.

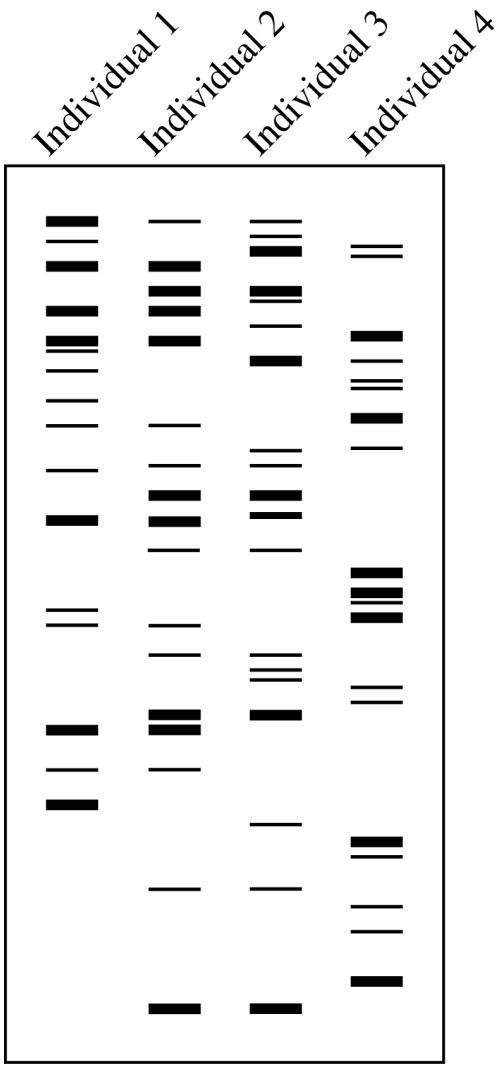
(B) Phosphorylated p53 stimulates transcription of p21, and the resulting p21 protein suppresses cell division until the DNA damage is repaired. ✓

(C) Phosphorylated p53 binds cyclin − Cdk complexes, and the resulting protein complex repairs the DNA damage.

(D) Phosphorylated p53 activates p21 proteins, and the p21 proteins in turn repair the DNA damage.

6 Gene Expression and Regulation MCQ

2.



Samples of DNA were isolated from four different individuals and each sample was digested by the same restriction enzymes. Gel electrophoresis was used to separate the resulting DNA fragments and the results are shown above. These data best support which of the following hypotheses?

(A) Individual 1 is the offspring of 2 and 3.

(B) Individual 1 is the offspring of 3 and 4.

(C) Individual 2 is the offspring of 1 and 3. ✓

(D) Individual 2 is the offspring of 1 and 4.

(E) Individual 3 is the offspring of 1 and 4.

6 Gene Expression and Regulation MCQ

3. A genetic counselor is consulted by a young man who is worried about developing Huntington's disease, an inherited disorder caused by a dominant allele of a single gene. The young man explains that his cousin was recently diagnosed with Huntington's disease, and the news has caused him to consider his own risk of developing the disorder. Which of the following questions will best help the genetic counselor to evaluate the risk of the young man developing Huntington's disease and transmitting it to his children?

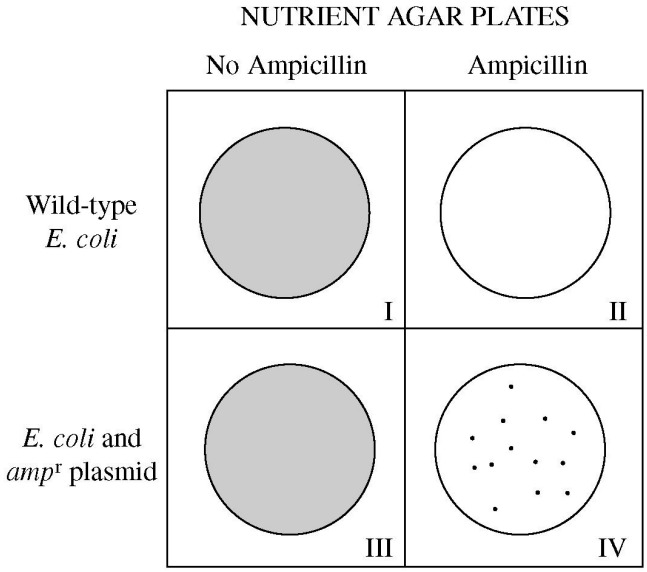
(A) Were you and your cousin born in the same geographical area?

(B) Were your parents or grandparents ever diagnosed with Huntington's disease? ✓

(C) Were you in physical contact with a person diagnosed with Huntington's disease?

(D) Were you ever exposed to substances that are suspected of being mutagens?

In a transformation experiment, a sample of *E. coli* bacteria was mixed with a plasmid containing the gene for resistance to the antibiotic ampicillin (*amp*^{r}). Plasmid was not added to a second sample. Samples were plated on nutrient agar plates, some of which were supplemented with the antibiotic ampicillin. The results of *E. coli* growth are summarized below. The shaded area represents extensive growth of bacteria; dots represent individual colonies of bacteria.



4. Plates that have only ampicillin-resistant bacteria growing include which of the following?

(A) I only

(B) III only

(C) IV only ✓

(D) I and II

5. Which of the following best explains why there is no growth on plate II?

6 Gene Expression and Regulation MCQ

(A) The initial *E. coli* culture was not ampicillin-resistant. ✓

(B) The transformation procedure killed the bacteria.

(C) Nutrient agar inhibits *E. coli* growth.

(D) The bacteria on the plate were transformed.

6. Plates I and III were included in the experimental design in order to

(A) demonstrate that the *E. coli* cultures were viable ✓

(B) demonstrate that the plasmid can lose its amp^r gene

(C) demonstrate that the plasmid is needed for *E. coli* growth

(D) prepare the *E. coli* for transformation

7. Which of the following statements best explains why there are fewer colonies on plate IV than on plate III?

(A) Plate IV is the positive control.

(B) Not all *E. coli* cells are successfully transformed. ✓

(C) The bacteria on plate III did not mutate.

(D) The plasmid inhibits *E. coli* growth.

8. In a second experiment, the plasmid contained the gene for human insulin as well as the amp^r gene. Which of the following plates would have the highest percentage of bacteria that are expected to produce insulin?

(A) I only

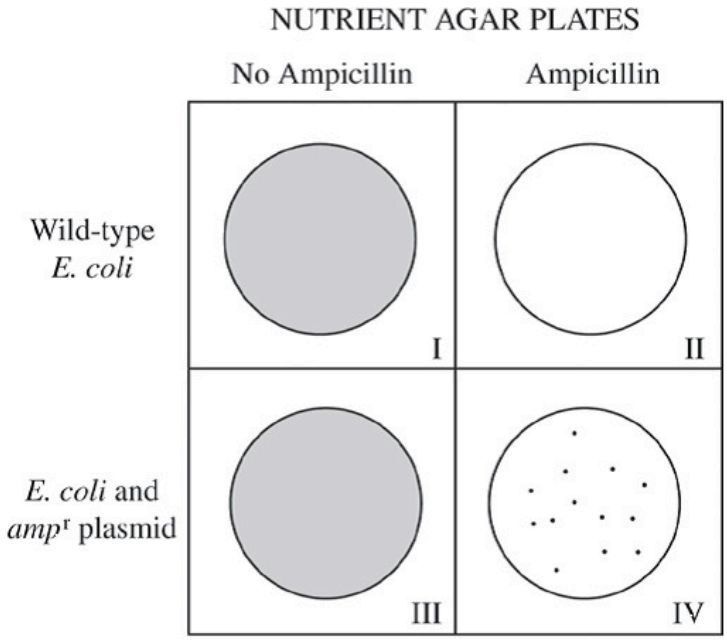
(B) III only

(C) IV only ✓

(D) I and III

6 Gene Expression and Regulation MCQ

In a transformation experiment, a sample of *E. coli* bacteria was mixed with a plasmid containing the gene for resistance to the antibiotic ampicillin (*amp*^{r}). Plasmid was not added to a second sample. Samples were plated on nutrient agar plates, some of which were supplemented with the antibiotic ampicillin. The results of *E. coli* growth are summarized below. The shaded area represents extensive growth of bacteria; dots represent individual colonies of bacteria.



9. Plates that have **only** ampicillin-resistant bacteria growing include which of the following?

(A) I only

(B) III only

(C) IV only ✓

(D) I and II

Answer C

This option is correct. Plate IV shows that only *E. coli* with the plasmid containing the gene for ampicillin-resistant plasmid grew individual colonies with ampicillin in the agar, thus showing the competent cells that transformed successfully.

10. Which of the following best explains why there is no growth on plate II?

6 Gene Expression and Regulation MCQ

(A) The initial *E. coli* culture was not ampicillin-resistant. ✓

(B) The transformation procedure killed the bacteria.

(C) Nutrient agar inhibits *E. coli* growth.

(D) The bacteria on the plate were transformed.

Answer A

This option is correct. The initial wild-type *E. coli* did not contain the plasmid containing the gene for resistance to the ampicillin (*amp*r) and, when exposed to ampicillin, did not grow on Plate II.

11. Plates I and III were included in the experimental design in order to

(A) demonstrate that the *E. coli* cultures were viable ✓

(B) demonstrate that the plasmid can lose its amp^r gene

(C) demonstrate that the plasmid is needed for *E. coli* growth

(D) prepare the *E. coli* for transformation

Answer A

This option is correct. Both Plates I and III had extensive *E. coli* growth, showing that the cells were viable before and after the transformation procedure.

12. Which of the following statements best explains why there are fewer colonies on plate IV than on plate III?

(A) Plate IV is the positive control.

(B) Not all *E. coli* cells are successfully transformed. ✓

(C) The bacteria on plate III did not mutate.

(D) The plasmid inhibits *E. coli* growth.

Answer B

This option is correct. Only the *E. coli* that have been transformed and contain the amp^r gene will grow and produce colonies.

6 Gene Expression and Regulation MCQ

13. In a second experiment, the plasmid contained the gene for human insulin as well as the *amp*^{r} gene. Which of the following plates would have the highest percentage of bacteria that are expected to produce insulin?

(A) I only

(B) III only

(C) IV only ✓

(D) I and III

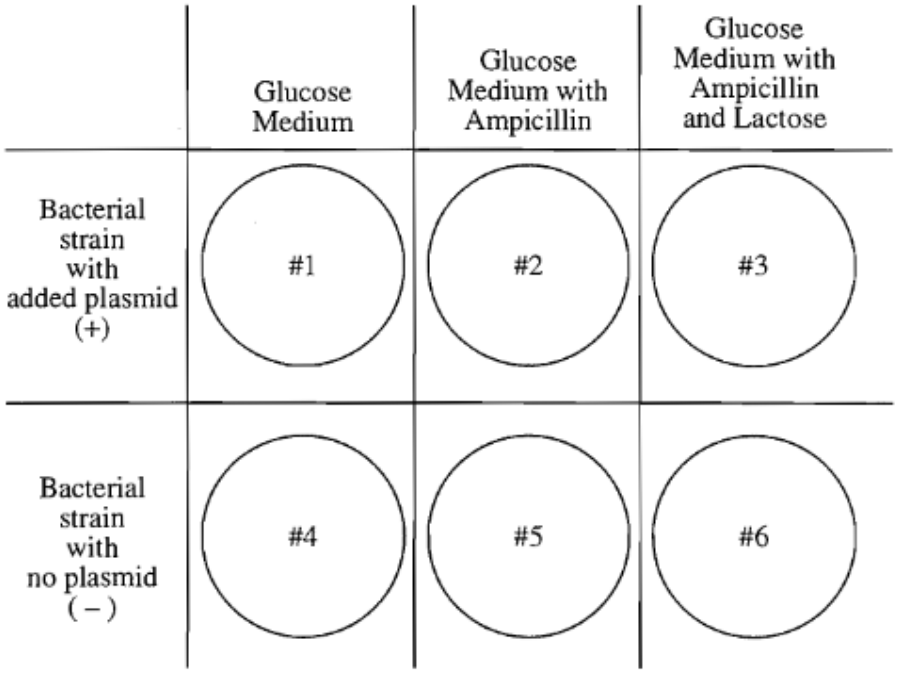
Answer C

This option is correct. In Plate IV the colonies of *E. coli* were successfully transformed and expressed the *amp*^{r} gene; adding another gene to the plasmid would express the human insulin gene the same way.

6 Gene Expression and Regulation MCQ

- **Directions:** Each qroup of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

A scientist is using an ampicillin-sensitive strain of bacteria that cannot use lactose because it has a nonfunctional gene in the *lac* operon. She has two plasmids. One contains a functional copy of the affected gene of the *lac* operon, and the other contains the gene for ampicillin resistance. Using restrictions enzymes and DNA ligase, she forms a recombinant plasmid containing both genes. She then adds a high concentration of the plasmid to a tube of the bacteria in a medium for bacterial growth that contains glucose as the only energy source. This tube (+) and a control tube (-) with similar bacteria but no plasmid are both incubated under the appropriate conditions for growth and plasmid uptake. The scientist then spreads a sample of each bacterial culture (+ and -) on each of the three types of plates indicated below.



14. If no new mutations occur, it would be most reasonable to expect bacterial growth on which of the following plates?

(A) 1 and 2 only
(B) 3 and 4 only
(C) 5 and 6 only
(D) 4, 5, and 6 only
(E) 1, 2, 3, and 4 only ✓

15. If the scientist had forgotten to use DNA ligase during the preparation of the recombinant plasmid, bacterial growth would most likely have occurred on which of the following?

6 Gene Expression and Regulation MCQ

(A) 1 and 2 only

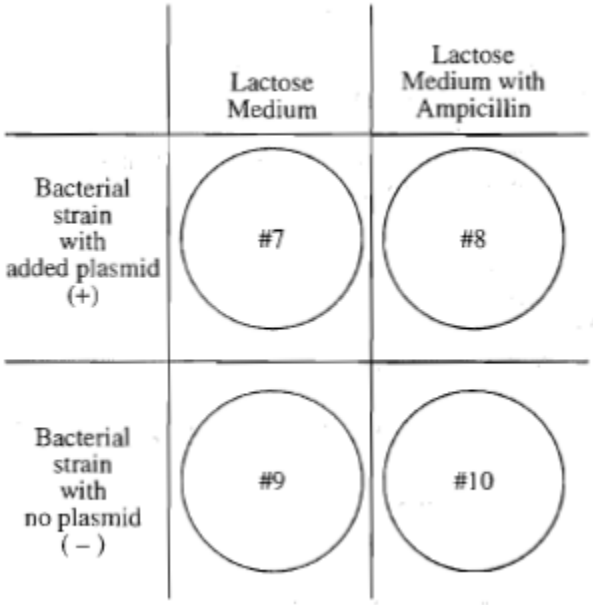
(B) 1 and 4 only ✓

(C) 4 and 5 only

(D) 1, 2, and 3 only

(E) 4, 5, and 6 only

16.



If the scientist used the cultures to perform another experiment as shown above, using medium that contained lactose as the only energy source, growth would most likely occur on which of the following plates?

(A) 10 only

(B) 7 and 8 only ✓

(C) 7 and 9 only

(D) 8 and 10 only

(E) 9 and 10 only

---

17. A new mutation that arose in one copy of gene *X* in a somatic cell resulted in the formation of a tumor. Which of the following pieces of evidence best describes how the new mutation directly caused the tumor?

(A) Protein X normally stimulates cell division, and the mutation created an overactive version of protein X. ✓

(B) Protein X normally activates a growth hormone receptor, and the mutation decreased the stability of protein X.

(C) Protein X normally prevents passage through the cell cycle, and the mutation created an overactive version of protein X.

(D) Protein X normally regulates gene expression, and the mutation created an underactive version of protein X that blocked the cell cycle.

6 Gene Expression and Regulation MCQ

Directions: Each group of questions below concerns an experimental or a laboratory situation. In each case, first study the description of the situation. Then choose the one best answer to each question following it.

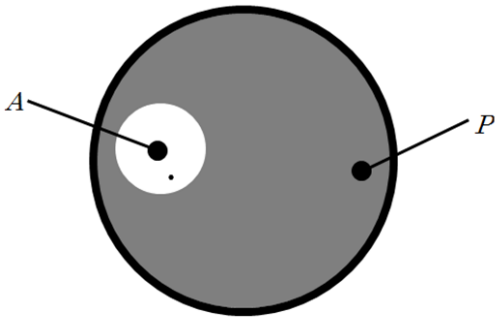


Plate I

A sterile agar plate, I, is streaked with a pure culture of bacteria by means of aseptic techniques. Paper discs treated with the antibiotics Aureomycin (*A*) and penicillin (*P*) are placed at opposite sides of the plate, as shown in the diagram above. The plate is examined after a 24-hour incubation period, and a clear ring is discovered around disc *A*, but not around disc *P*. Within the clear ring around disc *A*, a single bacterial colony with physical characteristics like those of the pure culture is observed. A second sterile agar plate, II, is then streaked with this single colony and also incubated with antibiotics.

18. The single colony found within the clear ring in plate I is most likely made up of the descendants of a bacterial cell that

(A) contaminated the agar plate

(B) contained information conferring resistance to Aureo-mycin ✓

(C) changed its response to Aureomycin as a result of being exposed to the antibiotic

(D) was susceptible to both penicillin and Aureomycin

(E) emigrated from another area of the agar plate

19. Which of the following would most likely be observed in plate II after 24 hours?

(A) A clear ring larger than that around disc *A* in plate I would appear around disc *A* only.

(B) A clear ring larger than that around disc *A* in plate I would appear around disc *P* only.

(C) A clear ring smaller than that around disc *A* in plate I would appear around disc *P* only.

(D) There would be a clear ring around both disc *A* and disc *P*.

(E) There would not be a clear ring around either disc *A* or disc *P*. ✓

6 Gene Expression and Regulation MCQ

20. Antigens are foreign proteins that invade the systems of organisms. Vaccines function by stimulating an organism's immune system to develop antibodies against a particular antigen. Developing a vaccine involves producing an antigen that can be introduced into the organism being vaccinated and which will trigger an immune response without causing the disease associated with the antigen. Certain strains of bacteria can be used to produce antigens used in vaccines.

Which of the following best explains how bacteria can be genetically engineered to produce a desired antigen?

(A) The gene coding for the antigen can be inserted into plasmids that can be used to transform the bacteria. ✓

(B) The bacteria need to be exposed to the antigen so they can produce the antibodies.

(C) The DNA of the antigen has to be transcribed in order for the mRNA produced to be inserted into the bacteria.

(D) The mRNA of the antigen has to be translated in order for the protein to be inserted into the bacteria.

Answer A

Correct. The bacterial plasmid containing the antigen gene would be taken up by bacterial cells through transformation. This gene would be transcribed and translated by the bacteria to produce the desired antigen.

Tay-Sachs disease is a rare inherited disorder caused by an autosomal recessive allele of the *HEXA* gene. Affected individuals exhibit severe neurological symptoms and do not survive to reproductive age. Individuals who inherit one copy of the allele (Tay-Sachs carriers) typically show no symptoms of the disorder. The frequencies of Tay-Sachs carriers in the general population of North America and in three different subpopulations are presented in the table.

FREQUENCY OF TAY-SACHS CARRIERS IN DIFFERENT POPULATIONS

Population	Frequency of Tay-Sachs Carriers
General population	0.004
Subpopulation I	0.037
Subpopulation II	0.035
Subpopulation III	0.020

21. A researcher claims that Tay-Sachs carriers are protected against the infectious disease tuberculosis (TB). Which of the following observations about the annual incidence of tuberculosis in subpopulation II could best be used to support the researcher's claim?

6 Gene Expression and Regulation MCQ

(A) The incidence of TB in subpopulation II is roughly equal to the incidence of TB in the general population.

(B) The incidence of TB in subpopulation II is greater than the incidence of TB in the general population.

(C) The incidence of TB in subpopulation II is lower than the incidence of TB in the general population. ✓

(D) The incidence of TB in subpopulation II is roughly equal to the incidence of Tay-Sachs disease in the general population.

22. Based on the information presented, which of the following best explains the difference in phenotype between Tay-Sachs carriers and homozygous recessive individuals?

(A) Tay-Sachs carriers received a vaccination that homozygous recessive individuals did not receive.

(B) Tay-Sachs carriers inherited an extra chromosome that homozygous recessive individuals did not inherit.

(C) Tay-Sachs carriers have access to a critical nutrient that homozygous recessive individuals did not inherit.

(D) Tay-Sachs carriers synthesize an essential enzyme that homozygous recessive individuals cannot synthesize. ✓

23. Which of the following is an ethical question about Tay-Sachs disease that cannot be answered using scientific methods?

(A) Would a difference in the *HEXA* alleles in subpopulations I and II affect the severity of the neurological symptoms?

(B) Should genetic testing be required before individuals in subpopulation III are advised to not have children? ✓

(C) Could the frequency of Tay-Sachs carriers in subpopulation I be a consequence of a genetic bottleneck?

(D) Should a statistical test be used to evaluate whether the general population is in Hardy-Weinberg equilibrium?

6 Gene Expression and Regulation MCQ

24. Arsenic is a toxic element found in both aquatic and terrestrial environments. Scientists have found genes that allow bacteria to remove arsenic from their cytoplasm. Arsenic enters cells as arsenate that must be converted to arsenite to leave cells. Figure 1 provides a summary of the arsenic resistance genes found in the operons of three different bacteria. *E. coli* R773 is found in environments with low arsenic levels. *Herminiimonas arsenicoxydans* and *Ochrobactrum tritici* are both found in arsenic-rich environments.

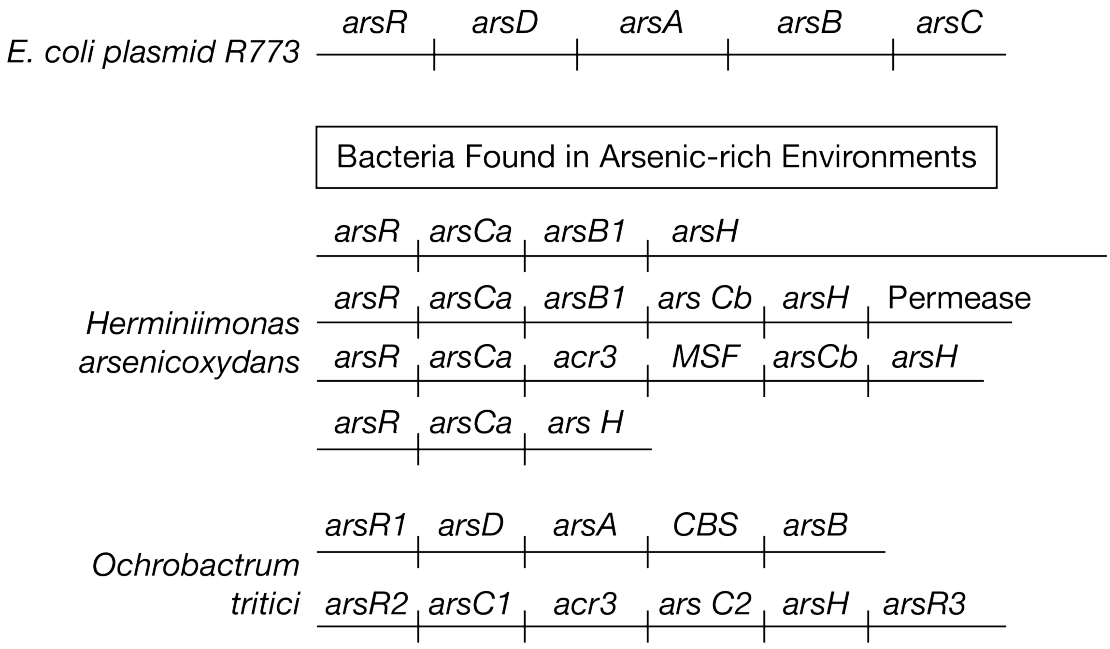


Figure 1. Operons found in three selected bacteria for arsenic removal

Researchers claim that bacteria that live in environments heavily contaminated with arsenic are more efficient at processing arsenic into arsenite and removing this toxin from their cells. Justify this claim based on the evidence shown in Figure 1.

(A) There are multiple operons controlling the production of proteins that process and remove arsenite from cells in both *H. arsenicoxydans* and *O. tritici*. In contrast, *E. coli* has only one operon devoted to arsenic removal. ✓

(B) Both *H. arsenicoxydans* and *O. tritici* contain the $arsR$ gene that codes for a repressor that turns on the operon to eliminate arsenite from the cell.

(C) Both *O. tritici* and *E. coli* contain the $arsD$ gene, which codes for a protein that helps remove arsenite from the cell.

(D) Both *H. arsenicoxydans* and *O. tritici*. have more arsenic resistance genes than has *E. coli*.

Answer A

Correct. Species found in arsenic-rich environments have several operons allowing for several pathways to process and remove arsenite. Also, this allows for a greater capacity of gene regulation.

6 Gene Expression and Regulation MCQ

Ascorbic acid (vitamin C) is an organic molecule necessary for the health of plants and animals. The majority of animals, including most mammals, synthesize ascorbic acid from organic precursors, but some primates are unable to synthesize ascorbic acid and must instead acquire it from dietary sources, such as certain fruits and vegetables.

The *L-gulonolactone oxidase (GULO)* gene encodes an enzyme that catalyzes a required step in the biosynthesis of ascorbic acid. Most mammals carry a functional copy of the *GULO* gene, but some primates carry only a *GULO* pseudogene, which is a nonfunctional variant.

A comparison of *GULO* genes and *GULO* pseudogenes from different animals can provide insight into the evolutionary relatedness of the animals. In Table I, selected members of some mammalian groups are listed, along with an indication of their ability to synthesize ascorbic acid. Table II shows an alignment of amino acid coding sequences from homologous regions of the *GULO* genes and *GULO* pseudogenes of the organisms listed in Table I. Figure 1 represents the universal genetic code.

TABLE I: SELECTED MAMMALIAN GROUPS

Group	Selected Members	Biosynthesis of Ascorbic Acid
Nonprimate mammals	Elephant, mouse	Yes
Primate mammals	Lemur	Yes
	Orangutan, chimpanzee	No
	Human	No

TABLE II: DNA SEQUENCE ALIGNMENT*

	Relative Positions of Nucleotides in Nontemplate (Coding) Sequence																										
	1 (5′)	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27 (3′)
Elephant	G	A	C	A	C	C	C	A	T	C	T	G	A	A	G	A	A	G	T	C	G	G	A	A	T	A	C
Mouse	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	A	G	T	C	T	G	A	G	T	A	C
Lemur	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	G	G	T	C	C	G	A	G	T	A	C
Orangutan	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Chimp	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Human	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C

*For each DNA segment, the alternating shaded and unshaded nucleotides indicate the triplet codons of the open reading frame, shown from left (5′) to right (3′) as the nontemplate (coding) strand. An "–" indicates the absence of a nucleotide.

6 Gene Expression and Regulation MCQ

Second Base

5′ First Base	U	C	A	G	3′ Third Base
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA stop	UGA stop	A
	UUG Leu	UCG Ser	UAG stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

Figure 1. Universal genetic code.

25. Comparison of DNA sequences in Table II suggests that a functional *GULO* gene in lemurs can have a G, C, or T at position 21 but only a G at position 22. Which of the following pairs of predictions is most helpful in explaining the discrepancy?

6 Gene Expression and Regulation MCQ

(A)

A substitution at position 21 would result in	A substitution at position 22 would result in
No change to the protein	A premature stop codon or an amino acid with different biochemical characteristics

(B)

A substitution at position 21 would result in	A substitution at position 22 would result in
A different amino acid	A premature stop codon or an amino acid with different biochemical characteristics

(C)

A substitution at position 21 would result in	A substitution at position 22 would result in
No change to the protein	A frame shift producing an inactive protein

(D)

A substitution at position 21 would result in	A substitution at position 22 would result in
An amino acid with different biochemical characteristics	No transcription of the gene

26. Which of the following is the most likely effect of the mutation at nucleotide position 7 in the *GULO* gene of humans?

6 Gene Expression and Regulation MCQ

(A) The mutation results in the deletion of the *GULO* gene, so no polypeptide can be translated.

(B) The deletion of the single nucleotide causes a frame shift, changing the primary structure downstream of the mutation and resulting in a nonfunctional protein. ✓

(C) The point mutation causes a substitution of the amino acid isoleucine (Ile)for histidine (His) at position 7, resulting in a protein with higher than normal activity.

(D) The substitution of a single nucleotide in the *GULO* coding region results in a stop codon. This results in a smaller nonfunctional protein.

27. Lemurs are primates that live on the island of Madagascar off the coast of Africa. Lemurs have a functional *GULO* gene and are able to produce ascorbic acid. However, primates that live in other places (e.g., humans, chimpanzees, and orangutans) have a *GULO* pseudogene and are unable to produce ascorbic acid. Which of the following best explains the genetic variation among primate species?

(A) Food sources on the island of Madagascar were deficient in ascorbic acid, which caused the primates to migrate to the mainland where ascorbic acid–rich foods are more widely available.

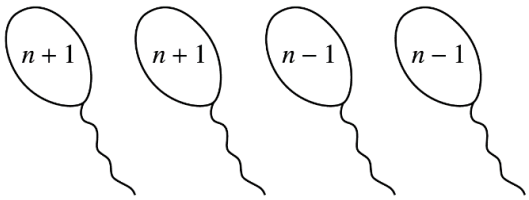
(B) Human activity on Madagascar, including poaching, drove the other primates into extinction on the island.

(C) Food sources where non-Madagascar primates lived provided ample ascorbic acid in the diet, which removed the selective pressure for maintaining a functional *GULO* gene. ✓

(D) Food sources on the island of Madagascar were deficient in ascorbic acid, which caused lemurs to adapt by developing the ability to produce an active GULO protein.

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28. A spermatocyte produces the following four sperm cells.



These cells are the result of nondisjunction during which of the following phases?

(A) Interphase (G1 or G2)

(B) Interphase (S)

(C) Mitosis

(D) Meiosis I ✓

(E) Meiosis II

29. Which of the following is a correct statement about mutations?

6 Gene Expression and Regulation MCQ

(A) They are a source of variation for evolution. ✓

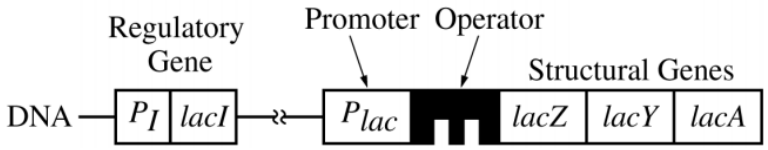
(B) They drive evolution by creating mutation pressures.

(C) They are irreversible.

(D) They occur in germ cells but not in somatic cells.

(E) They are most often beneficial to the organisms in which they occur.

LAC OPERON STRUCTURE



The functions of the loci of the *lac* operon shown in the diagram are described in the table below.

Locus	Function
P_I	Attachment site for RNA polymerase
lacI	Encodes a repressor protein that prevents transcription of the structural genes of the *lac* operon
P_{lac}	Attachment site for RNA polymerase
Operator	Binding site for the repressor protein
lacZ	Encodes beta-galactosidase, the enzyme that digests lactose to glucose and galactose
lacY	Encodes lactose permease, the channel through which lactose moves into the cell
lacA	Encodes galactoside transacetylase

The diagram above represents a segment of the *E. coli* chromosome that contains the *lacI* gene and part of the *lac* operon, a coordinately regulated set of genes that are required for the metabolism of lactose. The presence of lactose, which causes the repressor to be released from the operator, results in increased transcription of the *lac* operon.

30. Bacterial cells that contain green fluorescent protein (GFP) will fluoresce under ultraviolet light. Which of the following is the most likely outcome of replacing the *lacZ* gene in the *E. coli lac* operon with the gene encoding GFP?

(A) Bacteria growing in the presence of lactose will fluoresce under ultraviolet light. ✓

(B) Beta-galactosidase will be made only when bacteria are cultured under ultraviolet light.

(C) Ultraviolet light will cause a bond to form between glucose and galactose monomers.

(D) Ultraviolet light will cause a duplication of the *lac* operon.

31. Which of the following best explains the contribution of the *lac* operon to the metabolic efficiency of a bacterial cell?

6 Gene Expression and Regulation MCQ

(A) Expression of the *lacI* gene requires lactose.

(B) RNA polymerase is rapidly degraded by the product of the *lacP* locus.

(C) The repressor binds to DNA only when the cellular concentration of glucose is low.

(D) The *lacZ* gene is highly expressed only when lactose is available. ✓

32. Which of the following is the most likely consequence of a mutation at the operator locus that prevents binding of the repressor protein?

(A) Expression of the structural genes will be repressed, even in the presence of lactose.

(B) Beta-galactosidase will be produced, even in the absence of lactose. ✓

(C) RNA polymerase will attach at the P_{lac} locus, but transcription will be blocked.

(D) The operator locus will code for a different protein and thereby prevent transcription of the structural gene.

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33. Living cells typically have biosynthetic pathways to synthesize at least some of the amino acids used in making proteins. Some strains of *E. coli*, a prokaryote, can synthesize the amino acid tryptophan, while other *E. coli* strains cannot. Similarly, some strains of the yeast *S. cerevisiae*, a eukaryote, can synthesize tryptophan, while other *S. cerevisiae* strains cannot.

Which of the following describes the most likely source of genetic variation found in the tryptophan synthesis pathways of both species?

(A) Exchange of genetic information occurs through crossing over.

(B) Viral transmission of genetic information required to synthesize tryptophan occurs.

(C) Random assortment of chromosomes leads to genetic variation.

(D) Errors in DNA replication lead to genetic variation. ✓

Answer D

This option is correct. The most likely source of genetic variation found in the tryptophan synthesis pathways of both species would have occurred primarily with the imperfect nature of DNA replication, which could have led to an increase in genetic variation in both species.

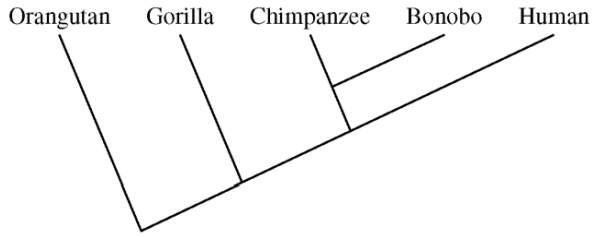
6 Gene Expression and Regulation MCQ

The *TAS2R38* gene encodes a receptor protein that influences the ability to taste bitterness. The gene has two alleles: a dominant, wild-type allele that enables an individual (taster) to taste bitterness and a recessive, mutant allele that interferes with the ability of an individual (nontaster) to taste bitterness. Three single nucleotide mutations in the coding region of the *TAS2R38* gene are associated with the nontaster allele. The nucleotides present at the three positions are shown in the table below.

NUCLEOTIDE VARIATION IN THE *TAS2R38* GENE

	Position in the Nucleotide Sequence		
	145	785	886
Human nontaster	G	T	A
Human taster	C	C	G
Bonobo	C	C	G
Chimpanzee	C	C	G
Gorilla	C	C	G

A cladogram representing the evolutionary relatedness of selected primates is shown below.



34. The TAS2R38 receptor protein has been detected on the surface of cells from individuals who are homozygous for the nontaster allele of the *TAS2R38* gene. Which of the following is the most likely effect of the mutations associated with the nontaster allele on *TAS2R38* gene expression?

(A) The mutations change the primary structure of the encoded receptor protein. ✓

(B) The mutations increase the stability of the *TAS2R38* mRNA.

(C) The mutations prevent transcription of the *TAS2R38* gene.

(D) The mutations prevent translation of the *TAS2R38* mRNA.

6 Gene Expression and Regulation MCQ

The following questions refer to the following DNA strand and table of codons.

	3′ TAG	TTC	AAA	CCG	CGT	AAC	ATT 5′
Triplet Number	1	2	3	4	5	6	7

Second Letter

First Letter (5′ End)	U	C	A	G	Third Letter (3′ End)
U	UUU phe	UCU ser	UAU tyr	UGU cys	U
	UUC phe	UCC ser	UAC tyr	UGC cys	C
	UUA leu	UCA ser	UAA stop	UGA stop	A
	UUG leu	UCG ser	UAG stop	UGG trp	G
C	CUU leu	CCU pro	CAU his	CGU arg	U
	CUC leu	CCC pro	CAC his	CGC arg	C
	CUA leu	CCA pro	CAA gln	CGA arg	A
	CUG leu	CCG pro	CAG gln	CGG arg	G
A	AUU ile	ACU thr	AAU asn	AGU ser	U
	AUC ile	ACC thr	AAC asn	AGC ser	C
	AUA ile	ACA thr	AAA lys	AGA arg	A
	AUG met	ACG thr	AAG lys	AGG arg	G
G	GUU val	GCU ala	GAU asp	GGU gly	U
	GUC val	GCC ala	GAC asp	GGC gly	C
	GUA val	GCA ala	GAA glu	GGA gly	A
	GUG val	GCG ala	GAG glu	GGG gly	G

35. Which of the following modifications of the DNA would produce the greatest change in the primary structure of the polypeptide chain?

(A) Deleting the first T in the second triplet ✓

(B) Changing the second triplet to read 3’ CTC 5’

(C) Changing the third triplet to read 3’ AAC 5’

(D) Changing the fourth triplet to read 3’ CCA 5’

(E) Deleting the sixth triplet

36. In which of the following would there NOT be a change in the amino acid sequence of the peptide coded for by this DNA?

(A) Changing 3’ AAA 5’ to read 3’ AAG 5’ ✓

(B) Changing 3’ TTC 5’ to read 3’ ATC 5’

(C) Changing 3’ CCG 5’ to read 3’ GGC 5’

(D) Deleting the first A from 3’ AAA 5’

(E) Deleting the last triplet

6 Gene Expression and Regulation MCQ

37. The mRNA transcribed from the DNA would read

(A) 5’ TAG TTC AAA CCG CGT AAC AAT 3’

(B) 5’ ATC AAG TTT GGC GCA TTG TAA 3’

(C) 5’ AUC AAG UUU GGC GCA UUG UAA 3’ ✓

(D) 5’ AAU CAA UGC GCC AAA CUU GAU 3’

(E) 5’ AUU GUU ACG CGG UUU GAA CUA 3’

---

38. Red-green color blindness in humans is caused by a recessive allele located on the X chromosome. Figure 1 shows the potential offspring of a female who is red-green color-blind and a male with full-color vision. All of the possible male offspring would be color-blind, and all of the possible female offspring would have full-color vision.

If during the production of male gametes an error in meiosis occurred, sperm containing both an X and a Y chromosome could be produced.

Potential Offspring With No Error in Meiosis				Potential Offspring With Error in Meiosis		
	X^B	Y				$\mathrm{X}^B\mathrm{Y}$
X^b	$\mathrm{X}^B\mathrm{X}^b$	$\mathrm{X}^b\mathrm{Y}$	$\mathrm{X}^B = \text{Full color vision}$ $\mathrm{X}^b = \text{Red-green colorblind}$	X^b	X^b	$\mathrm{X}^b\mathrm{X}^B\mathrm{Y}$
X^b	$\mathrm{X}^B\mathrm{X}^b$	$\mathrm{X}^b\mathrm{Y}$		X^b	X^b	$\mathrm{X}^b\mathrm{X}^B\mathrm{Y}$

Figure 1. Possible offspring of a female who is red-green color-blind and a male who has full-color vision.

How would the extra chromosome affect the male offspring produced by the gamete?

(A) None of the potential offspring would be male, because the potentially male zygote would have two X chromosomes, and the Y chromosome would be ignored.

(B) The male offspring would all be red-green color-blind, because of interference from alleles on the Y chromosome.

(C) The male offspring would have full-color vision, because of the presence of the extra X chromosome. ✓

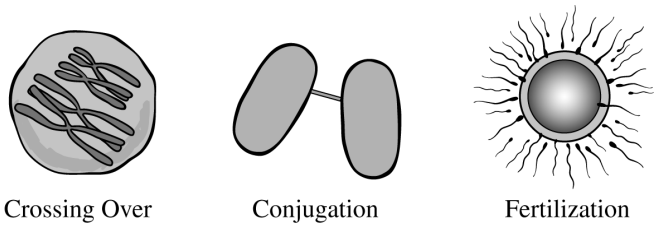
(D) There would be no change to the phenotypes of the possible offspring, because the extra X chromosome would not be active.

Answer C

Correct. The presence of the Y chromosome would trigger the development of a male, and the extra X chromosome would provide an allele for full-color vision, a dominant trait.

6 Gene Expression and Regulation MCQ

39.



Crossing Over Conjugation Fertilization

The processes illustrated in the models depicted above all result in which of the following?

(A) Transcription

(B) An increase in genetic variation ✓

(C) An increase in the chromosome number

(D) Horizontal gene transfer

40. Cystic fibrosis (CF) is a progressive genetic disease that causes persistent lung infections and affects the ability to breathe. CF is inherited in an autosomal recessive manner, caused by the presence of mutations in both copies of the gene for the cystic fibrosis transmembrane conductance regulator (CFTR) protein. Partial nucleotide sequences and the corresponding amino acid sequences for an unaffected individual and an affected individual are modeled in Figure 1.

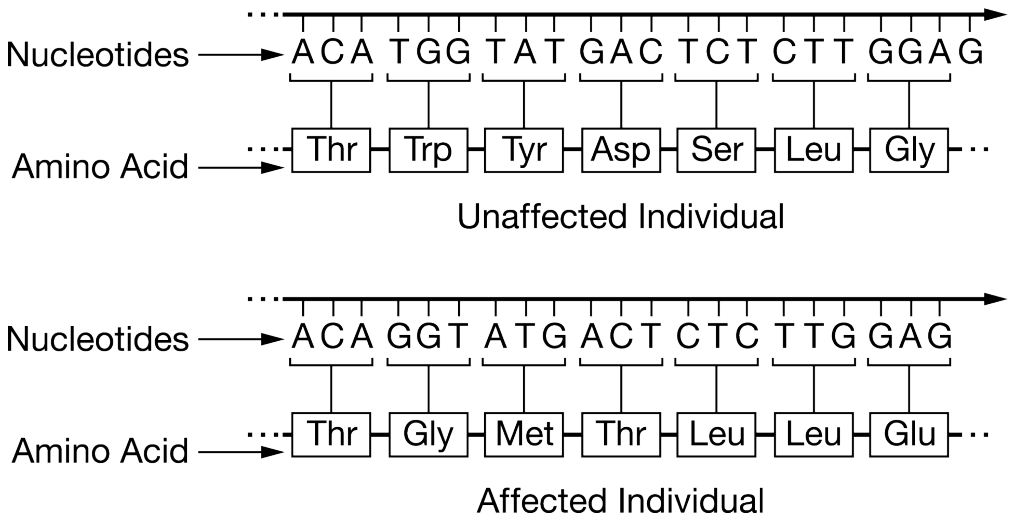


Figure 1. CFTR protein sequences in unaffected and affected individuals

Based on the information in Figure 1, which type of mutation explains the nature of the change in DNA that resulted in cystic fibrosis in the affected individual?

(A) Substitution, because the amino acid tryptophan is replaced with glycine.

(B) Insertion, because an extra guanine is present, which changes the reading frame.

(C) Deletion, because a thymine is missing, which changes the reading frame. ✓

(D) Duplication, because the amino acid leucine occurs twice, which changes the reading frame.

6 Gene Expression and Regulation MCQ

Answer C

Correct. Deletion of the thymine nucleotide results in a frameshift that alters the triplet code sequence and changes the rest of the amino acid sequence.



Figure 1. Domain structure of two proteins

Investigators studied the interactions between two different proteins, protein I and protein II. Each protein has two structural domains (Figure 1) that represent different functional parts of the protein. Different combinations of the protein domains were tested for their ability to bind to a known DNA sequence, interact with each other, and activate transcription. The results are shown in Table 1.

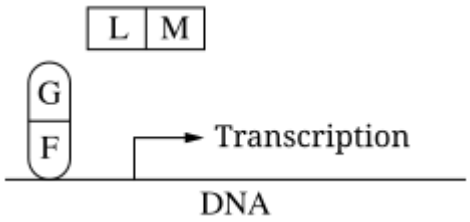
TABLE 1. INTERACTION AND FUNCTION OF COMBINATIONS OF STRUCTURAL VARIANTS OF PROTEINS I AND II

Test	Domain(s) of Protein I Used	Domain(s) of Protein II Used	DNA Binding	Protein-Protein Interaction	Transcription Activation
1	FG	LM	+	+	+
2	FG	None	+	−	−
3	None	LM	−	−	−
4	F	LM	+	−	−
5	G	LM	−	+	−
6	FG	L	+	−	−
7	FG	M	+	+	+

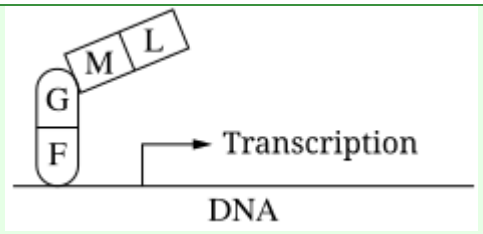
6 Gene Expression and Regulation MCQ

41. Which of the following best models the interaction of protein I and protein II that results in the transcription of target genes?

(A)



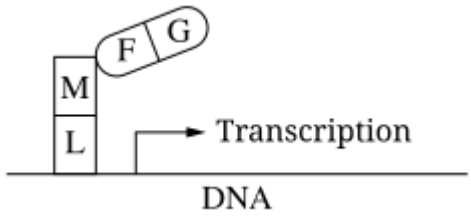
(B)



(C)



(D)



Answer B

Correct. Based on the data for test 4 (in comparison with the data for tests 1, 2, and 5), the F domain of protein I is responsible for DNA binding. The data for tests 5 and 7 (in comparison with the data for tests 4 and 6) demonstrate that the G domain of protein I interacts with the M domain of protein II.

6 Gene Expression and Regulation MCQ

Dystrophin is a protein that is expressed in certain muscle cells. In combination with other cellular proteins, dystrophin strengthens protein fibers in muscle cells to allow muscles to contract without injury.

Nucleotide deletions in the gene that encodes dystrophin are associated with the genetic disorder Duchenne Muscular Dystrophy (DMD). Individuals with DMD do not produce functional dystrophin and, as a result, the protein fibers, and then entire muscle cells, become damaged.

The history of DMD for three generations of a family is shown in Figure 1.

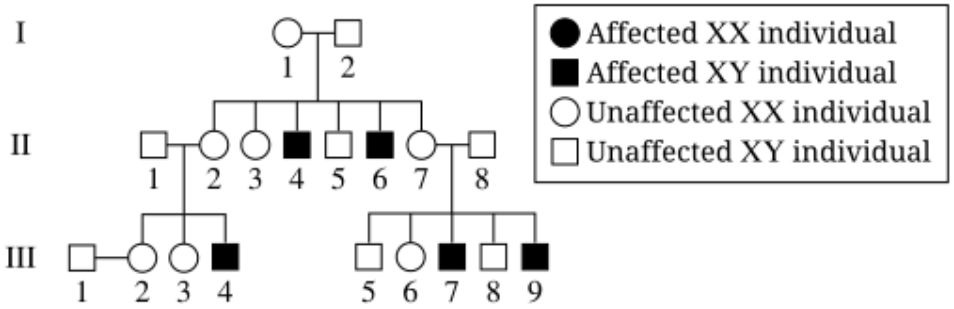


Figure 1. Pedigree showing instances of DMD in a family

Individuals III–1 and III–2 plan to have children and wish to first determine whether individual III–2 is a carrier of DMD. Individual III–2 undergoes genetic testing to determine whether individual III-2 carries a particular allele for the mutated dystrophin that is associated with the disorder in this family. The results of gel electrophoresis analysis of the individual's dystrophin alleles and the alleles of several family members are shown in Figure 2.

6 Gene Expression and Regulation MCQ

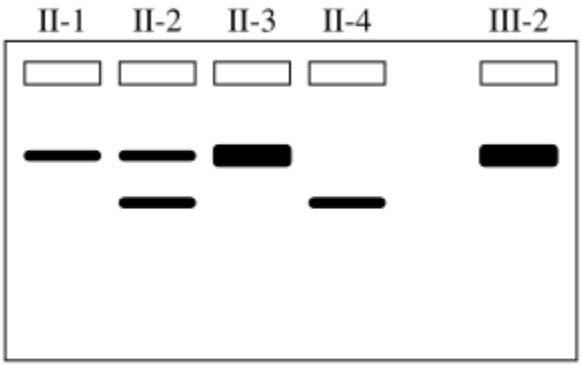


Figure 2. Gel electrophoresis showing the dystrophin alleles present in five members of the family

42. Which of the following best explains how the dystrophin protein is synthesized by certain muscle cells of an individual but not by other cell types?

(A) Each cell has different genes as a result of changes that occur during development and cell specialization.

(B) The presence of different transcription factors in different cell types results in differential gene expression. ✓

(C) A cell can only translate mRNA transcripts that have gene-specific sequences that are recognized by the ribosomes.

(D) The promoter region of the gene is necessary for expression to occur, and this region is only present in certain cell types.

Answer B

Correct. Transcription factors, in general, determine which parts of the genome will be transcribed. Tissue- and cell-specific transcription factors result in differential gene expression that leads to different cell types synthesizing different proteins.

43. Another form of muscular dystrophy, BMD, has a less severe phenotype. Individuals with BMD produce less dystrophin protein or abnormal dystrophin protein with reduced function, whereas individuals with DMD produce no dystrophin protein. Both BMD and DMD are commonly caused by the deletion of a portion of the dystrophin gene.

Which of the following best explains why there is a difference in phenotype between BMD and DMD?

6 Gene Expression and Regulation MCQ

(A) BMD can be caused by the deletion of exons, while DMD is caused by the deletion of introns.

(B) BMD can be caused by a deletion that affects a single amino acid, while DMD is caused by a deletion that alters all the amino acids that follow the mutation. ✓

(C) BMD can be caused by a deletion that removes the transcription start site, while DMD is caused by a deletion that removes the transcription stop signal.

(D) BMD can be caused by the inheritance of two alleles containing deletions, while DMD is caused by the inheritance of one allele containing the deletion and one normal allele.

Answer B

Correct. A single amino acid change may alter the function of the protein but would be expected to be less detrimental than a frameshift mutation that alters every amino acid added after the mutation.

44. Over 2,000 different mutations in the gene encoding dystrophin are associated with different forms of muscular dystrophy. The mutations include deletions of a portion of the gene, duplications of a portion of the gene, and sometimes substitutions of base pairs in certain regions of the gene.

Which of the following correctly predicts the movement of a mutant allele compared with the movement of the normal allele when analyzed by gel electrophoresis?

(A) An allele with base pair substitutions in an exon will travel farther than the normal allele does.

(B) An allele with a duplication in an exon will travel a shorter distance than the normal allele does. ✓

(C) An allele with base pair deletions in an exon will travel the same distance as the normal allele does.

(D) An allele with a duplication in an exon will travel a longer distance than the normal allele does.

Answer B

Correct. An allele with a duplication will contain more nucleotides than in a normal allele. Given the same charge and amount of time, longer DNA sequences with a larger molecular mass will not move as far as shorter DNA sequences will.

6 Gene Expression and Regulation MCQ

45. An evolutionary biologist hypothesizes that two morphologically similar plant species are not closely related. To test the hypothesis, the biologist collects DNA samples from each of the two plant species and then uses restriction enzymes to cut the DNA samples into fragments, which are then subjected to gel electrophoresis. The results are shown in Figure 1.

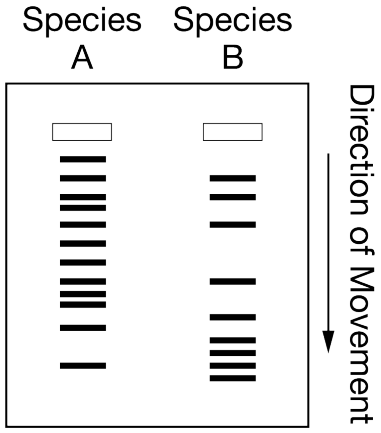


Figure 1. DNA analysis of two species

Given the results shown in Figure 1, which of the following correctly explains a relationship between the two species?

(A) Species B is the ancestor of species A because it has fewer bands.

(B) Species A is more complex than species B because it has more bands.

(C) Species B has more short fragments of DNA than species A does. ✓

(D) Species A has more short fragments of DNA than species B does.

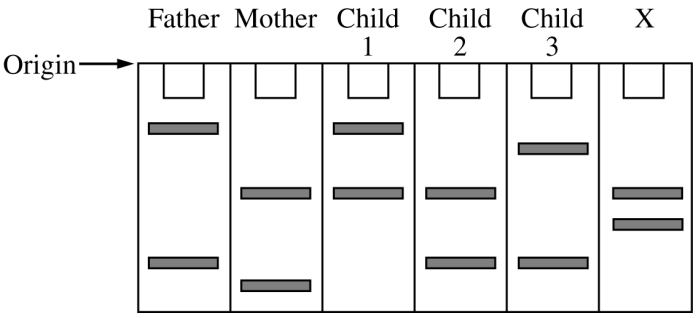
Answer C

Correct. Electrophoresis separates DNA fragments by size and charge. The smaller fragments will move more quickly through the gel. In this example, species B has more fragments that move further through the gel. Thus, species B has more short fragments than species A does.

6 Gene Expression and Regulation MCQ

For following group of questions first study the description of the situation or data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

The diagram below illustrates the results of electrophoresis of DNA sequences obtained from a family of two adults and three children, and amplified using PCR. The bands represent short repeating sequences of variable length. Results for another female (X) are included for comparison.



46. The banding patterns of the DNA fragments reveal that

(A) child 1 and child 2 cannot be biological siblings

(B) child 1 and child 3 probably look like the mother

(C) the mother cannot be the biological parent of all three children ✓

(D) the mother's DNA has the same DNA sequence as the father's DNA

(E) child 2 and child 3 inherited all of their DNA from the father

47. Which of the following is the best explanation for the fragment pattern for individual X ?

(A) She has only one member of this chromosome pair.

(B) She has only one living parent.

(C) She is homozygous for this particular DNA fragment.

(D) She is the mother's child from another marriage. ✓

(E) She is not related to any member of the family being tested.

48. Gel electrophoresis can be used to directly determine which of the following?

(A) The amino acid sequence of a protein

(B) The number of base pairs in a fragment of DNA ✓

(C) The organelle in a cell toward which a protein moves after its initial synthesis

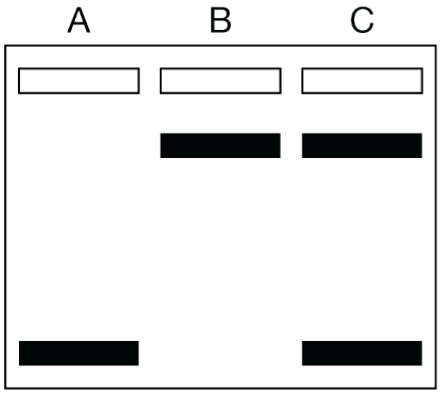
(D) Whether one allele of a gene is more subject to genetic drift than are other alleles of the same gene

6 Gene Expression and Regulation MCQ

Answer B

Correct. By using an electrophoresis gel, the number of base pairs in a fragment of DNA can be estimated by comparing the unknown DNA fragment to a set of standard DNA fragments with a known number of base pairs.

49. Scientists conducted a transformation experiment using *E. coli bacteria* and the pTru plasmid. Samples of the pTru plasmid (lane A) and the chromosomal DNA from two different *E. coli* strains that the scientists attempted to transform (lane B and lane C) were compared using gel electrophoresis. The results are shown in Figure 1.



Lane	Sample
A	pTru
B	*E. coli* strain I
C	*E. coli* strain II

Figure 1. Results of *E. coli* transformation with pTru plasmid

Which of the following statements best explains the experimental results observed in Figure 1 ?

(A) *E. coli* in both lanes B and C have been successfully transformed and contain additional genetic information.

(B) *E. coli* in lane B have been successfully transformed and contain additional genetic information.

(C) *E. coli* in lane C have been successfully transformed and contain additional genetic information. ✓

(D) Which *E. coli* have been transformed cannot be determined from this gel.

Answer C

Correct. A successful transformation would contain a band of DNA from the pTru plasmid and a band of DNA from the *E. coli* chromosome. Lane C has both bands correctly identified and is experimental evidence of a successful bacteria transformation of *E. coli* with the pTru plasmid.

6 Gene Expression and Regulation MCQ

50. Students subjected three samples of five different molecules to gel electrophoresis as shown in Figure 1.

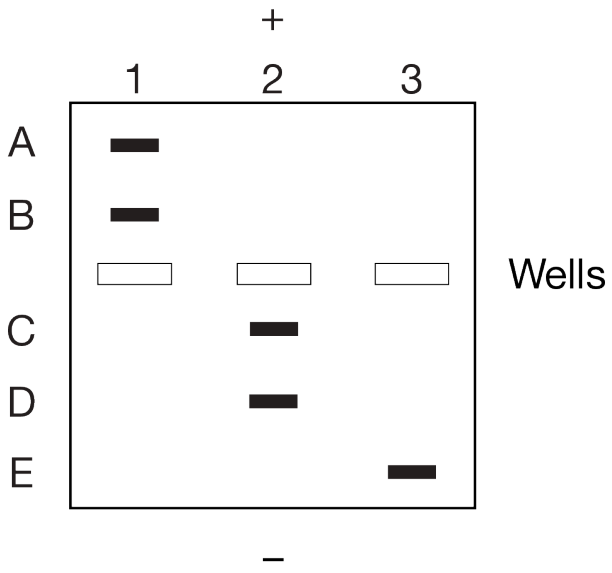


Figure 1. Gel electrophoresis of three prepared samples

Which of the following statements best explains the pattern seen on the gel with regard to the size and charge of molecules A and B?

(A) Molecules A and B are positively charged, and molecule A is smaller than molecule B.

(B) Molecules A and B are positively charged, and molecule A is larger than molecule B.

(C) Molecules A and B are negatively charged, and molecule A is smaller than molecule B. ✓

(D) Molecules A and B are negatively charged, and molecule A is larger than molecule B.

Answer C

Correct. Only negatively charged molecules will move toward the positive electrode. The smaller molecule A will migrate further from the well than the larger molecule B will.

6 Gene Expression and Regulation MCQ

51. Mutations in the *MYO6* and *POU4F3* genes have been associated with a form of hereditary hearing loss in humans. Researchers studying the genes have proposed that *POU4F3* encodes a transcription factor that influences the regulation of *MYO6*.

Which of the following questions will best help guide the researchers toward a direct test of their proposal?

(A) Have mutations in other genes also been associated with hearing loss?

(B) In what types of cells are the mutant forms of the *POU4F3* gene expressed?

(C) Are mutations in the *MYO6* and *POU4F3* genes also found in mice?

(D) Do mutations in the *POU4F3* gene affect *MYO6* mRNA levels in cells? ✓

52. Which of the following is most likely to create genetic variation in a population?

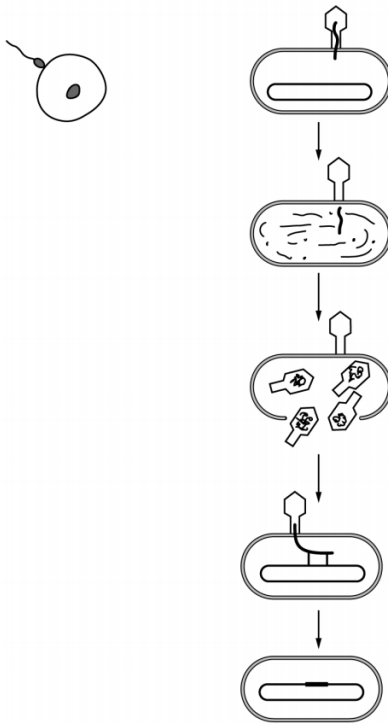
(A) RNA polymerase errors during transcription

(B) Helicase failure to unwind DNA during DNA replication

(C) DNA polymerase errors during replication ✓

(D) Misincorporation of amino acids by tRNA during translation

53.



Which of the following correctly compares the two processes shown above?

(A) Both processes increase genetic variation. ✓

(B) Both processes represent aspects of sexual reproduction.

(C) Both processes require partitioning of genetic material and organelles.

(D) The amount of genetic material per cell remains constant in both processes.

6 Gene Expression and Regulation MCQ

54. Gibberellic acid stimulates the cells of germinating grass seeds to produce mRNA molecules that code for hydrolytic enzymes. In this case the role of gibberellic acid can best be described as that of

(A) a regulator of gene activity ✓

(B) a stimulator of hydrolase secretion

(C) a stimulator of DNA replication

(D) an allosteric activator of hydrolase

(E) an activator of translation

55. Huntington's disease has been traced to the number of CAG repeats in the *HTT* gene, which is located on chromosome 4. The phenotypic influence of individual alleles with different numbers of repeats is shown in Table 1.

Table 1: Effect of CAG repeat number on Huntington's disease expression

CAG Repeats	Expression
< 27	No symptoms
27–35	No symptoms but increased risk for offspring
36–39	May develop symptoms in old age
> 40	Will express symptoms from an early age

Which of the following is most likely the immediate cause of the first appearance of Huntington's disease in a person?

(A) A point mutation occurs in the *HTT* gene.

(B) The first appearance of the CAG repeat occurs in the *HTT* gene.

(C) An allele with more than 39 CAG repeats was inherited by the affected person. ✓

(D) The person inherited two alleles that each contained 20 CAG repeats.

Answer C

Correct. At some point, all alleles for the *HTT* gene contain less than 39 CAG repeats. Only after this threshold is crossed will a person inherit an allele that will cause symptoms of Huntington's disease.

6 Gene Expression and Regulation MCQ

56. Genetic engineering techniques can be used when analyzing and manipulating DNA and RNA. Scientists used gel electrophoresis to study transcription of gene L and discovered that mRNA strands of three different lengths are consistently produced.

Which of the following explanations best accounts for this experimental result?

(A) Gel electrophoresis can only be used with DNA (not mRNA), so experimental results are not interpretable.

(B) RNA polymerase consistently makes the same errors during transcription of gene L.

(C) Gene L is mutated, so RNA polymerase does not always transcribe the correct sequence.

(D) Pre-mRNA of gene L is subject to alternative splicing, so three mRNA sequences are possible. ✓

Answer D

Correct. Excision of different introns and splicing and retention of different exons can generate different versions of the resulting mRNA, which may have different lengths.

57. Insulin is a hormone produced by some pancreatic cells. Scientists have isolated the DNA sequence that codes for human insulin production.

Which of the following best predicts the effect of inserting this gene into the DNA of a bacterial cell?

(A) The recombinant bacterium will produce human insulin using its own transcription and translation machinery. ✓

(B) The recombinant bacterium will not transcribe the human insulin gene because this gene is not normally found in the bacterial genome.

(C) The recombinant bacterium will transcribe the gene but will be unable to translate the mRNA.

(D) The recombinant bacterium will die because it has been exposed to foreign DNA.

Answer A

Correct. DNA is the carrier of genetic information, and its insertion into the bacterium is sufficient to produce human insulin. The bacterium can transcribe the gene with RNA polymerase and translate the mRNA with its ribosomes.

6 Gene Expression and Regulation MCQ

Iron is an essential nutrient that is acquired by organisms from the environment. When intracellular levels of iron are relatively high, living cells synthesize an iron-storage protein called ferritin.

The induction of ferritin synthesis by iron was investigated in rats. Figure 1 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and actinomycin D, a drug that inhibits transcription.

Figure 2 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and cycloheximide, a drug that inhibits translation.

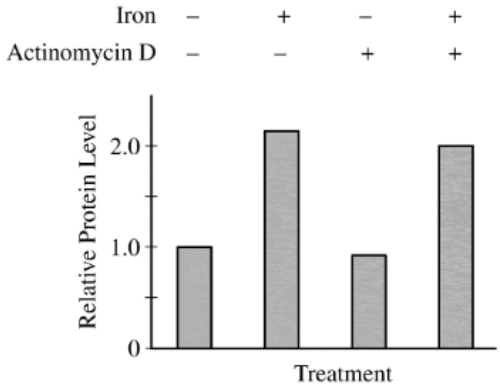


Figure 1. Ferritin Protein Levels

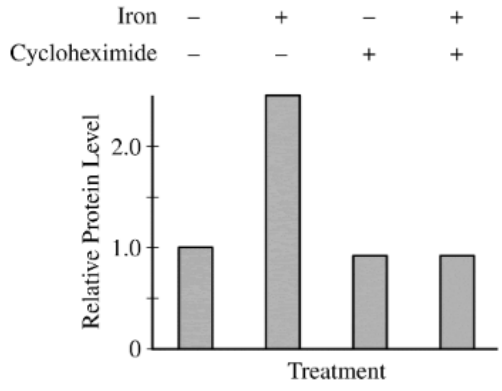


Figure 2. Ferritin Protein Levels

58. Which of the following biotechnology approaches could be used to identify ferritin mRNA in a sample of total cellular RNA?

(A) RNA samples could be directly cloned into a DNA plasmid, grown in bacteria, and tested for the ability to bind iron.

(B) RNA samples could be separated by size using agarose gel electrophoresis and incubated with labeled single-stranded DNA molecules that are complementary to the ferritin mRNA. ✓

(C) RNA samples could be converted to protein and subsequently cut with a restriction endonuclease that recognizes DNA sequences.

(D) RNA samples could be examined under a high-power microscope to visually identify the ferritin mRNA.

59. In individuals with iron-rich diets, intracellular levels of iron are higher than in individuals with iron-poor diets. Which of the following best explains how organisms regulate ferritin protein levels in response to available iron?

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(A) Translation of ferritin mRNA is inhibited in the presence of iron.

(B) Translation of ferritin mRNA is stimulated by iron. ✓

(C) Ferritin protein is more stable in the presence of iron.

(D) Ferritin protein is less stable in the presence of iron.

60. The data can best be used to support which of the following claims about the mechanism for regulating ferritin gene expression?

(A) Iron increases ribosome binding to ferritin mRNA. ✓

(B) Iron causes the ferritin mRNA to be degraded.

(C) Iron increases the effect of actinomycin D on RNA polymerase.

(D) Iron decreases the ability of ferritin mRNA to bind to ribosomes.

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61. The enzyme lactase aids in the digestion of lactose, a sugar found in the milk of most mammals. In most mammal species, adults do not produce lactase. Continuing to produce lactase into adulthood in people is called lactase persistence. A number of different alleles have been identified that result in lactase persistence. Figure 1 shows the percentage of people in different geographic areas parts of the Old World that exhibit lactase persistence.

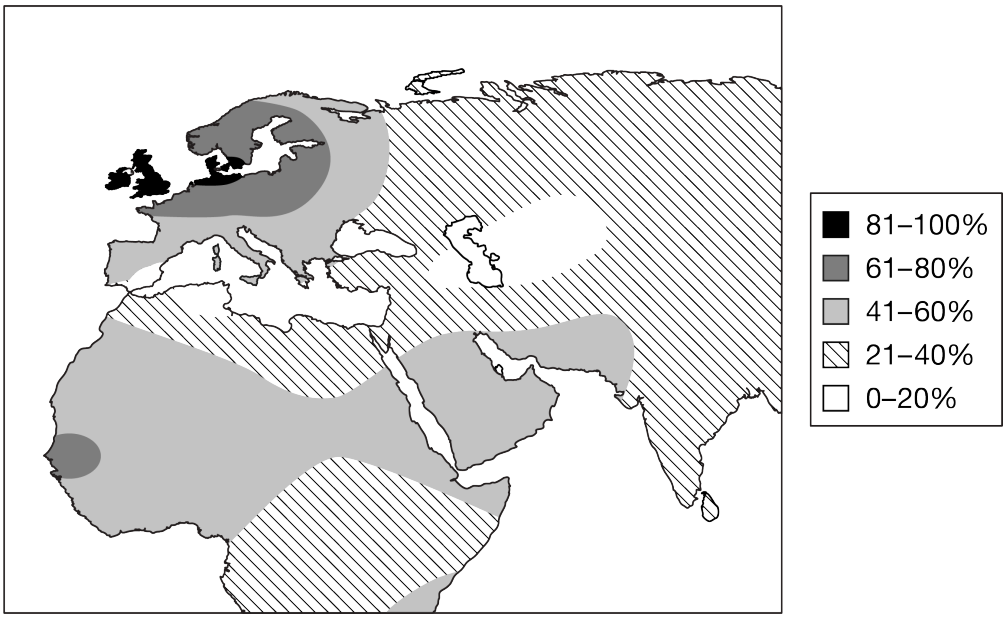


Figure 1. Distribution of lactase persistence in Europe, North Africa, and parts of Asia

Which of the following best explains the distribution of lactase persistence in the areas shown in Figure 1 ?

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(A) Lactase persistence developed because people were malnourished in Europe.

(B) Lactase persistence alleles are present in all human populations and are expressed when lactose is consumed.

(C) Mutations conferring lactase persistence likely arose independently in different geographic areas and offered a selective advantage. ✓

(D) The mutations that cause lactase persistence are detrimental to humans and will eventually disappear from the gene pool.

Answer C

Correct. The map shows two widely separated areas in which a high percentage of the population shows lactase persistence. Based on the distribution, it is likely that different mutations arose in these two different areas. In areas where dairy products are a significant portion of the adult diet, individuals with a lactase persistence allele are more likely to survive, reproduce, and pass on the mutation.

62. Both liver cells and lens cells have the genes for making the proteins albumin and crystalline. However, only liver cells express the blood protein albumin and only lens cells express crystalline, the main protein in the lens of the eye. Both of these genes have enhancer sequences associated with them.

The claim that gene regulation results in differential gene expression and influences cellular products (albumin or crystalline) is best supported by evidence in which of the following statements?

(A) Liver cells possess transcriptional activators that are different from those of lens cells. ✓

(B) Liver cells and lens cells use different RNA polymerase enzymes to transcribe DNA.

(C) Liver cells and lens cells possess the same transcriptional activators.

(D) Liver cells and lens cells possess different general transcription factors.

Answer A

Correct. The specific regulatory transcription factors produced in a given cell determines which genes are actually expressed. For example, the transcriptional activators that allow for substantial transcription of the albumin gene exist only in the liver cells.

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63. Lynch syndrome is an inherited condition associated with an increased risk for colon cancer, as well as certain other cancers. Mutations in one of several genes involved in DNA repair during DNA replication have been associated with Lynch syndrome. DNA sequencing was performed for an individual. The results indicated that the individual carries one of the dominant alleles that has been associated with Lynch syndrome.

Which of the following best explains how the results should be interpreted?

(A) The individual does not have an increased risk of developing cancer because one dominant allele is insufficient to cause the disease.

(B) The individual has an increased risk of developing colon cancer. ✓

(C) Because the person's DNA has the mutation, other family members must have cancer.

(D) Results cannot be interpreted until testing determines if additional mutated alleles are present.

Answer B

Correct. The individual may have an increased risk of developing colon cancer. Not all people with alleles associated with Lynch syndrome develop cancer.

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In mammals, the dark color of skin, hair, and eyes is due to a pigment called melanin. Melanin is produced by specialized skin cells called melanocytes. The melanin is then transferred to other skin cells called keratinocytes. Melanocytes synthesize melanin in a multistep metabolic pathway (Figure 1). The amount of melanin produced is dependent on the amount of the enzymes TYR, $\mathrm{TRP2}$, and $\mathrm{TRP1}$ present inside melanocytes.

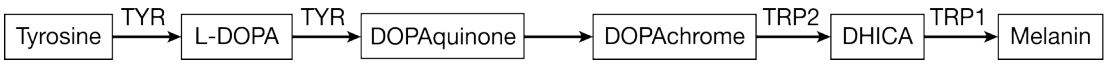


Figure 1. Melanin synthesis pathway

The peptide hormone α–melanocyte stimulating hormone (α–MSH) activates a signal transduction pathway leading to the activation of MITF. MITF is a transcription factor that increases the expression of the TYR, $TRP2$, and $TRP1$ genes (Figure 2).

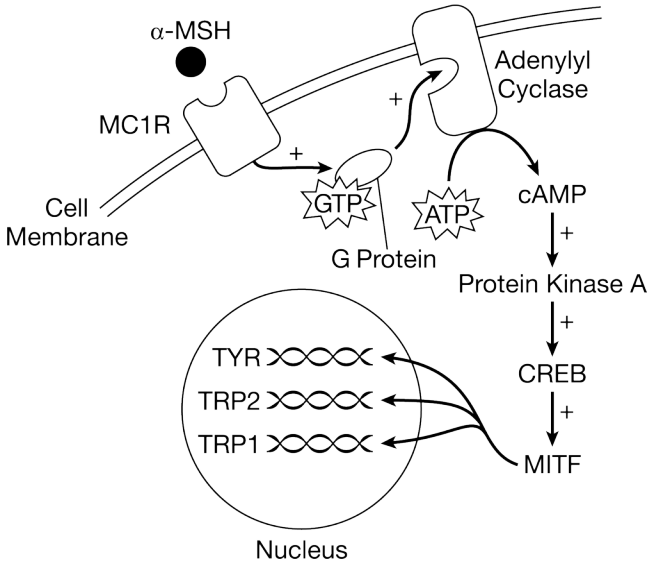


Figure 2. Activation of melanin synthesis genes in melanocytes

Some mammals increase melanin production in response to ultraviolet (UV) radiation. The UV radiation causes damage to DNA in keratinocytes, which activates the $\mathrm{p53}$ protein. $\mathrm{p53}$ increases the expression of the $POMC$ gene. The POMC protein is then cleaved to produce α–MSH. The keratinocytes secrete α–MSH, which signals nearby melanocytes. The increased melanin absorbs UV radiation, reducing further DNA damage.

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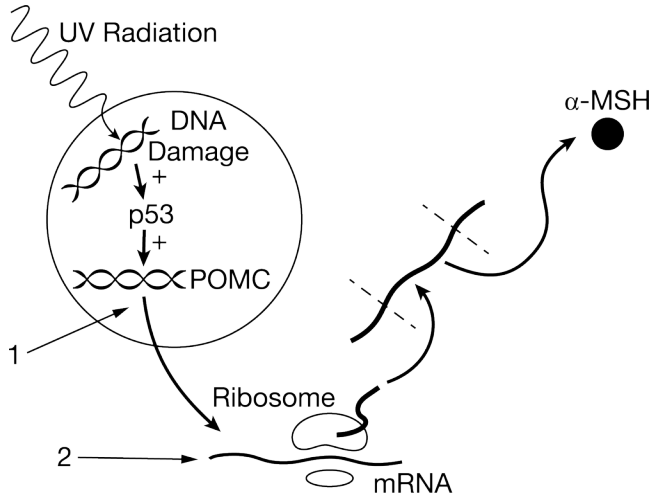


Figure 3. Production of α–MSH in keratinocytes in response to UV radiation

64. Mice have melanocytes in the skin on their ears and show a tanning response to UV radiation. Researchers were studying a mutant population of mice that do not show a tanning response. Genetic testing of these mutant mice showed that the pathway causing the production of α–MSH by keratinocytes in response to UV radiation was fully functional. Thus, the researchers claimed that the lack of tanning response was due a nonfunctional MC1R.

Which of the following pieces of evidence would best support the researchers' claim above?

(A) When researchers applied a drug that activates adenylyl cyclase to the mutant mice's ears, the level of melanin increased. ✓

(B) When researchers viewed sections of mutant mouse ears under the microscope, they found melanocyte numbers comparable to nonmutant mice.

(C) When researchers exposed the mutant mice to UV radiation, the amount of POMC mRNA in keratinocytes did not change.

(D) When researchers exposed the mutant mice to UV radiation, the level of melanin production did not change.

Answer A

Correct. Application of a drug that activates adenylyl cyclase would increase the production of cAMP, which would activate protein kinase A, which would activate CREB, which would activate MITF, which would activate the *TYR*, *TYR2*, and *TRP1* genes for the enzymes required to synthesize melanin. This evidence demonstrates that the parts of the signal transduction pathway from adenylyl cyclase onward are functional and that the enzymes required for melanin production are functional.

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65. Based on the information provided in Figure 1 and Figure 2, which of the following best explains the effects of a mutation in the promoter of the TYR gene that prevents it from being transcribed?

(A) DNA damage due to UV radiation will be strongly inhibited, resulting in a positive selection pressure.

(B) DNA damage due to UV radiation will be strongly inhibited, resulting in a negative selection pressure.

(C) Skin pigmentation will not be able to change, resulting in a positive selection pressure.

(D) Skin pigmentation will not be able to change, resulting in a negative selection pressure. ✓

Answer D

Correct. As shown in Figure 1, TYR is required for two steps in the melanin synthesis pathway. If it is not expressed properly, then this will prevent melanocytes from synthesizing melanin regardless of exposure to UV radiation. This would result in a negative selection pressure.

66. Which of the following best explains a process occurring between point 1 and point 2 in Figure 3 ?

(A) α–MSH is produced.

(B) The TYR gene is transcribed.

(C) Polypeptides are removed from a protein.

(D) A poly-A tail is added to RNA. ✓

Answer D

Correct. Eukaryotic mRNA is modified after transcription and a poly-A tail is added before it is translated at the ribosome.

67. Researchers discovered a mutant form of the TYR gene with a deletion of a single guanine nucleotide in the beginning of the coding sequence.

Which of the following best predicts the phenotype of an individual who is homozygous for this TYR mutation?

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(A) The mutation will cause a single amino acid change in the TYR protein, which will not be enough to disrupt its function. Therefore, those with this mutation will produce melanin in the hair, skin, and eyes and tan in response to UV radiation.

(B) The mutation will cause a single amino acid change in the TYR protein, leading to a nonfunctional TYR protein. Therefore, those with this mutation will lack melanin in the hair, skin, and eyes and will not tan in response to UV radiation.

(C) The mutation will change all subsequent amino acids in the TYR protein, leading to nonfunctional TYR protein. Since the *TRP1* and *TRP2* genes were not affected, the TRP1 and TRP2 proteins will fill the role of the TYR protein. Therefore, those with this mutation will produce melanin in the hair, skin, and eyes in response to UV radiation.

(D) The mutation will change all subsequent amino acids in the TYR protein, leading to nonfunctional TYR protein. Individuals with this mutation will lack melanin in their hair, skin, and eyes and will not tan in response to UV radiation. ✓

Answer D

Correct. Without a functional TYR enzyme to catalyze the first two steps of the melanin synthesis pathway, melanin will not be produced. Without melanin, individuals will not have a tanning response to UV radiation. Note that UV radiation exposure will still lead to the production of α–MSH in keratinocytes. The α–MSH will then activate a signal transduction pathway in melanocytes, leading to the production of MITF, which will activate the*TYR*, *TYR2*, and *TYR1* genes. However, the TYR enzyme will still be nonfunctional because of the major frameshift mutation.

68. Which of the following claims about the *TYR*, *TRP2*, and *TRP1* mammalian genes is most likely to be accurate?

(A) The *TYR*, *TRP2*, and *TRP1* genes are located next to each other on a single chromosome and are organized into an operon.

(B) The *TYR*, *TRP2*, and *TRP1* genes may be located on different chromosomes but are activated by the same transcription factor. ✓

(C) The *TYR*, *TRP2*, and *TRP1* genes are identical genes since they are activated by the same transcription factor.

(D) The *TYR*, *TRP2*, and *TRP1* genes may be located on different chromosomes but with identical operator sequences.

Answer B

Correct. Related eukaryotic genes are often located on different chromosomes. In humans, *TYR* is on chromosome 11, *TRP2* is on chromosome 13, and *TRP1* is on chromosome 9. The melanocytes control these genes together, as the same transcription factor, MITF, activates the transcription of all

three genes.

69. Which of the following claims best explains why keratinocytes do not produce melanin?

(A) Keratinocytes do not contain the TYR, $TRP2$, and $TRP1$ genes.

(B) Keratinocytes do not contain the $MC1R$ gene.

(C) Keratinocytes do not express the $MITF$ gene. ✓

(D) Keratinocytes do not express the $POMC$ gene.

Answer C

Correct. MITF stands for melanocyte inducing transcription factor, and its expression during embryonic development is critical to the differentiation of melanocytes. The genes activated by MITF include the TYR, $TRP2$, and $TRP1$ genes that are involved in producing melanin. In keratinocytes, MITF is not expressed, so it cannot activate transcription of genes involved in melanin production.

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70. Methicillin-resistant *Staphylococcus aureus* (MRSA) can be a serious threat to human health. There is evidence that *S. aureus* infections are common in hospitals and that MRSA have become resistant to other antibiotics besides methicillin. This suggests that the rapid evolution of resistance in the bacteria poses a serious public-health challenge. Which of the following best explains the ability of MRSA to evade existing drug therapies?

(A) MRSA have very long generation times and very large population sizes.

(B) MRSA develop new alleles by intentionally introducing specific mutations that will give them a selective advantage over other bacteria.

(C) MRSA metabolize many drugs in their lysosomes and therefore evolve resistance at a high rate.

(D) MRSA exchange genetic material with other antibiotic-resistant bacteria, which can spread resistance in the *S. aureus* population. ✓

71. Small single-stranded RNA molecules called microRNAs (miRNAs) are capable of base pairing with specific binding sites in the $3'$ untranslated region of many mRNA transcripts. Transcription of gene *Q* yields an mRNA transcript that contains such an miRNA binding site, which can associate with miRNA-delta, a specific miRNA molecule.

Which of the following best supports the claim that binding of miRNA-delta to the miRNA binding site inhibits translation of gene *Q* mRNA?

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(A) When the promoter for gene *Q* is altered, transcription is inhibited.

(B) Translation of *Q* mRNA is inhibited regardless of whether the miRNA binding site sequence is altered.

(C) Translation of *Q* mRNA is inhibited in the absence of miRNA-delta.

(D) When the miRNA binding site sequence is altered, translation of *Q* mRNA occurs in the presence of miRNA-delta. ✓

Answer D

Correct. If the miRNA binding site sequence is altered, it cannot base-pair with miRNA-delta because they will no longer be complementary with one another. If miRNA-delta cannot bind to the miRNA binding site, translation of *Q* mRNA will proceed.

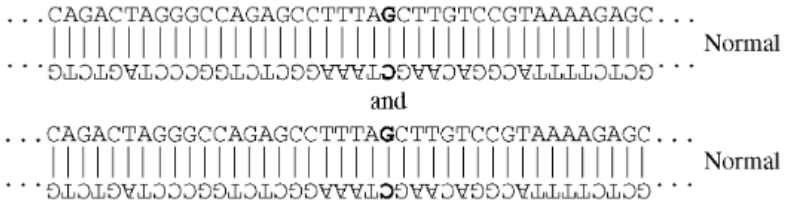
72. Mutations in DNA are usually caused by chemical or radiation damage to DNA molecules, followed by imperfect repair of the damage. Immediately after this kind of imperfect repair, there may be a mismatched base pair in the DNA. The illustration below shows an example of a mismatch, with the relevant pair bases in bold.



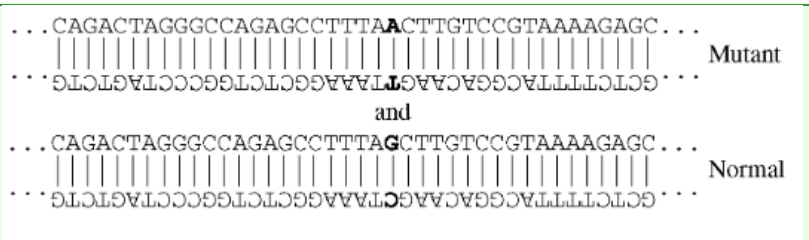
Which of the following best represents the DNA of the two daughter cells produced when a bacterial cell with this particular mismatch replicates its DNA and divides?

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(A)



(B) ✓



(C)



(D)



6 Gene Expression and Regulation MCQ

73. Figure 1 illustrates the X and Y chromosomes during meiosis I and meiosis II of normal spermatogenesis in a mammal species.

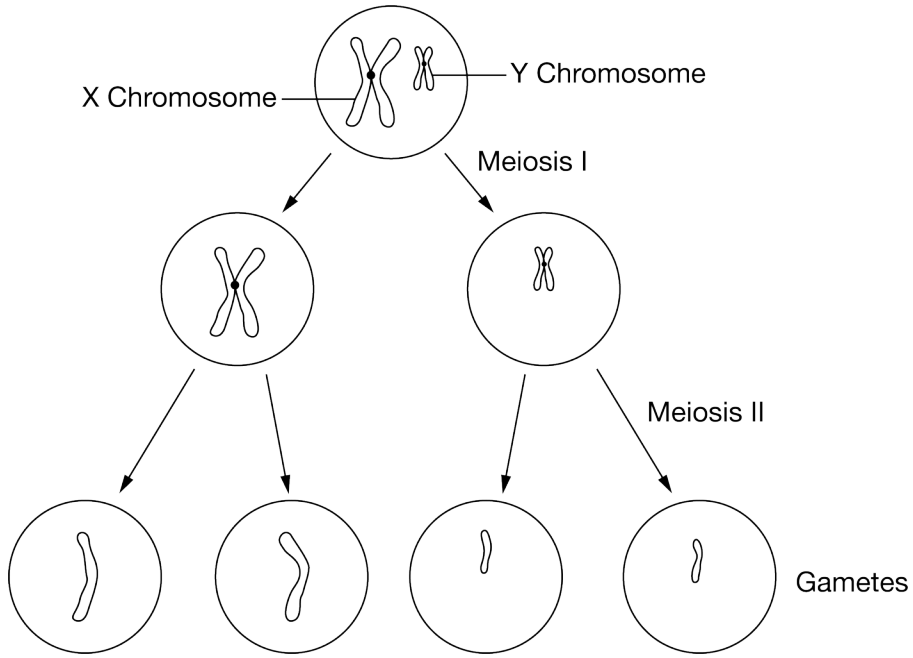


Figure 1. X and Y chromosomes during meiosis I and meiosis II

If the normal spermatogenesis is disrupted, the gametes can have different chromosomes than expected. Which of the following is the most likely cause of one of the four gametes having two X chromosomes and one having neither an X nor a Y chromosome?

(A) Nondisjunction of the chromosomes during meiosis I

(B) Nondisjunction of both the X and Y chromosomes during meiosis II

(C) Nondisjunction of the Y chromosome during meiosis II

(D) Nondisjunction of the X chromosome during meiosis II ✓

Answer D

Correct. A nondisjunction event of the X chromosome in meiosis II would result in both of the X chromatids going to one gamete and no X chromatids going to the other gamete.

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74. Nondisjunction during meiosis can negatively affect gamete formation. A model showing a possible nondisjunction event and its impact on gamete formation is shown in Figure 1.

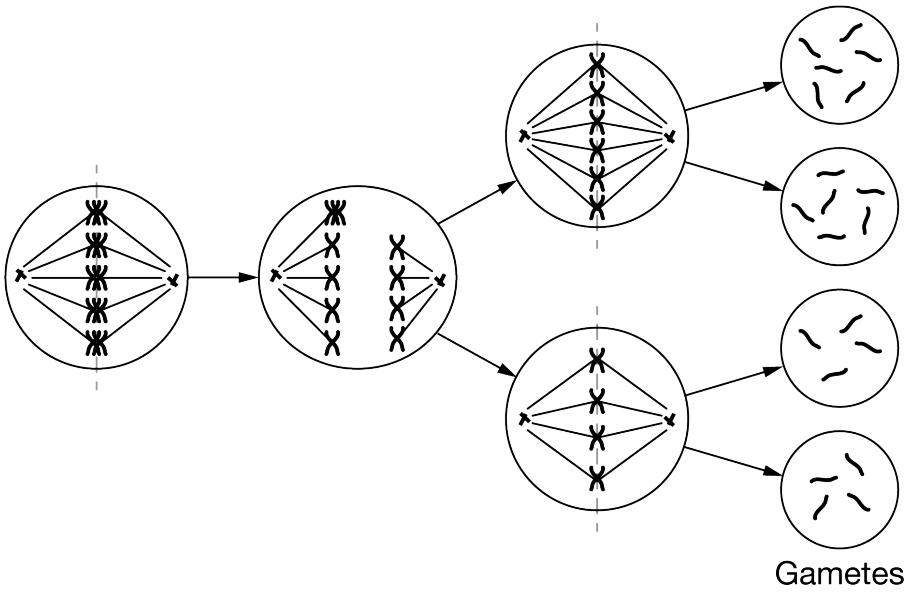


Figure 1. Model of a nondisjunction event

Which of the following best describes the most likely impact on an individual produced from fertilization between one of the daughter cells shown and a normal gamete?

(A) Because nondisjunction occurred in anaphase I, all gametes will be normal and the resulting individual will be phenotypically normal.

(B) Because nondisjunction occurred in anaphase I, all gametes will have an abnormal chromosome number and the individual will likely exhibit phenotypic evidence of the nondisjunction event. ✓

(C) Because nondisjunction occurred in anaphase II, all gametes will be normal and the resulting individual will be phenotypically normal.

(D) Because nondisjunction occurred in anaphase II, all gametes will have an abnormal chromosome number and the individual will likely exhibit phenotypic evidence of the nondisjunction event.

Answer B

Correct. Nondisjunction occurred in anaphase I and will result in an abnormal chromosome number, as the resulting chromosome numbers in the daughter cells are either $n+1$ or $n-1$.

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75. Phytochromes are molecules that change light stimuli into chemical signals, and they are thought to target light-activated genes in plants. A study was conducted to determine how certain cell proteins were made in a plant cell using a phytochrome. Figures 1 and 2 represent findings from the study.

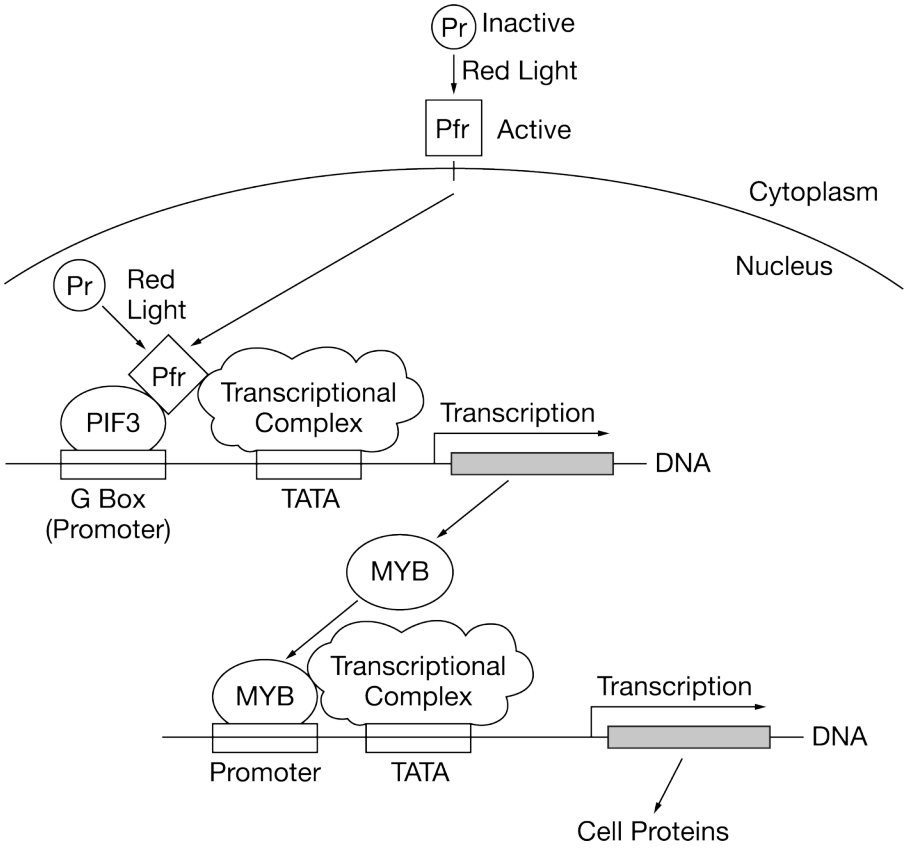


Figure 1. Phytochrome response to exposure to red light

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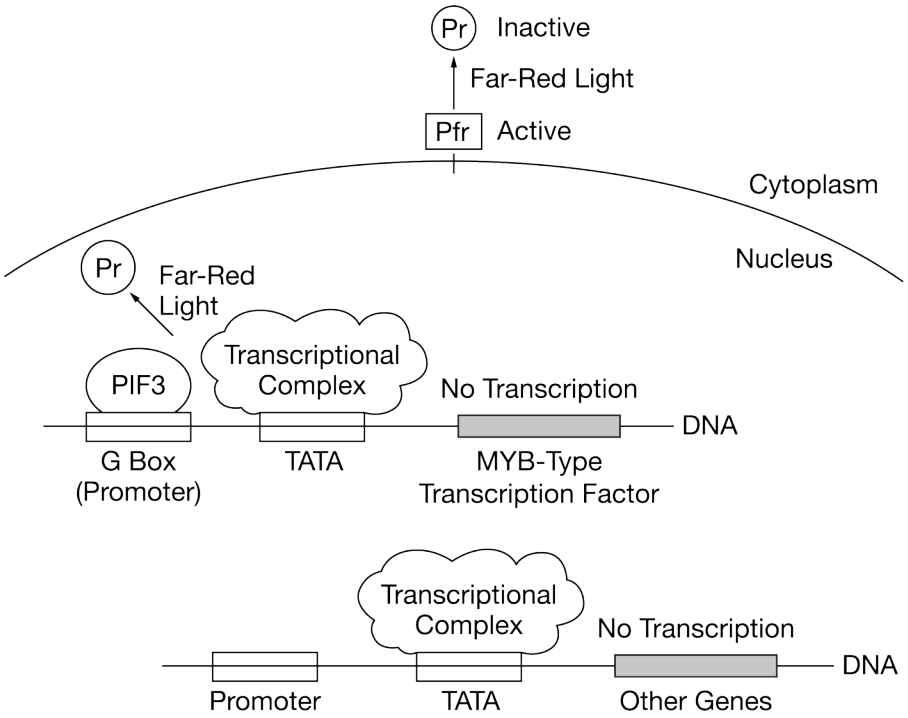


Figure 2. Phytochrome response to exposure to far-red light

Use the response models shown in Figures 1 and 2 to justify the claim that phytochromes regulate the transcription of genes leading to the production of certain cellular proteins.

(A) When inactive phytochrome Pr is activated by red light to become phytochrome Pfr, it is transported into the nucleus where it binds to the transcription factor PIF3 at the promoter. This stimulates transcription, ultimately leading to protein production. Far-red light inactivates the phytochrome, which will turn transcription off by not binding to PIF3. ✓

(B) Far-red light activates phytochrome Pr, causing it to travel to the nucleus where it binds to PIF3 at the promoter. This stimulates transcription, ultimately leading to protein production. Red light inactivates the phytochrome, which will turn transcription off by not binding to PIF3.

(C) MYB, and not Pfr, is activated by red light, causing it to bind to the promoter and stimulate transcription and translation of cellular proteins.

(D) PIF3 binds to the promoter only in the presence of red light and Pfr. Any time PIF3 is bound to the promoter, MYB is transcribed, initiating transcription of various other proteins in the cell.

Answer A

Correct. The diagram indicates the phytochrome is light sensitive, becoming activated by red light. Activated Pfr goes into the nucleus and binds to PIF3, which activates the transcriptional complex, thus turning on the genes responsible for making cellular proteins.

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- **Directions:** This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

(A) Transcription
(B) Translation
(C) Transformation
(D) Replication
(E) Reverse transcription

76. Process in which naked DNA is taken up by a bacterial or yeast cell

(A) A
(B) B
(C) C ✓
(D) D
(E) E

77. Process in which RNA is produced by using a DNA template

(A) A ✓
(B) B
(C) C
(D) D
(E) E

78. Process in which a protein is assembled at a ribosome

(A) A
(B) B ✓
(C) C
(D) D
(E) E

79. Which of the following describes a process that increases genetic diversity in prokaryotic cells?

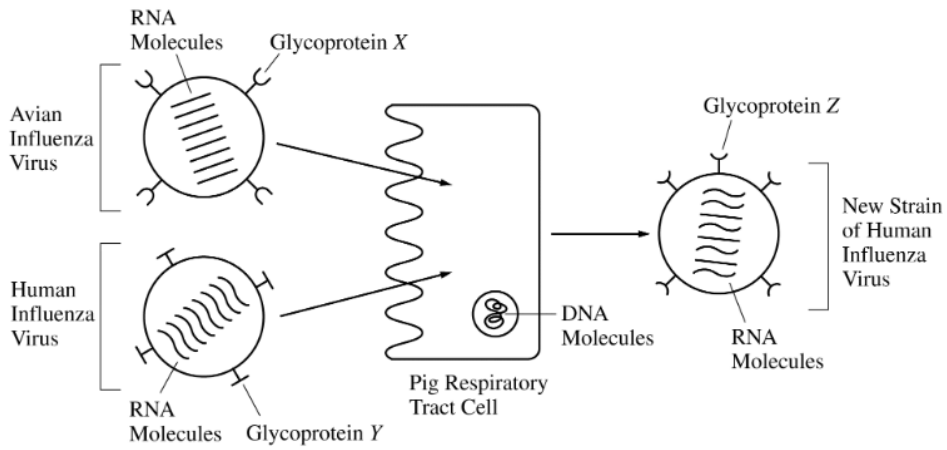
(A) Chromosomes segregate independently and form different combinations in daughter cells.
(B) Homologous chromosomes exchange genetic material during meiosis.
(C) Conjugation transfers DNA from one cell to the other cell. ✓
(D) A cell infected with two genetically different viruses can produce viruses with new allele combinations.

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Answer C

Correct. When a cytoplasmic bridge forms between two prokaryotic cells, DNA can be exchanged between the cells, increasing genetic diversity. This process is called conjugation.

80.



The cells lining the respiratory tract of pigs have receptors for both avian and human influenza viruses. Based on the model above, which of the following best explains the origin of the new strain of human influenza virus?

(A) The new viral strain inherited a mixture of genetic material from both avian influenza virus and human influenza virus. ✓

(B) The new viral strain inherited RNA molecules from the avian influenza virus and packaged them inside the human influenza virus membrane.

(C) The new viral strain inherited a mutant DNA molecule from the pig respiratory tract cell.

(D) The new viral strain inherited an RNA molecule that had recombined with a DNA molecule from the pig respiratory tract cell.

81. Sickle-cell anemia is associated with a mutation in the gene encoding the beta subunit of hemoglobin that results in a change from glutamic acid to valine at position 6. All other amino acids are identical to a normal hemoglobin molecule.

Based on the information above, which of the following mutations is the most likely cause of sickle-cell anemia?

(A) A single base-pair substitution in the gene encoding the beta subunit ✓

(B) A single base-pair insertion in the gene encoding the beta subunit

(C) A single base-pair deletion in the gene encoding the beta subunit

(D) A translocation of DNA from one chromosome to another

6 Gene Expression and Regulation MCQ

82. Sickle-cell anemia results from a point mutation in the *HBB* gene. The mutation results in the replacement of an amino acid that has a hydrophilic R-group with an amino acid that has a hydrophobic R-group on the exterior of the hemoglobin protein. Such a mutation would most likely result in altered

(A) properties of the molecule as a result of abnormal interactions between adjacent hemoglobin molecules ✓

(B) DNA structure as a result of abnormal hydrogen bonding between nitrogenous bases

(C) fatty acid structure as a result of changes in ionic interactions between adjacent fatty acid chains

(D) protein secondary structure as a result of abnormal hydrophobic interactions between R-groups in the backbone of the protein

Answer A

This option is correct. The mutation results in the replacement of an amino acid, which alters the properties of the R-group both structurally and functionally; thus, the interactions between adjacent hemoglobin molecules would also be altered. In this case, hemoglobin molecules would tend to stick together as a result of hydrophobic interactions between amino acids on the surface.

83. Sickle-cell anemia results from a point mutation in the *HBB* gene. The mutation results in the replacement of an amino acid that has a hydrophilic R-group with an amino acid that has a hydrophobic R-group on the exterior of the hemoglobin protein. Such a mutation would most likely result in altered

(A) properties of the molecule as a result of abnormal interactions between adjacent hemoglobin molecules ✓

(B) DNA structure as a result of abnormal hydrogen bonding between nitrogenous bases

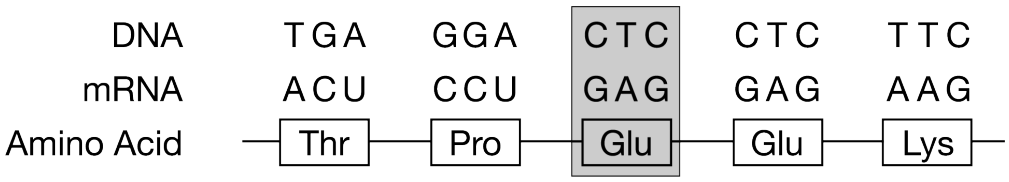
(C) fatty acid structure as a result of changes in ionic interactions between adjacent fatty acid chains

(D) protein secondary structure as a result of abnormal hydrophobic interactions between R-groups in the backbone of the protein

6 Gene Expression and Regulation MCQ

84. Sickle-cell anemia is an inherited blood disorder in which one of the hemoglobin subunits is replaced with a different form of hemoglobin. Partial DNA sequences of the *HBB* gene for normal hemoglobin and for sickle-cell hemoglobin are shown in Figure 1.

Partial sequence for normal hemoglobin:



Partial sequence for sickle-cell hemoglobin:

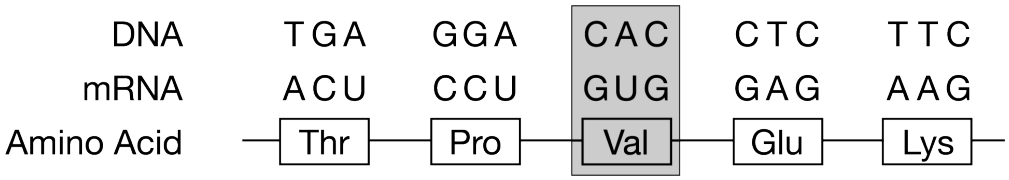


Figure 1. Comparison of partial DNA sequences for normal hemoglobin and hemoglobin with a sickle-cell mutation

Which of the following best describes the type of mutation shown in Figure 1 that leads to sickle-cell anemia?

(A) Insertion

(B) Deletion

(C) Substitution ✓

(D) Frameshift

Answer C

Correct. The change in the DNA sequence from thymine to adenine resulted in a different amino acid in the protein, valine instead of glutamic acid.

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Snowshoe hares are small mammals that live in forests throughout Canada and the northern United States. Hares feed primarily on vegetation and are preyed on by lynx and owls.

In the fall, the hares shed their brown coat (molt), and a new white coat grows in over a period of 40 days. In the spring, the hares molt again and their white coat is replaced with a brown coat over a period of approximately 30 to 50 days.

In a recent study, researchers observed that the majority of hares in an isolated population began the molting process during the same week in the fall, even though the ground was not snow covered. The majority of hares also began the molting process during the same week in the spring, even though the ground was still snow covered. The researchers formulated a hypothesis that the timing of the molting process is influenced by seasonal changes in the daylight period.

85. A second isolated hare population has white coats all year, although the area is not covered by snow.

Which of the following is a testable question the researchers can ask to explain the origin of the different coat color of the second population?

(A) Do the genes associated with coat color in both populations have the same sequence of DNA nucleotides? ✓

(B) Are hares that have a white coat all year round better able to find food than those that change coat color?

(C) Is the reproductive fitness greater in hares that have a white coat all year round than in those that change coat colors?

(D) Is the same genetic code used to produce the enzymes involved in the production of coat color pigments in both populations?

Answer A

Correct. A constant all-white coat color would strongly suggest a change in the genetic makeup of the hare population. To determine whether this is the case, sequencing of the coat color genes in both populations would indicate whether the genes are the same or different.

86. Some strains of the bacterium *Streptococcus pyogenes* secrete poisonous substances called exotoxins. The genes encoding the exotoxins are thought to have originated in bacteriophages, which are viruses that infect bacteria.

Which of the following is the most likely mechanism by which the *S. pyogenes* acquired the ability to produce the exotoxins?

(A) Bacteriophages engulfed cellular debris from dead bacteria.

(B) Bacteriophages in the environment activated bacterial cell division.

(C) Bacteriophage DNA became integrated in the bacterial chromosome. ✓

(D) Bacteriophage proteins were absorbed into bacteria cells by endocytosis.

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87. Trisomy 21 is a condition in which a child is born with an extra chromosome in pair 21. Researchers assessed the frequency of children born with trisomy 21 by age of the mothers at birth (maternal age) and primary cause of the error leading to trisomy 21. The findings are presented in Figure 1.

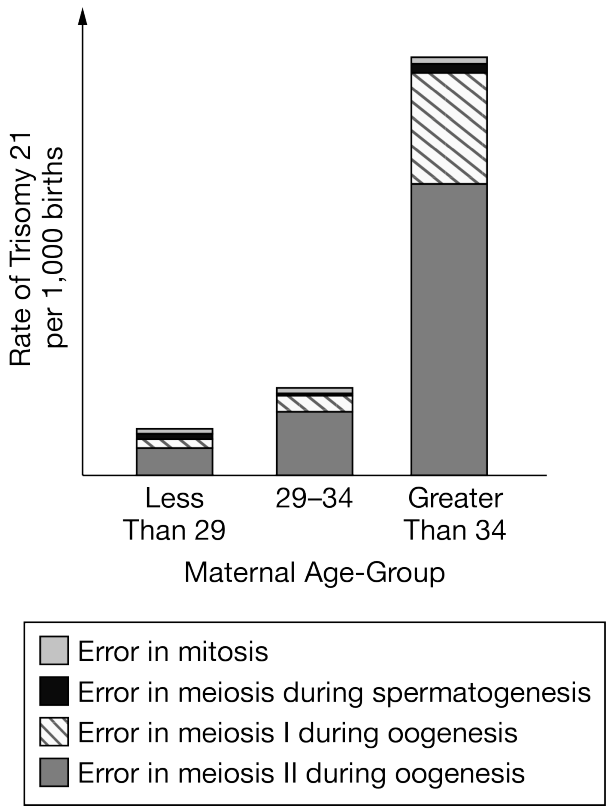


Figure 1. Incidence and primary cause of trisomy 21 by maternal age-group

Based on the data in Figure 1, which of the following is most likely the primary cause of the pattern of frequency of trisomy 21 births in the selected maternal age-groups?

(A) At older maternal ages, there is an increase in the number of errors during mitosis, which leads to an increase in nondisjunction during egg production.

(B) The incidence of nondisjunction errors in meiosis during sperm production is positively correlated with increasing maternal age.

(C) At older maternal ages, the incidence of errors in meiosis during egg production increases, which leads to an increase in nondisjunction. ✓

(D) Errors in meiosis leading to nondisjunction are more likely to occur during meiosis I than during meiosis II.

Answer C

Correct. Maternal age is correlated with a significant increase in meiotic errors during egg maturation.

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Figure 1 shows that most of trisomy 21 births are due to these errors, with the number of these errors observed increasing significantly with maternal age.

88. The bacterium *Vibrio cholerae* is harmless unless a lysogenic bacteriophage provides the gene coding for the cholera toxin, which converts the bacterium to the virulent form that causes cholera. Which of the following best explains how the gene encoding cholera toxin becomes part of the bacterial genome?

(A) The bacteriophage inserts the toxin gene into the host cell DNA, and the gene is expressed with the rest of the host cell's genes. ✓

(B) The bacteriophage makes copies of the toxin gene and expresses the copies inside the bacteriophage.

(C) The bacteriophage converts its toxin gene into mRNA, which is then translated by the host cell.

(D) The bacteriophage transforms itself into a self-replicating protein that can survive inside the host cell.

89. The trp operon in *E. coli* is an example of a repressible operon that consists of genes coding for enzymes used to synthesize tryptophan. When tryptophan levels are high, the operon is turned off and these genes are not transcribed. However, it is also known that tryptophan does not bind directly to the operator DNA sequence. A regulatory gene called $trpR$ has also been discovered although it is not part of the trp operon. The proposed model of how tryptophan acts as a corepressor is shown in Figure 1.

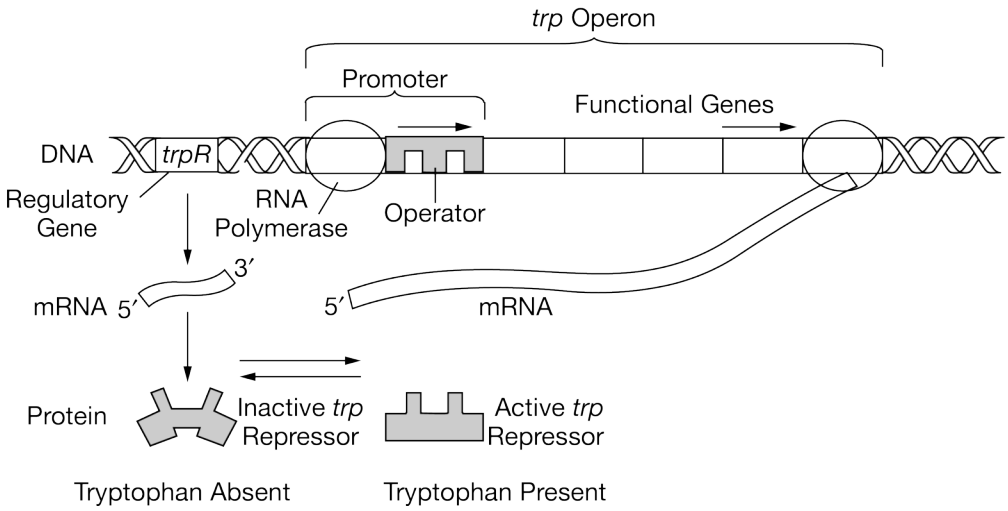


Figure 1. Model of proposed regulation of the trp operon by corepressors trp repressor and tryptophan

Which of the following evidence best supports a claim that tryptophan functions as a corepressor?

(A) Normal expression of $trpR$ causes the trp operon to be transcribed regardless of tryptophan levels.

(B) When the operator sequence is mutated, the trp operon is not transcribed.

(C) The $trpR$ gene codes for a repressor protein that has a DNA binding domain.

(D) When $trpR$ is mutated, the trp operon is transcribed regardless of tryptophan levels. ✓

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Answer D

Correct. As tryptophan accumulates in the cell, it inhibits its own production by activating the trp repressor protein (coded by the $trpR$ gene), which then binds to the operator and blocks transcription of the trp operon. If the $trpR$ gene is mutated, then tryptophan cannot bind to the allosteric binding site to activate trp repressor protein. Thus, no matter how high tryptophan levels get, the trp operon will not be repressed.

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In prokaryotes, the genes of the trp operon encode enzymes used in the synthesis of tryptophan. The genes of the trp operon are actively expressed (Figure 1A) unless abundant tryptophan is present in the environment. When tryptophan is abundant, some tryptophan binds to the trp repressor protein, which in turn causes the repressor to bind to the operator and block transcription of the genes (Figure 1B). The repressor protein is encoded by the $trpR$ gene which is not a part of the trp operon.

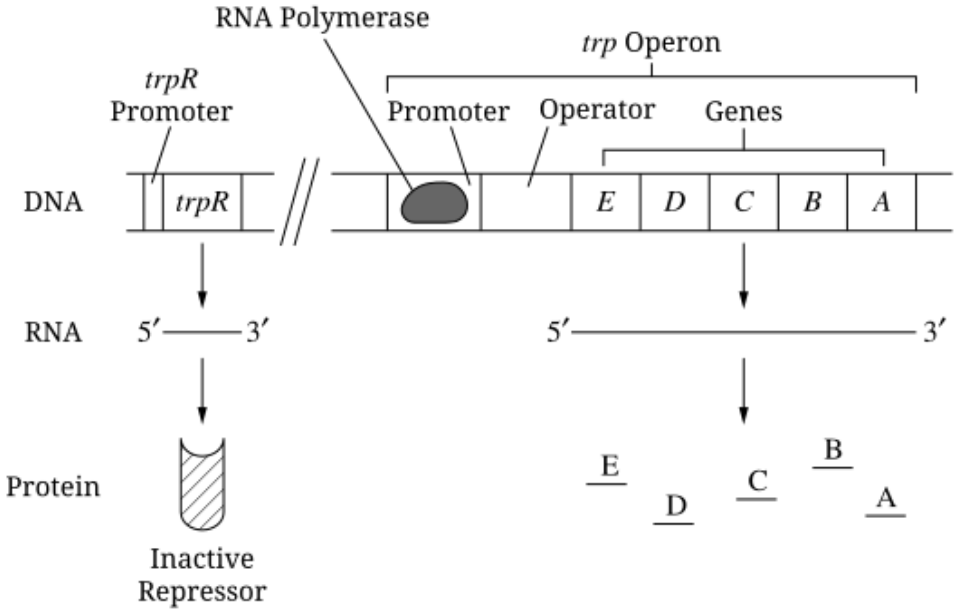


Figure 1A. When tryptophan is absent, the trp operon is active.

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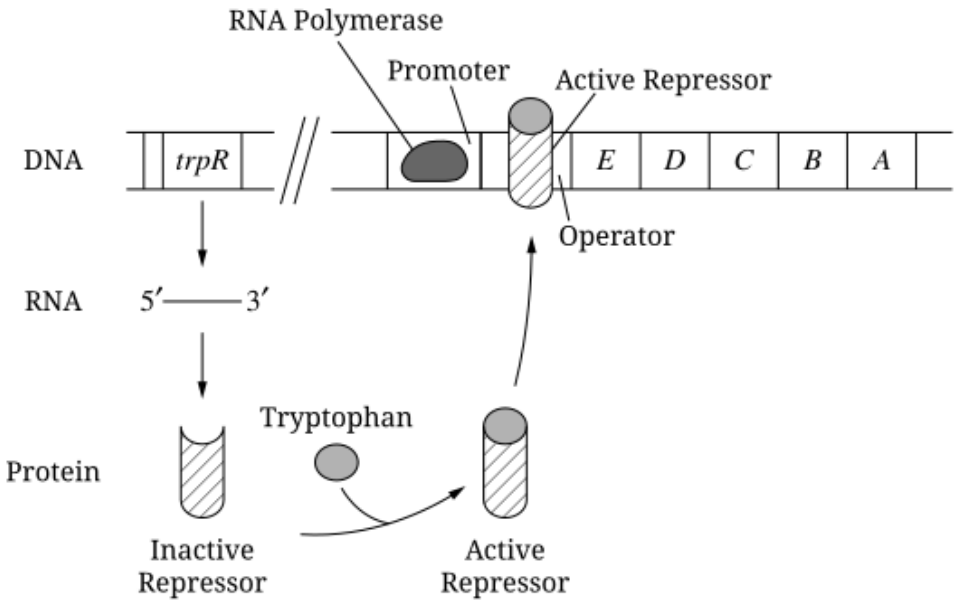


Figure 1B. When tryptophan is abundant, the *trp* operon is inactivated by the binding of the repressor protein to the operator.

90. A scientist claimed that an *E. coli* strain had either a mutated *trpR* gene or a mutated operator. Which of the following observations most likely supports the claim?

(A) Transcription from the operon occurred only in the presence of abundant tryptophan.

(B) The strain of *E. coli* required more tryptophan for its metabolic processes than does a strain of *E. coli* with typical tryptophan regulatory controls.

(C) Enzymes required for the synthesis of tryptophan were continuously produced whether tryptophan was absent or present in large quantities. ✓

(D) The cells died when they were grown in nutrient medium that lacked tryptophan.

Answer C

Correct. Under high tryptophan levels, the wild-type repressor protein binds to tryptophan and the complex binds to the operator, blocking transcription and ultimately the production of the enzymes that produce more tryptophan. In this mutant, high tryptophan levels do not reduce the production of more tryptophan. This suggests either that the repressor protein is unable to bind to tryptophan or the operator or, alternatively, that the operator cannot interact with the repressor-tryptophan complex to repress transcription.

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91. Which of the following questions will most likely allow the scientist to design an experiment to determine whether unregulated tryptophan synthesis in a strain of *E. coli* is caused by a mutation in the $trpR$ gene or in the operator of the trp operon?

(A) Is normal regulation of tryptophan synthesis restored when the operator sequence of this strain of *E. coli* is duplicated?

(B) Is normal regulation of tryptophan synthesis restored when a gene encoding a normal repressor protein is introduced into and expressed by this strain of *E. coli*? ✓

(C) Is normal regulation of tryptophan synthesis restored when a plasmid that contains a normal operator sequence is introduced into this strain of *E. coli*?

(D) Is normal regulation of tryptophan synthesis restored when the repressor protein gene is deleted in this strain of *E. coli*?

Answer B

Correct. Providing a functional copy of the $trpR$ gene that encodes the repressor protein would allow the scientist to observe whether regulatory functionality is restored. Restored functionality would indicate that the unregulated tryptophan synthesis was due to a mutation in the $trpR$ gene. If functionality is not restored, that would suggest a mutant operator sequence.

92. Attenuation is another mechanism by which tryptophan synthesis is regulated when tryptophan is abundant in the environment. During attenuation, ribosomes quickly translate the 5′ end of the mRNA transcript, causing the transcript to fold into a loop that forces RNA polymerase to detach from the DNA.

Which of the following best explains why attenuation is a mechanism that is unique to prokaryotes?

(A) Prokaryotic genes do not contain introns.

(B) Only prokaryotes can translate an mRNA while the mRNA is being transcribed. ✓

(C) Eukaryotes require more amino acids than prokaryotes do and therefore do not shut down production of tryptophan.

(D) Eukaryotes do not have groups of genes that are regulated together.

Answer B

Correct. Attenuation requires simultaneous transcription and translation. In eukaryotes, mRNA is transcribed and processed in the nucleus before being transported to the cytosol for translation. Prokaryotes have no nuclear envelope to separate these processes.

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93. Which of the following best supports a claim that an *E. coli* strain has a mutation in gene E that results in the production of a nonfunctional enzyme?

(A) Cells of this strain synthesize very little of the repressor protein and produce a large amount of tryptophan.

(B) RNA polymerase does not bind to the promoter of the operon in cells of this strain unless tryptophan is added to the nutrient medium.

(C) The operator sequence of the operon always has repressor proteins bound to it, independent of the amount of tryptophan in the nutrient medium.

(D) mRNA is continuously transcribed from the operon, but the amount of tryptophan produced from the cells of this strain remains low. ✓

Answer D

Correct. Continuous transcription of the operon during low levels of tryptophan is expected, but the lack of resultant tryptophan production suggests that one of the enzymes in the tryptophan synthesis pathway is nonfunctional.

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94. Exposure to ultraviolet (UV) radiation is the leading cause of skin cancer in humans. Figure 1 shows a model of how UV exposure damages DNA.

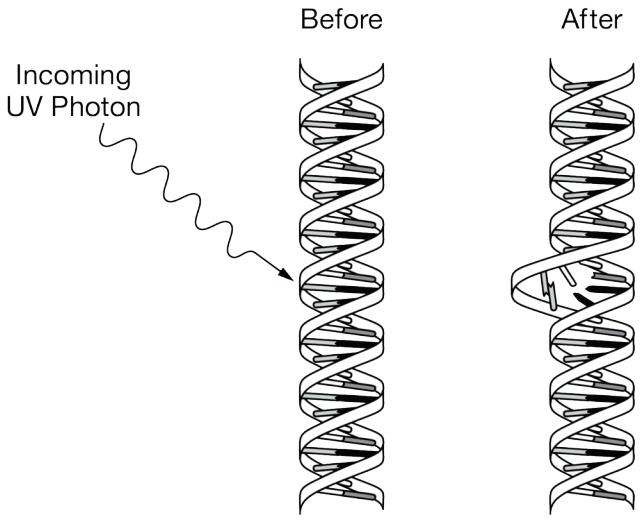


Figure 1. Model of damage to DNA caused by UV exposure

Which of the following statements best explains what is shown in Figure 1 ?

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(A) UV exposure triggers DNA replication, which results in rapid cell proliferation.

(B) Naturally occurring dimers are removed by the UV photons, causing misshapen DNA, which results in replication errors.

(C) The hydrogen bonds between base pairs absorb the UV photons, separating the two DNA strands, which results in rapid DNA replication.

(D) UV photons cause dimers to form, leading to misshapen DNA, which results in replication and transcription errors. ✓

Answer D

Correct. The model shows the partial separation of a segment of DNA caused by the structural change resulting from dimer formation. The presence of dimers can lead to errors in both transcription and replication.

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question.

In the 1940's, Avery MacCleod, and McCarty transformed nonencapsulated bacteria into encapsulated forms by growing the nonencapsulated cells in a culture containing an extract made from dead encapsulated cells. The transformed cells produced colonies of encapsulated bacteria. Three different procedures and their results are outlined below.

Procedure I:

Extract made from dead encapsulated cells added to culture medium.
Nonencapsulated bacteria added to culture medium.

Results: Both nonencapsulated and encapsulated bacteria grow.

Procedure II:

Extract made from dead encapsulated cells treated with protein-degrading enzymes before adding extract to culture medium.
Nonencapsulated bacteria added to culture medium.

Results: Both nonencapsulated and encapsulated bacteria grow.

Procedure III:

Extract made from dead encapsulated cells treated with DNAse (an enzyme that selectively destroys DNA) before adding extract to culture medium.
Nonencapsulated bacteria added to culture medium.

Results: Only nonencapsulated bacteria grow.

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95. A resonable conclusion to draw from the results of the experiment is that

(A) DNA is the genetic material ✓

(B) DNA replication is semiconservative

(C) DNA is a double helix

(D) DNA is translated into protein

(E) mutation is a change in the genetic material

96. What was the purpose of treating the extract with DNAse in Procedure III ?

(A) To remove the encapsulated bacteria from the extract

(B) To serve as a positive control by demonstrating that a protein in the extract is the transforming factor

(C) To serve as a negative control by demonstrating that transformation does not occur without DNA ✓

(D) To destroy enzymes in the extract

(E) To destroy any capsules that might be in the extract

97. What was the purpose of treating the extract with protein-degrading enzymes in Procedure II ?

(A) To demonstrate that the transforming factor is an enzyme

(B) To demonstrate that the transforming factor is not a protein ✓

(C) To destroy nuclei acids in the extract

(D) To destroy any capsules in the extract

(E) To prevent the extract from being contaminated by nonencapsulated bacteria

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98. Erwin Chargaff investigated the nucleotide composition of DNA. He analyzed DNA from various organisms and measured the relative amounts of adenine (A), guanine (G), cytosine (C), and thymine (T) present in the DNA of each organism. Table 1 contains a selected data set of his results.

Table 1. Nucleotide composition of sample DNA from selected organisms

Organism	Relative Amounts of Bases					
	%A	%G	%C	%T	%(G+C)	%(A + T)
Octopus	33.2	17.6	17.6	31.6	35.2	64.8
Chicken	28.0	22.0	21.6	28.4	43.7	56.4
Rat	28.6	21.4	20.5	28.4	42.9	56.0
Grasshopper	29.3	20.5	20.7	29.3	41.2	58.6
Wheat	27.3	22.7	22.8	27.1	45.5	54.4

Which of the following statements best explains the data set?

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(A) Since the $\%\mathrm{A}$ and the $\%\mathrm{G}$ add up to approximately 50 percent in each sample, adenine and guanine molecules must pair up in a double-stranded DNA molecule.

(B) Since the $\%\mathrm{A}$ and the $\%\mathrm{T}$ are approximately the same in each sample, adenine and thymine molecules must pair up in a double-stranded DNA molecule. ✓

(C) Since the $\%(\mathrm{A}+\mathrm{T})$ is greater than the $\%(\mathrm{G}+\mathrm{C})$ in each sample, DNA molecules must have a poly-A tail at one end.

(D) Since the $\%\mathrm{C}$ and the $\%\mathrm{T}$ add up to approximately 50 percent in each sample, cytosine and thymine molecules must both contain a single ring.

Answer B

Correct. The percentages of adenine and thymine are approximately the same because adenine aligns with thymine in double-stranded DNA and they are therefore present in each organism in the same amounts.

99. Cells that contain only circular chromosomes are most probably which of the following?

(A) Protist cells
(B) Fungal cells
(C) Bacterial cells ✓
(D) Plant cells
(E) Animal cells

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the following groups of biological compounds.

(A) Proteins
(B) Carbohydrates
(C) Nucleic acids
(D) Lipids
(E) Steroids

100. Used to carry the genetic code

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(A) Proteins
(B) Carbohydrates
(C) Nucleic acids ✓
(D) Lipids
(E) Steroids

---

101. A scientist separated a sample of cells of the same type into two dishes, dish X and dish Y. Each dish had the same number of cells. After a period of time, the scientist measured the quantity of histone proteins in each dish (Table 1).

Table 1. Histone Protein Quantities

Dish	Histone Proteins (\(\text{ng}\))
X	6.62
Y	13.57

Which of the following statements best explains the greater quantity of histone proteins in dish Y shown in Table 1?

(A) Prokaryotic cells are preparing for cell division, because prokaryotes must double the quantity of histone proteins each time they replicate their DNA.
(B) Eukaryotic cells are preparing for cell division, because eukaryotes must double the quantity of histone proteins each time they replicate their DNA. ✓
(C) Prokaryotic cells are preparing for protein synthesis, because prokaryotes must double the quantity of histone proteins when they manufacture proteins.
(D) Eukaryotic cells are preparing for protein synthesis, because eukaryotes must double the quantity of histone proteins when they manufacture proteins.

102. Huntington's disease, an autosomal dominant disorder, is caused by a mutation in the HTT gene. The HTT gene contains multiple repeats of the nucleotide sequence CAG. A person with fewer than 35 CAG repeats in the HTT gene is unlikely to show the neurological symptoms of Huntington's disease. A person with 40 or more CAG repeats almost always becomes symptomatic. Due to errors in meiosis, an individual without symptoms of Huntington's disease can produce gametes with a larger number of CAG repeats than there are in their somatic cells.

A woman develops Huntington's disease. Her father had the disorder. Her mother did not, and there is no history of the disorder in the mother's family. Which of the following best explains how the woman inherited Huntington's disease?

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(A) She inherited an allele with fewer than 40 CAG repeats in the *HTT* gene because her mother did not have Huntington's disease.

(B) She inherited an allele with more than 40 CAG repeats in the *HTT* gene from her father. ✓

(C) Her mother produced eggs that all have more than 40 repeats in the *HTT* gene.

(D) Her mother produced eggs that all have fewer than 40 CAG repeats in the *HTT* gene.

Answer B

Correct. If her father was heterozygous for Huntington's disease, half of his sperm would carry an *HTT* allele with more than 40 CAG repeats. His children would have a fifty-fifty chance of inheriting the allele responsible for Huntington's disease.

103. Antibiotics can be used to kill the specific pathogenic bacterium, *Mycobacterium tuberculosis*, that causes tuberculosis. The appearance of antibiotic-resistant strains has made it more difficult to cure *M. tuberculosis* infections. These antibiotic-resistant bacteria survive and pass on the genes to their offspring, making the resistant phenotype more common in the population.

DNA analysis indicates that the genes for antibiotic resistance are not normally present in bacterial chromosomal DNA.

Which of the following statements best explains how the genes for antibiotic resistance can be transmitted between bacteria without the exchange of bacterial chromosomal DNA?

(A) The antibiotic-resistant bacteria release a hormone that signals neighboring bacteria to become resistant.

(B) The genes for antibiotic resistance are located on a plasmid that can be passed to neighboring bacteria. ✓

(C) The antibiotic-resistant bacteria are the result of bacteria that specifically modify their own chromosomal DNA to neutralize the antibiotics.

(D) The antibiotic alters the bacterial genome of each bacterium, which results in an antibiotic-resistant population.

Answer B

Correct. Bacteria can exchange and express the genes found on plasmids, which are foreign, extrachromosomal loops of DNA that they pick up. Genes for antibiotic resistance are located on these plasmids.

104. Which of the following statements best explains the structure and importance of plasmids to prokaryotes?

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(A) Plasmids are circular, single-stranded RNA molecules that transfer information from the prokaryotic chromosome to the ribosomes during protein synthesis.

(B) Plasmids are circular, double-stranded DNA molecules that provide genes that may aid in survival of the prokaryotic cell. ✓

(C) Plasmids are single-stranded DNA molecules, which are replicated from the prokaryotic chromosome, that prevent viral reproduction within the prokaryotic cell.

(D) Plasmids are double-stranded RNA molecules that are transmitted by conjugation that enable other prokaryotic cells to acquire useful genes.

Answer B

Correct. Plasmids are small, circular, double-stranded DNA molecules. Plasmids can be acquired and can be replicated by prokaryotes. Plasmids may carry genes, such as antibiotic resistance genes, that may help the prokaryote survive.

105. Which of the following best explains how the pattern of DNA arrangement in chromosomes could be used, in most cases, to determine if an organism was a prokaryote or a eukaryote?

(A)

Prokaryotic DNA	**Eukaryotic** DNA
Single circular chromosome	Multiple circular chromosomes

(B)

Prokaryotic DNA	**Eukaryotic** DNA
Multiple chromosomes	Single chromosome

(C)

Prokaryotic DNA	**Eukaryotic** DNA
Single linear chromosome	Multiple linear chromosomes

(D) ✓

Prokaryotic DNA	**Eukaryotic** DNA
Single circular chromosome	Multiple linear chromosomes

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Answer D

Correct. This describes the typical arrangement of chromosomes: prokaryotes have DNA arranged in a single circular chromosome and eukaryotes have DNA arranged in multiple linear chromosomes.

106. Nucleotide base pairing in DNA is universal across organisms. Each pair (T − A; C − G) consists of a purine and a pyrimidine. Which of the following best explains how the base pairs form?

(A) Ionic bonds join two double-ringed structures in each pair.

(B) Hydrogen bonds join two single-ringed structures in each pair.

(C) Hydrogen bonds join a double-ringed structure to a single-ringed structure in each pair. ✓

(D) Covalent bonds join a double-ringed structure to a single-ringed structure in each pair.

Answer C

Correct. Each pair consists of one single-ringed structure (pyrimidine) joined to a double-ringed structure (purine). The hydrogen bonds allow the strands to separate for DNA replication and for transcription from DNA to RNA.

107. A tobacco plant can be made to express a gene from fireflies, resulting in the emission of light. Which of the following is the basis for this phenomenon?

(A) Chloroplasts can be made to produce light if firefly proteins are injected into plant cells.

(B) Fireflies and tobacco plants share a recent common ancestor.

(C) Fireflies and tobacco plants are infected by the same kinds of bacteria.

(D) Transcription and translation are fundamen-tally similar in both fireflies and tobacco plants. ✓

(E) Most enzymes in fireflies have the same amino acid sequence as the enzymes in tobacco plants.

108. A student analyzed a viral genome and found that the genome had the following nucleotide composition.

- 28% adenine
- 20% thymine
- 35% cytosine
- 17% guanine

Which of the following best describes the structure of the viral genome?

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(A) Double-stranded DNA

(B) Single-stranded DNA ✓

(C) Double-stranded RNA

(D) Single-stranded RNA

109. A model of a process involving nucleic acids is shown in Figure 1.

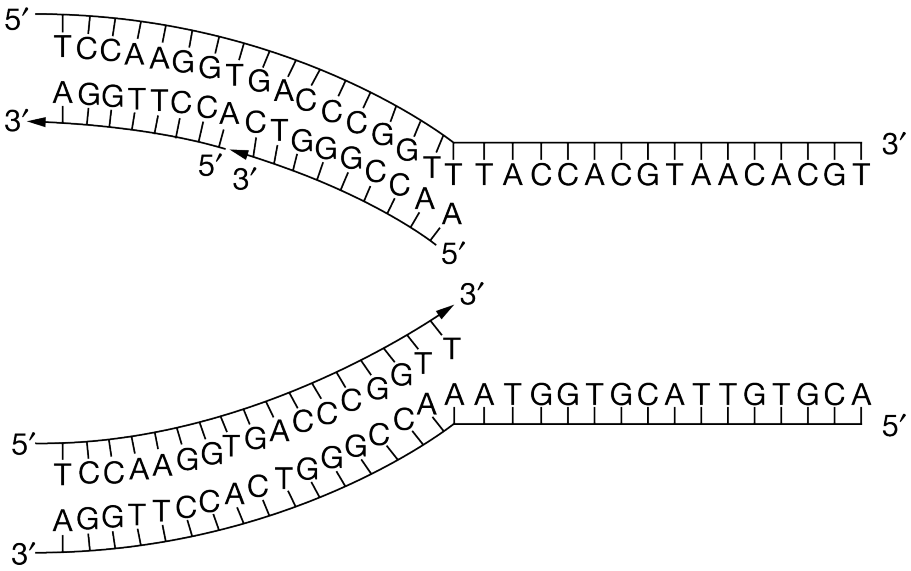


Figure 1. Model of a process involving nucleic acids

Which of the following best explains what process is represented in Figure 1 ?

(A) New DNA strands are being synthesized in the 3′ to 5′ direction from their DNA templates.

(B) New DNA strands are being synthesized in the 5′ to 3′ direction from their DNA templates. ✓

(C) A new RNA strand is being synthesized in the 3′ to 5′ end from its DNA template.

(D) Two new RNA strands are being synthesized in both directions from their DNA templates.

Answer B

Correct. New DNA strands are synthesized in the 5′ to 3′ direction from the DNA template, and this is depicted in the diagram. The leading strand is being synthesized on the bottom, while the lagging strand is being synthesized in fragments on top.

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110. Figure 1 shows some relevant details of a model of how a deoxynucleotide, in this case dTMP, is added to a growing strand of DNA.

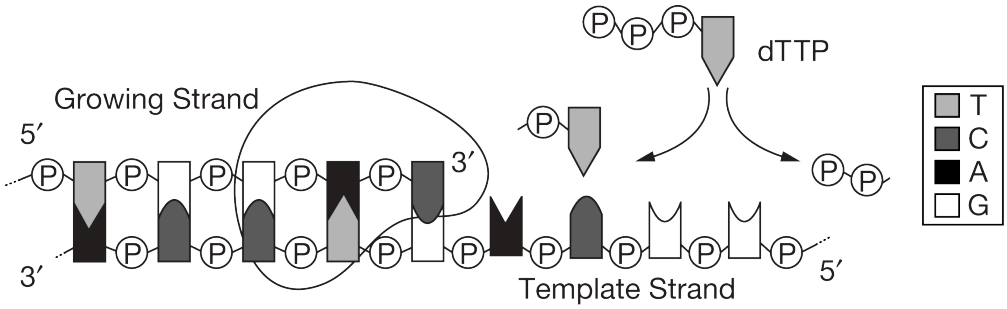


Figure 1. Model showing details of adding a deoxythymidine monophosphate (dTMP) nucleotide to a growing strand of DNA

The features of this model provide evidence for which explanation of why all growing strands are synthesized in a 5′ to 3′ direction?

(A) The two strands need to be antiparallel to bond properly.

(B) Thymine and adenine would not bond properly if the strand grew from 3′ to 5′.

(C) The translation of mRNA occurs in the 5′ to 3′ direction; therefore, the growing DNA strand must also grow in the 5′ to 3′ direction.

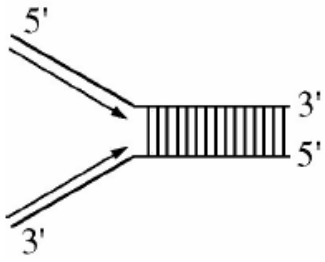
(D) The phosphate group, attached to the 5′ carbon of the dTMP, forms a covalent bond with the oxygen atom attached to the 3′ carbon of the growing strand. ✓

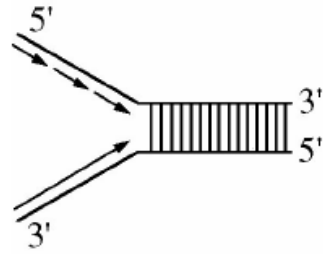
Answer D

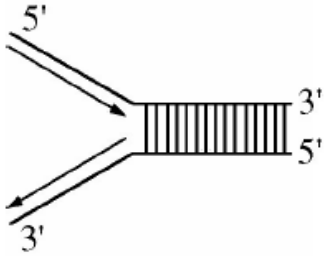
Correct. Synthesizing a growing strand requires the interaction of the phosphate groups found at the 5′ location on the new deoxynucleotide with the OH group attached to the 3′ carbon of the existing end of the growing strand. These specific interactions and the release of diphosphate explain the directionality requirement for a growing strand of DNA.

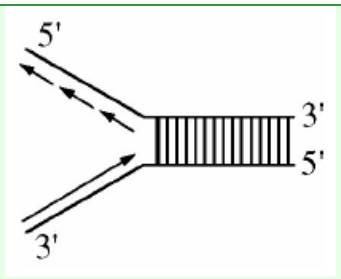
111. When DNA replicates, each strand of the original DNA molecule is used as a template for the synthesis of a second, complementary strand. Which of the following figures most accurately illustrates enzyme-mediated synthesis of new DNA at a replication fork?

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(A) 

(B) 

(C) 

(D) 

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112. Figure 1 depicts a simplified model of a replication bubble.

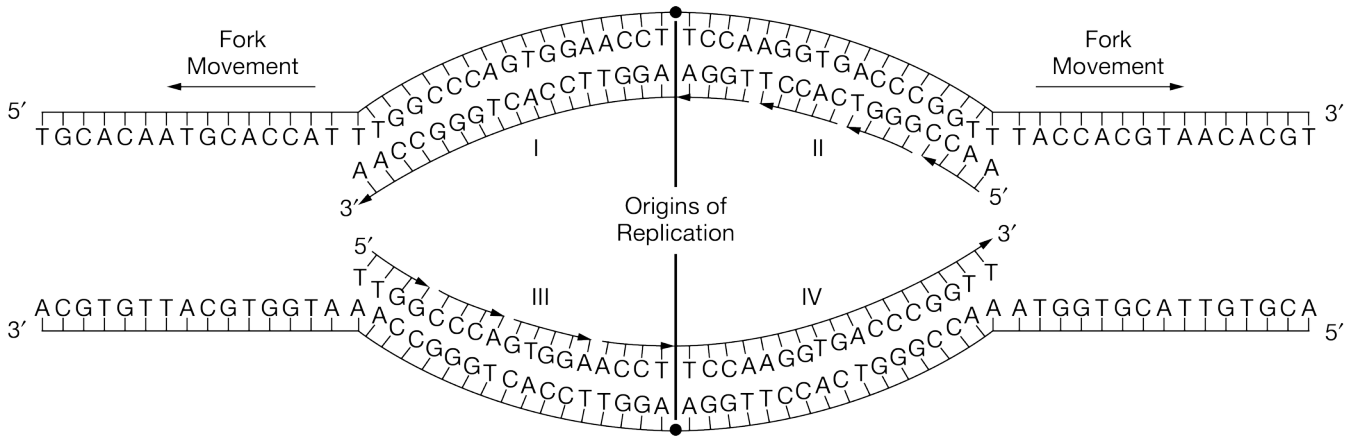


Figure 1. DNA replication bubble.

Which of the following best explains how this model illustrates DNA replication of both strands as a replication fork moves?

(A) I and IV are synthesized continuously in the 5′ to 3′ direction.

(B) II and III are synthesized in segments in the 3′ to 5′ direction.

(C) I is synthesized continuously in the 5′ to 3′ direction, and III is synthesized in segments in the 5′ to 3′ direction. ✓

(D) II is synthesized in segments after DNA polymerase is released from synthesizing strand IV.

Answer C

Correct. As a replication fork moves, DNA polymerase can continuously replicate one strand in the 5′ to 3′ direction as the double stranded DNA is separated. The other strand (lagging strand) must be replicated in segments as the fork moves, since DNA polymerase only replicates DNA in the 5′ to 3′ direction, and DNA ligase later joins these fragments together.

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Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the following enzymes.

(A) DNA ligase
(B) DNA polymerase
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase

113. Enzyme used during replication to attach Okazaki fragments to each other

(A) DNA ligase ✓
(B) DNA polymerase
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase

114. Enzyme used to position nucleotides during DNA replication

(A) DNA ligase
(B) DNA polymerase ✓
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase

115. Enzyme used in the synthesis of mRNA

(A) DNA ligase
(B) DNA polymerase
(C) RNA polymerase ✓
(D) Restriction enzyme
(E) Reverse transcriptase

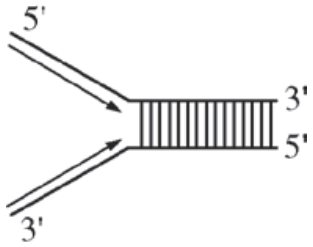
116. Enzyme found in retroviruses that produce DNA from an RNA template

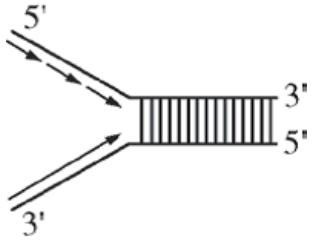
6 Gene Expression and Regulation MCQ

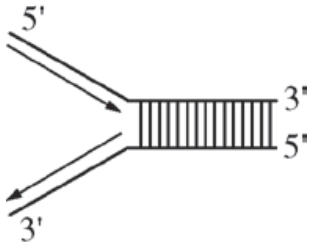
(A) DNA ligase
(B) DNA polymerase
(C) RNA polymerase
(D) Restriction enzyme
(E) Reverse transcriptase ✓

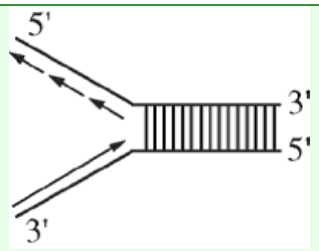
117. When DNA replicates, each strand of the original DNA molecule is used as a template for the synthesis of a second, complementary strand. Which of the following figures most accurately illustrates enzyme-mediated synthesis of new DNA at a replication fork?

6 Gene Expression and Regulation MCQ

(A) 

(B) 

(C) 

(D) 

Answer D

This option is correct. It demonstrates an understanding of the structure of DNA and the process of replication. The two strands of the DNA molecule run antiparallel to each other; the 5' end of one strand pairs with the 3' end of the other strand. Replication is semi-conservative, with each strand serving as the template for the creation of new complementary strands. Helicase unzips the DNA molecules between

6 Gene Expression and Regulation MCQ

the hydrogen bonds connecting the two strands. DNA polymerase “reads” each template strand in the 3'-to-5' direction and assembles the growing DNA chain in a 5'-to-3' direction. The “leading” strand is produced continuously, but on the “lagging” strand, Okazaki fragments are produced and are connected by ligase to produce the daughter molecule.

118. Figure 1 illustrates a model of the molecules involved in DNA replication and their placement relative to each other.

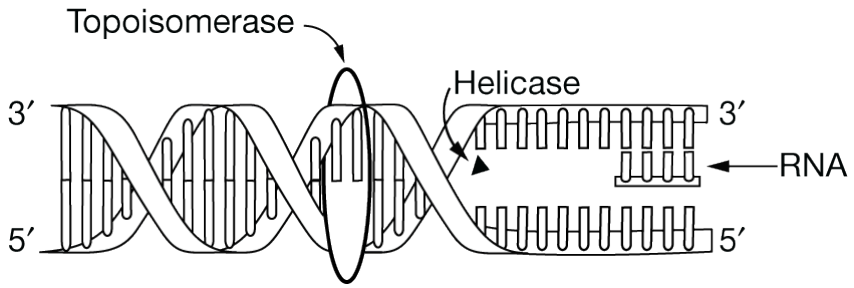


Figure 1. Model including molecules involved in DNA replication

Which of the following correctly explains where DNA replication will begin on the strand oriented 5′ → 3′, reading from left to right?

(A) DNA replication will be randomly initiated along the unwound portion of the DNA strand since base pairing will occur.

(B) DNA replication cannot occur since there is already RNA base pairing with the template strand.

(C) DNA replication will be initiated immediately to the left of the RNA, since DNA polymerase requires an RNA primer. ✓

(D) DNA replication will be initiated at the site of the topoisomerase since that is where DNA begins to uncoil.

Answer C

Correct. The RNA primer marks the beginning of DNA synthesis in the 5′ to 3′ direction. There is only one RNA primer present on the top strand in the diagram so DNA replication will begin there.

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119.

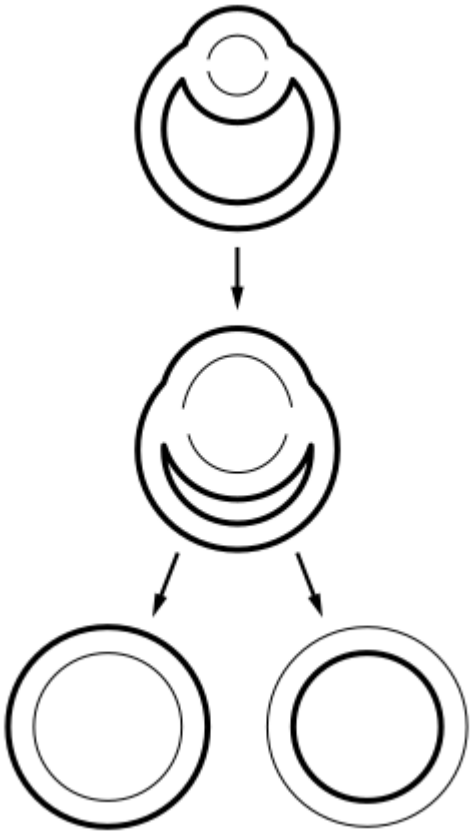


Figure 1. A cellular process

Which of the following processes is best represented by Figure 1 ?

(A) Cell signaling to increase gene expression

(B) Translation of an mRNA to produce a protein

(C) Regulation of gene expression by microRNA

(D) Replication of a plasmid ✓

Answer D

Correct. Plasmids are small circles of bacterial DNA. DNA replication starts by separating the two complementary, original strands of the plasmid, synthesizing two new strands that are complementary to each of the original strands, and finishing with two new circles of DNA, each of which contains one of the original plasmid strands and one new complementary strand.

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120. A simplified model of a DNA replication fork is represented in Figure 1. The protein labeled Enzyme 1 carries out a specific role in the DNA replication process.

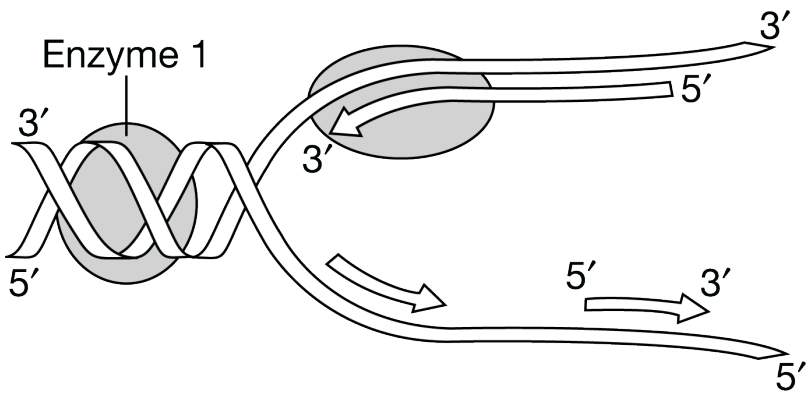


Figure 1. A simplified model of a DNA replication fork

Which of the following statements best explains the role of Enzyme 1 in the DNA replication process?

(A) Enzyme 1 is a DNA ligase that joins together the DNA fragments at a replication fork to form continuous strands.

(B) Enzyme 1 is a DNA primase that catalyzes the synthesis of RNA primers on the lagging strand of a replication fork.

(C) Enzyme 1 is a DNA polymerase that synthesizes new DNA by using the leading and lagging strands of a replication fork as templates.

(D) Enzyme 1 is a topoisomerase that relieves tension in the overwound DNA in front of a replication fork. ✓

Answer D

Correct. Because Enzyme 1 carries out its role in the DNA replication process ahead of the DNA replication fork, it is most likely a topoisomerase. Topoisomerases relieve tension in the overwound DNA in front of replication forks.

6 Gene Expression and Regulation MCQ

121. Labeled nucleotides were supplied to a cell culture before the cells began DNA replication. A simplified representation of the process for a short segment of DNA is shown in Figure 1. Labeled DNA bases are indicated with an asterisk (*).

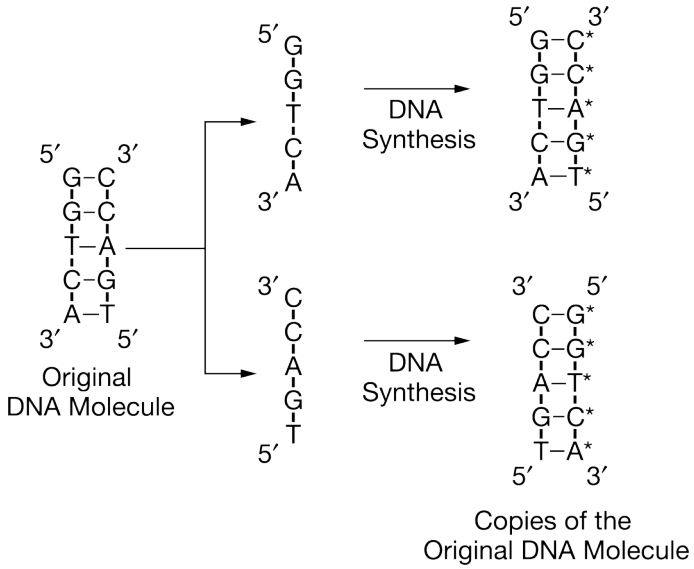


Figure 1. A simplified representation of the DNA replication process

Which of the following best helps explain how the process represented in Figure 1 produces DNA molecules that are hybrids of the original and the newly synthesized strands?

(A) Each template strand is broken down into nucleotides, which are then used to synthesize both strands of a new DNA molecule.

(B) Each template strand is broken into multiple fragments, which are randomly assembled into two different DNA molecules.

(C) Each newly synthesized strand is associated with another newly synthesized strand to form a new DNA molecule.

(D) Each newly synthesized strand remains associated with its template strand to form two copies of the original DNA molecule. ✓

Answer D

Correct. Figure 1 shows a simplified representation of semiconservative DNA replication. The process of semiconservative replication produces DNA molecules that consist of one template strand and one newly synthesized strand.

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122. Figure 1 represents part of a process essential to gene expression.

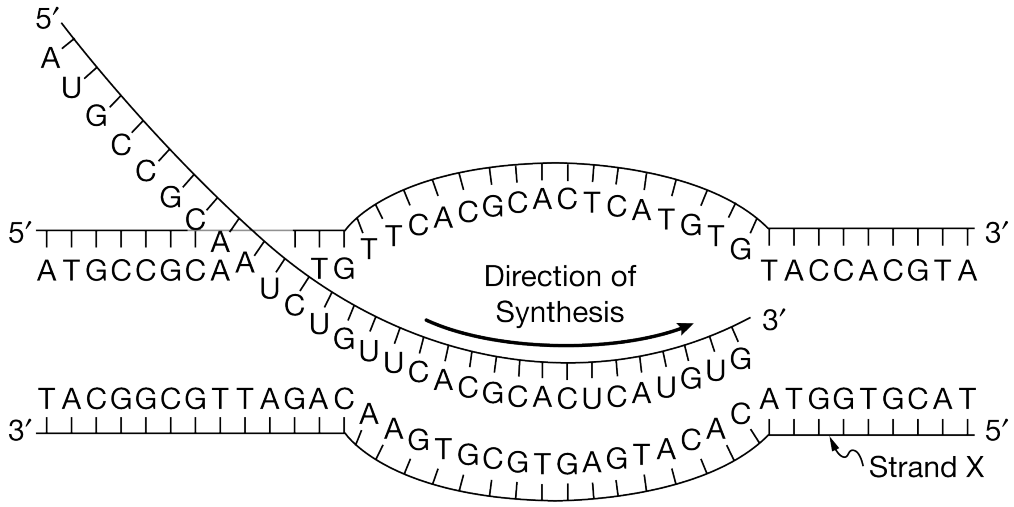


Figure 1. Model of process involved in gene expression.

Which of the following best explains what strand X represents?

(A) A complementary RNA sequence, because it contains thymine

(B) The coding strand in this process, because it is being read $3'$ to $5'$

(C) The antisense strand, because it is serving as a template ✓

(D) The pre-mRNA, because it does not yet have a GTP cap

Answer C

Correct. Strand X is serving as the template strand for the synthesis of a complementary strand of RNA. The template strand is also called the antisense strand.

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123. A model that represents a process occurring in a cell of a particular organism is shown in Figure 1.

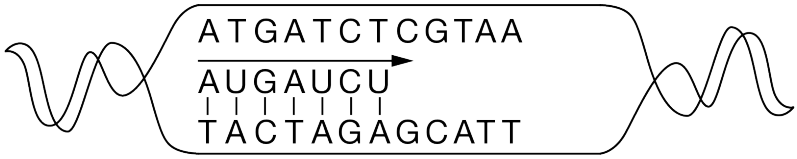


Figure 1. Process occurring in a cell

Which of the following correctly explains the process shown in Figure 1 ?

(A) DNA replication is occurring because replication is semi-conservative and the new strand is a copy of the template strand.

(B) Initiation of transcription is occurring because a strand of RNA is being produced from a DNA template strand. ✓

(C) Translation is occurring because the two strands have separated and a new strand is being produced.

(D) Alternative splicing of mRNA is occurring because the mRNA strand is being synthesized from only one strand of DNA.

Answer B

Correct. Transcription is occurring. Since uracil (U) is present in the new strand being formed, RNA is being formed from a DNA template.

124. The human *TPM1* gene encodes members of the tropomyosin family of cytoskeletal proteins. Which of the following best explains how different proteins can be made in different cell types from the one *TPM1* gene?

(A) Different introns are selectively converted to exons.

(B) Different exons are retained or spliced out of the primary transcript. ✓

(C) The GTP cap is selectively added to and activates different exons.

(D) Different portions of the primary transcript remain bound to the template DNA.

125.

Step 1 Step 2 Step 3

DNA ⟶ Pre-mRNA ⟶ mRNA ⟶ Protein

Which of the following best describes an event during step 2 in the simplified model above?

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(A) A new RNA molecule is synthesized using a DNA template.

(B) A new polypeptide is synthesized using an RNA template.

(C) Thymine nucleotides in an RNA molecule are replaced with uracil nucleotides.

(D) Noncoding sequences are removed from a newly synthesized RNA molecule. ✓

126. If a segment of DNA is 5'-TAC GAT TAG-3', the RNA that results from the transcription of this segment will be

(A) 3'-TAC GAT TAU-5'

(B) 3'-ATG CTA ATA-5'

(C) 3'-UAC GAU UAG-5'

(D) 3'-AAC GAU UAA-5'

(E) 3'-AUG CUA AUC-5' ✓

127. Eukaryotes transcribe RNA from DNA that contains introns and exons. Alternative splicing is one posttranscriptional modification that can create distinct mature mRNA molecules that lead to the production of different proteins from the same gene. Figure 1 shows a gene and the RNA produced after transcription and after alternative splicing.

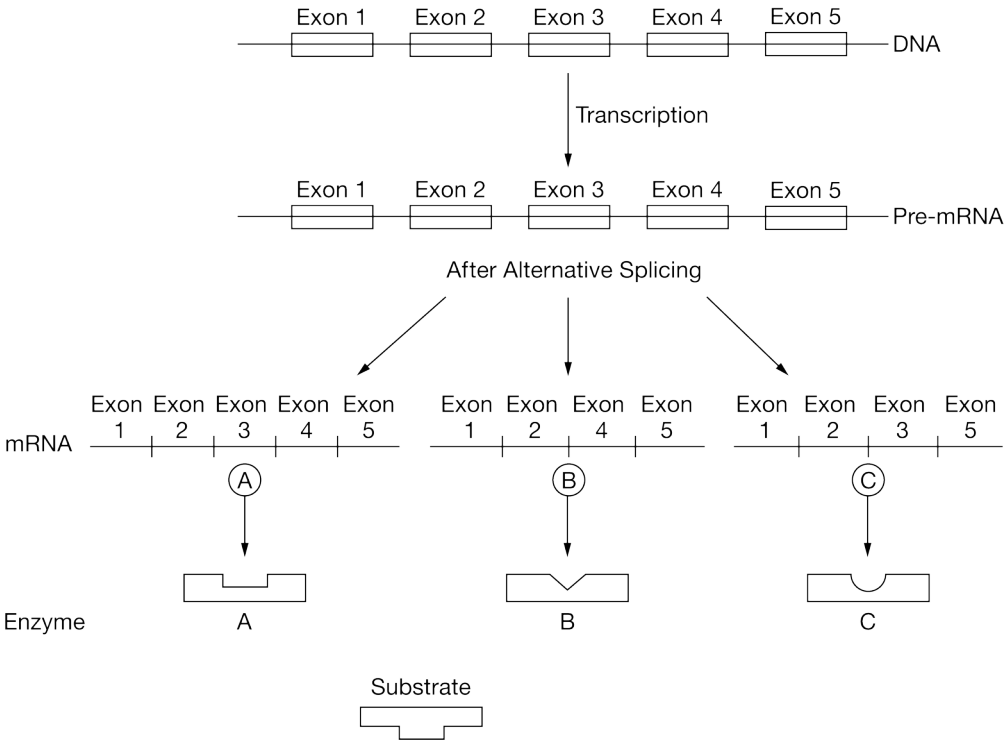


Figure 1. Model of posttranscriptional alternate splicing of mRNA

A cell needs to metabolize the substrate illustrated in Figure 1 for a vital cellular function. Which of the following best explains the long-term effect on the cell of splicing that yields only enzyme C mRNA?

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(A) The cell will die because it is unable to metabolize the substrate without enzyme A, which is structurally specific for the substrate shown. ✓

(B) The cell will remain healthy because all three of the above enzymes can metabolize the substrate, as they are from the same gene.

(C) The cell will remain healthy because the enzyme C mRNA will undergo alternative splicing again until it transformed into enzyme A mRNA.

(D) The cell will remain healthy because enzyme-substrate interactions are nonspecific and enzyme C will eventually metabolize the substrate.

Answer A

Correct. Alternative splicing allows all three enzymes to be produced. If only enzyme C mRNA is available for translation, then only enzyme C (which is not specific for the substrate) will be produced. Thus, the cell will not be able to perform a vital cellular function and it will die.

128.

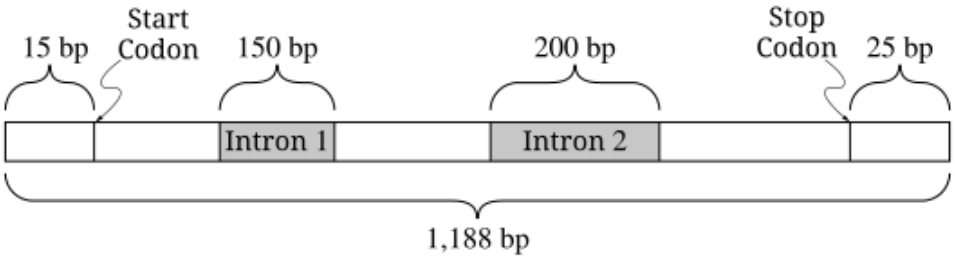


Figure 1. A model of the pre-mRNA transcribed from a gene

Which of the following is closest to the number of amino acids encoded by the mRNA produced from the pre-mRNA shown in Figure 1 ?

(A) 265 ✓

(B) 396

(C) 838

(D) 1,188

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Answer A

Correct. The total length of noncoding sequences [$150\ \text{bp} + 200\ \text{bp}$ for introns, $15\ \text{bp}$ for the 5' untranslated region (UTR), and $25\ \text{bp}$ for the 3' UTR $= 390$] as well as $3\ \text{bp}$ for the stop codon (which does not code for an amino acid) is subtracted from the total length of the $\text{pre} - \text{mRNA}$ ($1{,}188\ \text{bp}$). That total is then divided by 3 ($3\ \text{bp}$ per codon) to determine the total number of amino acids. $\frac{1{,}188-390-3}{3} = 265$

129. Which of the following normally leads to the production of functional messenger RNA in eukaryotic cells?

(A) A decrease in the rate of ribosome synthesis

(B) The removal of portions of RNA known as intervening sequences (introns) ✓

(C) A decrease in RNA polymerase activity

(D) The replication of new messenger RNA molecules from existing messenger RNA molecules

(E) The formation of peptide bonds between adjacent nucleotides

---

- **Directions**: Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

mRNA Codons	Amino Acids
AGA	arginine
GGA	glycine
AGC	serine
GCA	alanine
CAG	glutamine

130. . . . . glutamine-glutamine-glutamine . . .

. . . serine-serine-serine . . .

Which of the following messenger RNA sequences could code for both of the two amino acid sequences above, simply by a shift in the reading frame?

(A) . . . AGCAGCAGCAGC . . . ✓

(B) . . . AGUAGUAGUAGU . . .

(C) . . . CAACAACAACAA . . .

(D) . . .GCUGCUGCUGCU . . .

(E) . . . GCAAGCGCAAGC . . .

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131. . . . . glycine-serine-glycine . . .

Which of the following DNA strands will code for the amino acid sequence shown above?

(A) . . . ACTCCTTCT . . .

(B) . . . TCTCCGTCG . . .

(C) . . . CCGTCGACT . . .

(D) . . . CCTCCGTCG . . .

(E) . . . CCTTCGCCT . . . ✓

132.

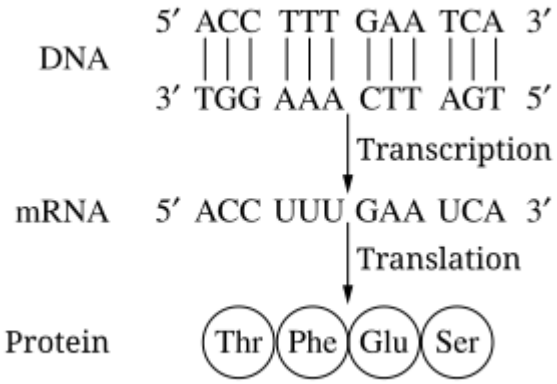


Figure 1. Model of a cellular process

Which of the following best describes the model shown in Figure 1 ?

(A) The model shows how genetic information directs the synthesis of polypeptides. ✓

(B) The model shows how a mutation in the DNA results in a nonfunctional protein.

(C) The model shows how information in the mRNA can be used to make both protein and DNA.

(D) The model shows how proteins are needed for the synthesis of both DNA and RNA.

Answer A

Correct. The model shows the central dogma of molecular biology in which DNA is transcribed to RNA, which then directs the formation of a polypeptide chain.

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133. Figure 1 represents part of a process that occurs in eukaryotic cells. There are untranslated regions (UTR) in this sequence.

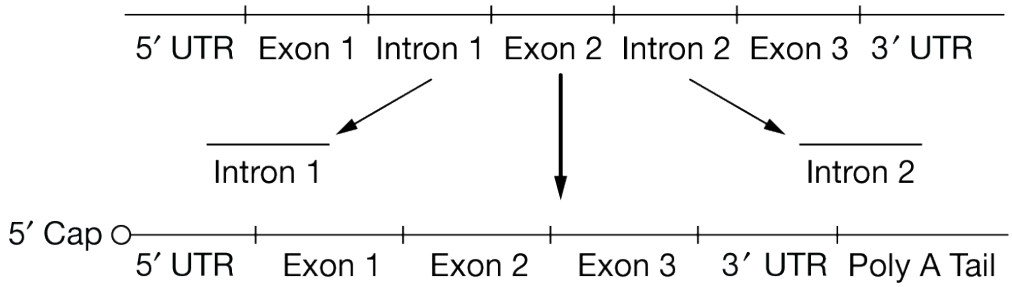


Figure 1. Cellular process involving nucleic acids

Which of the following best explains the process represented by Figure 1 ?

(A) The synthesis of mRNA in the 5′ to 3′ direction from DNA

(B) The modification of a protein to produce a functional form of that protein

(C) The translation of an mRNA molecule into a polypeptide

(D) The enzyme-regulated processing of pre-mRNA into mature mRNA ✓

Answer D

Correct. The removal of introns and the addition of a 3′ poly-A tail and a 5′GTP cap are associated with mRNA processing in eukaryotic cells, and the diagram reflects these changes.

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134. All cells must transcribe rRNA in order to construct a functioning ribosome. Scientists have isolated and identified rRNA genes that contribute to ribosomal structure for both prokaryotes and eukaryotes. Figure 1 compares the transcription and processing of prokaryotic and eukaryotic rRNA.

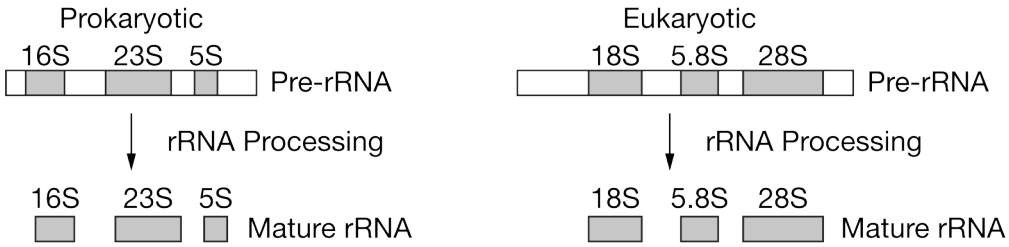


Figure 1. Comparison of rRNA processing in prokaryotes and eukaryotes

Which of the following statements provides the best explanation of the processes illustrated in Figure 1 ?

(A) Introns are removed from the pre-rRNA, and the mature rRNA molecules are joined and then translated to produce the protein portion of the ribosome.

(B) Introns are removed from the pre-rRNA, and each mature rRNA molecule is translated to produce the proteins that make up the ribosomal subunits.

(C) Sections of the pre-rRNA are removed, and the mature rRNA molecules are available to combine with proteins to form the ribosomal subunits. ✓

(D) Sections of the pre-rRNA are removed, and the mature rRNA molecules are available to bring different amino acids to the ribosome.

Answer C

Correct. In both prokaryotes and eukaryotes, the mature rRNA molecules are removed from a longer transcript. These mature rRNA molecules, together with proteins, form the large and small ribosomal subunits.

135. Which of the following statements concerning a gene is correct?

(A) A gene can code for a specific protein. ✓

(B) A gene can exist in alternate forms called introns.

(C) A gene undergoes crossing-over during DNA replication.

(D) A gene that is very similar in sequence in a human and in a bacterium is probably a recent mutation.

(E) A gene that is expressed in every offspring of every generation is recessive.

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136. The following the DNA sequence is a small part of the coding (nontemplate) strand from the open reading frame of *β-hemoglobin* gene. Given the codon chart listed below, what would be the effect of a mutation that deletes the G at the beginning of the DNA sequence?

5'- GTT TGT CTG TGG TAC CAC GTG GAC TGA - 3'

Second Base in Codon

First Base in Codon	U	C	A	G	Third Base in Codon
U	UUU, UUC: Phe; UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr; UAA Stop; UAG Stop	UGU, UGC: Cys; UGA Stop; UGG Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAC: His; CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile; AUG: Met or Start	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn; AAA, AAG: Lys	AGU, AGC: Ser; AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp; GAA, GAG: Glu	GGU, GGC, GGA, GGG: Gly	U C A G

(A) The mutation precedes the gene, so no changes would occur.

(B) Lysine (lys) would replace glutamine (gln), but there would be no other changes.

(C) The first amino acid would be missing, but there would be no other change to the protein.

(D) The reading frame of the sequence would shift, causing a change in the amino acid sequence after that point. ✓

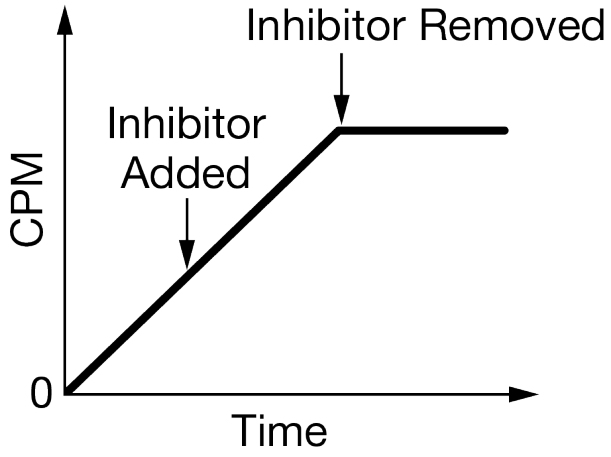
137. Cycloheximide (CHX) is a eukaryote protein synthesis inhibitor. It is used in biomedical research to inhibit protein synthesis in eukaryotic cells studied *in vitro*. Its effects are rapidly reversed by simply removing it from the culture medium.

In a translation experiment using a fungus culture, radiolabeled amino acids were added to the culture, allowing the researchers to measure the growth of a single polypeptide chain by measuring counts per minute (CPM). As the chain grew, the CPM increased. After a certain amount of time, CHX was added to the mixture, and the experiment continued. After an additional amount of time, the CHX was removed from the culture medium.

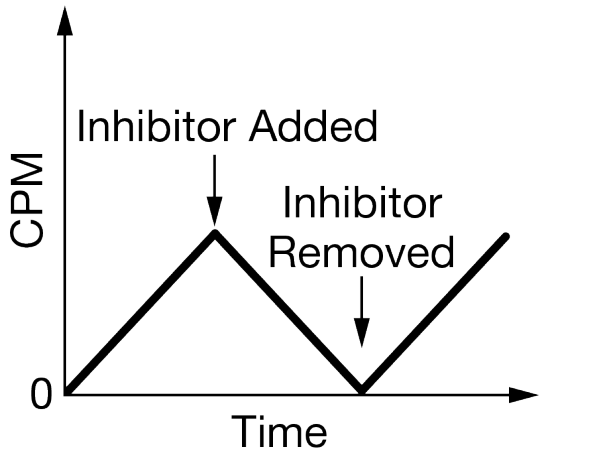
Which of the following graphs best predicts the data collected during the experiment?

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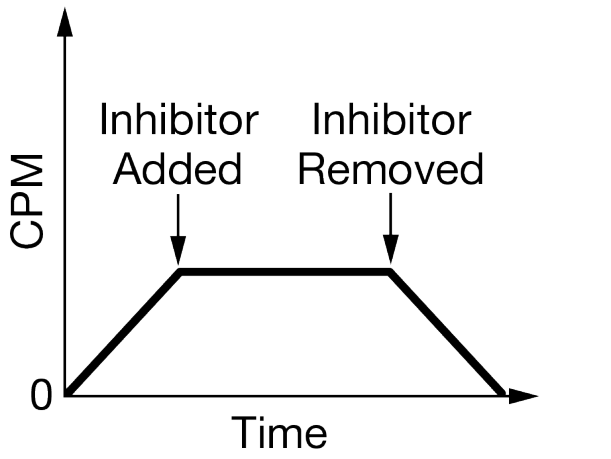
(A)



(B)

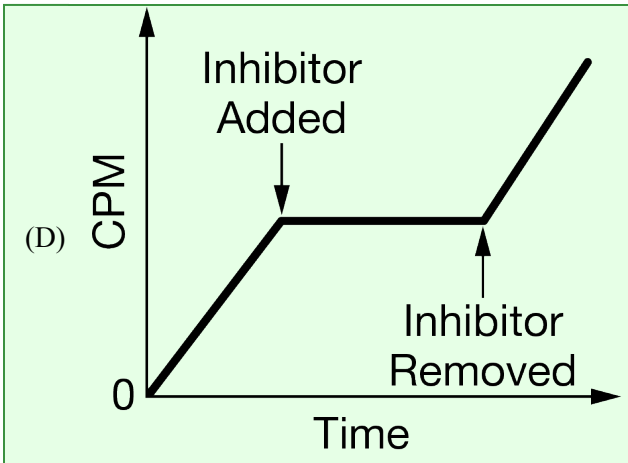


(C)



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(D)



Answer D

Correct. In the presence of the drug, protein synthesis halted and the polypeptide chain stopped growing. When the drug was removed, however, elongation resumed and the polypeptide chain started growing again.

138.

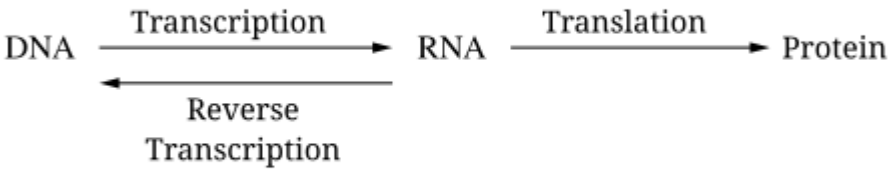


Figure 1. A model of a biological process

Which of the following processes requires all of the steps represented in Figure 1 ?

6 Gene Expression and Regulation MCQ

(A) Cell signaling to increase gene expression in an animal cell

(B) Replication of a retrovirus after it enters a host cell ✓

(C) Regulation of gene expression in a cell by microRNA

(D) Production of plasmids in a bacterial cell

Answer B

Correct. Retroviruses insert viral RNA and the enzyme reverse transcriptase into a host cell. The reverse transcriptase copies the RNA into DNA, and then the cell's own transcription and translation machinery synthesizes new viral proteins in the process of producing new viruses.

139. Analysis of DNA sequences from two individuals of the same species results in a greater estimate of genetic variability than does analysis of amino acid sequences from the same individuals because

(A) different DNA sequences can code for the same amino acid ✓

(B) some amino acid variations cannot be detected by protein electrophoresis

(C) DNA sequencing is a more reliable technique than protein electrophoresis

(D) proteins are more easily damaged than is DNA

(E) DNA is more heat-sensitive and therefore varies more

140. During the infection cycle for a typical retrovirus, which uses RNA as genetic material, the genetic variation in the resulting population of new virus particles is very high because of

(A) damage to the virus particle from envelope loss during infection

(B) errors introduced in the DNA molecule through reverse transcription ✓

(C) errors in the protein molecules produced in translation

(D) recombination of the genomes of free virus particles

Answer B

This option is correct. The process of reverse transcription does not involve proofreading.

6 Gene Expression and Regulation MCQ

141.

TABLE 1. RATE OF POLYPEPTIDE ELONGATION

Species	Number of Amino Acids Added	Duration of Measurement (seconds)
E. coli	1,200	60
S. cerevisiae	800	80

During translation of an mRNA, amino acids are added one-by-one to the growing end of a polypeptide. Researchers wanted to determine whether there is a difference in the rate of amino acid addition in *E. coli*, a prokaryote, and *S. cerevisiae*, a single-celled eukaryote (Table 1).

Based on the data in Table 1, how does the rate of polypeptide elongation in *E. coli* compare with the rate of polypeptide elongation in *S. cerevisiae*?

(A) The rate in *E. coli* is one-tenth the rate in *S. cerevisiae*.

(B) The rate in *E. coli* is half the rate in *S. cerevisiae.*

(C) The rate in *E. coli* is one-and-one-half times the rate in *S. cerevisiae.*

(D) The rate in *E. coli* is twice the rate in *S. cerevisiae*. ✓

Answer D

Correct. The rate of elongation in *E. coli* is $20 \text{ amino acids/second}$, while the rate of elongation in *S. cerevisiae* is $10 \text{ amino acids/second}$. Thus, *E. coli* has a rate that is twice that of *S. cerevisiae*.

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142. Molecular biologists are studying the processes of transcription and translation and have found that they are very similar in prokaryotes and eukaryotes, as summarized in Table 1.

Table 1. Comparison of Selected Features of Transcription and Translation

Feature of Protein Synthesis	Prokaryotes	Eukaryotes
Transcription		
RNA polymerase	Present	Present
mRNA	Synthesized	Synthesized
Location	Cytoplasm	Nucleus
Translation		
Large Ribosomal Subunit	50S	60S
Small Ribosomal Subunit	30S	40S
tRNA	Present	Present
Location	Cytoplasm	Cytoplasm

Based on the information in Table 1, which of the following best predicts a key difference in prokaryotes and eukaryotes with regard to transcription and translation?

(A) The two processes will occur simultaneously in prokaryotes but not eukaryotes. ✓

(B) Prokaryotic mRNA is shorter than eukaryotic mRNA.

(C) Eukaryotic mRNA contains more coding regions than prokaryotic DNA.

(D) The processing of mRNA by eukaryotes is required for the mRNA to leave the nucleus.

Answer A

Correct. Since transcription and translation both occur in the cytoplasm, they can occur simultaneously in prokaryotes. Note that the spatial separation of these two processes in eukaryotes increases the capacity for gene regulation.

143. Retroviruses have an RNA genome. HTLV-1 is a lysogenic retrovirus that establishes a latent infection in human cells. By which of the following mechanisms does infection by a retrovirus such as HTLV-1 most likely cause long-lasting genetic changes to host cells?

(A) The host-cell ribosomes translate the viral RNA genome that enters the cell upon initial viral infection.

(B) The viral RNA polymerase that transcribes host genes has a high error rate.

(C) The RNA viral genome is reverse transcribed into DNA that integrates into the host genome. ✓

(D) The RNA viral genome integrates into the host genome.

6 Gene Expression and Regulation MCQ

144. Retroviruses such as HIV and hepatitis B virus use RNA as their genetic material rather than DNA. In addition, they contain molecules of reverse transcriptase, an enzyme that uses an RNA template to synthesize complementary DNA.

Which of the following best predicts what will happen when a normal cell is exposed to a retrovirus?

(A) The reverse transcriptase will cut the host DNA into fragments, destroying the host cell.

(B) The reverse transcriptase will insert the viral RNA into the host's genome so it can be transcribed and translated.

(C) The reverse transcriptase will produce DNA from the viral RNA, which can be incorporated into the host's genome and then transcribed and translated. ✓

(D) The reverse transcriptase will force the host ribosomes to translate the viral RNA prior to polypeptide assembly.

Answer C

Correct. Reverse transcriptase will transcribe DNA from RNA. The DNA integrates into the host genome and is transcribed and translated for the assembly of new viral progeny.

145. Antibiotics interfere with prokaryotic cell functions. Streptomycin is an antibiotic that affects the small ribosomal subunit in prokaryotes. Specifically, streptomycin interferes with the proper binding of tRNA to mRNA in prokaryotic ribosomes.

Which of the following best predicts the most direct effect of exposing prokaryotic cells to streptomycin?

(A) Amino acid synthesis will be inhibited.

(B) No mRNA will be transcribed from DNA.

(C) Posttranslational modifications will be prevented.

(D) Synthesis of polypeptides will be inhibited. ✓

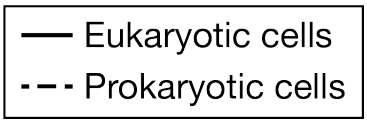
Answer D

Correct. Translation of mRNA involves assembling polypeptides from amino acids attached to tRNA specific for each amino acid. Streptomycin prevents the ribosome from interacting properly with tRNA, preventing the synthesis of needed proteins and leading to the death of the prokaryote.

6 Gene Expression and Regulation MCQ

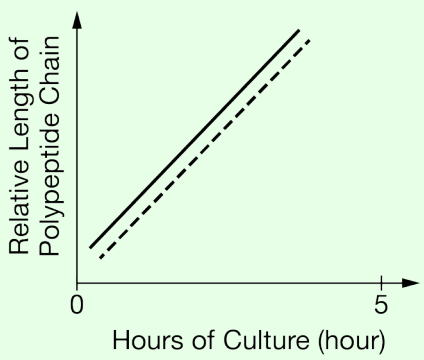
146. Nuclear pores regulate the passage of substances into and out of the nucleus. Antibodies such as mAb414 have been used to inhibit the movement of substances through the nuclear pores of rat liver cells. Scientists cultured rat liver cells (eukaryotic) and bacteria cells (prokaryotic) in separate dishes with radioactively labeled amino acids. A specific gene in both cell types was engineered to synthesize identical polypeptide chains, and translation of this gene was measured. The procedure was repeated with mAb414 (the inhibitor) added to each of the two cell cultures, and translation was monitored again.

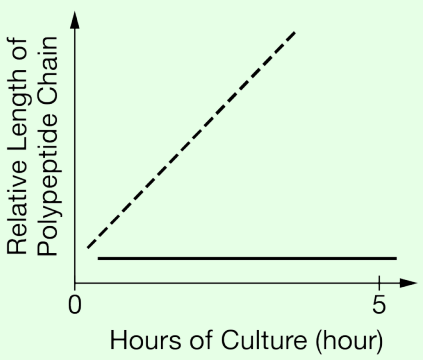
Which of the following sets of graphs best summarizes the results of the experiments?



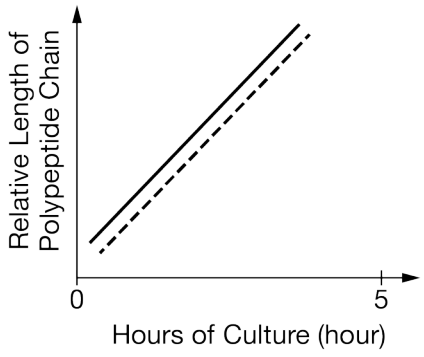
6 Gene Expression and Regulation MCQ

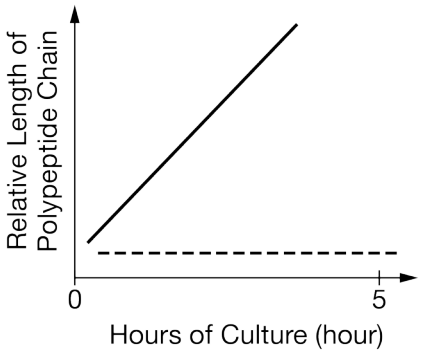
(A)



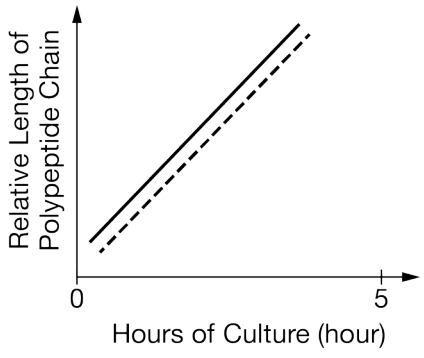


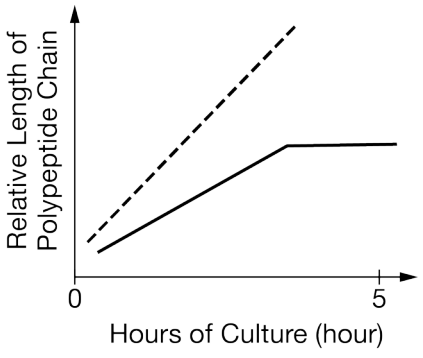
(B)



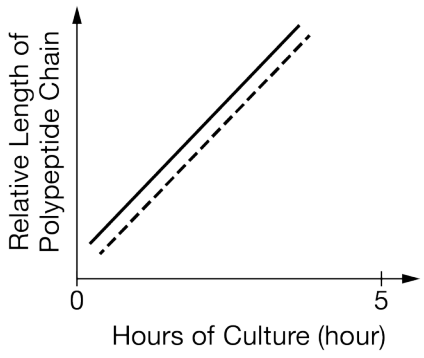


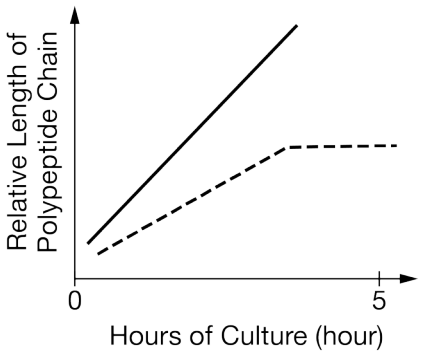
(C)





(D)





6 Gene Expression and Regulation MCQ

Answer A

Correct. In eukaryotes, transcription and translation occur at different areas in the cell, whereas in prokaryotes, they occur simultaneously. In the experimental dish with the eukaryotes, the drug inhibited the translocation of mRNA to the cytoplasm, thereby preventing the translation, so no polypeptide chain was formed. However, in the prokaryotes, with no nucleus and both processes occurring simultaneously, translation occurred and the polypeptide chain was formed.

147.

Second Base in Codon

First Base in Codon	U	C	A	G	Third Base in Codon
U	UUU, UUC: Phe; UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr; UAA Stop; UAG Stop	UGU, UGC: Cys; UGA Stop; UGG Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAC: His; CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile; AUG: Met or Start	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn; AAA, AAG: Lys	AGU, AGC: Ser; AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp; GAA, GAG: Glu	GGU, GGC, GGA, GGG: Gly	U C A G

5′-GACCGCAUGGUGACGAAAUUUGGCCAUUAA-3′

Based on the universal genetic code, which of the following represents the correct polypeptide that will result from translation of the mRNA molecule shown, beginning with the first available start codon?

(A) Asp – Arg – Met – Val – Thr – Lys – Phe – Gly – His

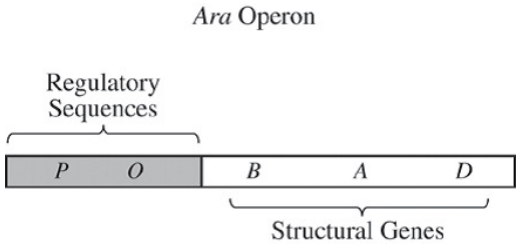
(B) Met – Arg – Asp – Stop – His – Gly – Phe – Lys – Thr – Val

(C) Met – Val – Thr – Lys – Phe – Gly – His ✓

(D) Val – Thr – Lys – Phe – Gly – His

6 Gene Expression and Regulation MCQ

148.



The regulatory sequences of the operon controlling arabinose metabolism (*ara* operon) were studied to determine whether bacteria can respond to changes in nutrient availability. It is predicted that if those regulatory sequences are functioning properly, the bacteria will produce the enzymes involved in arabinose metabolism (structural genes *B*, *A,* and *D*) in the presence of arabinose.

If a gene that encodes a green fluorescent protein (GFP) is substituted for the structural genes of the operon, activation of the regulatory sequences can be assayed by GFP expression. A culture of *E. coli* cells underwent a transformation procedure with a plasmid containing the regulatory sequences of the *ara* operon directly upstream of the gene encoding the GFP. The plasmid also confers ampicillin resistance to bacteria. Samples were then plated on different types of culture media. (Note: The GFP fluoresces only under UV light, not under white light.) The table below shows the results.

Transformation Results

Type of Culture Media			Color of Colonies Under White Light	Color of Colonies Under UV Light
Agar	Ampicillin	Arabinose		
+	-	-	All white	All white
+	+	-	All white	All white
+	-	+	All white	Mostly white, some green
+	+	+	All white	All green

+ Indicates the presence of the indicated substance in the culture media
- Indicates the absence of the indicated substance in the culture media

Which of the following can best be used to justify why the GFP is expressed by *E. coli* cells after transformation with the plasmid?

(A) The presence of arabinose in the nutrient agar activated the expression of the genes located downstream of the *ara* operon regulatory sequences. ✓

(B) The combination of ampicillin and arabinose in the nutrient agar inhibited the expression of certain gene products, resulting in the increased expression of the GFP.

(C) The nutrient agar without arabinose but with ampicillin activated the expression of the genes located downstream of the *ara* operon regulatory sequences.

(D) Both arabinose and ampicillin were required in the nutrient agar to activate the expression of genes located downstream of the *ara* operon regulatory sequences.

6 Gene Expression and Regulation MCQ

Answer A

This option is correct. Arabinose will bind with the repressor protein allowing expression of the genes controlled by the *ara* regulatory sequences. Because the genes for ampicillin resistance and GFP were inserted downstream of the regulatory sequences, these genes will be expressed and the products will cause the resultant colonies to be green under UV light. This answer indicates that the student is able to describe the connection between the regulation of gene expression and observed differences between different kinds of organisms.

149. Arctic foxes typically have a white coat in the winter. In summer, when there is no snow on the ground, the foxes typically have a darker coat.

Which of the following is most likely responsible for the seasonal change in coat color?

(A) The decrease in the amount of daylight in winter causes a change in gene expression, which results in the foxes growing a lighter-appearing coat. ✓

(B) The diet of the foxes in summer lacks a particular nutrient, which causes the foxes to lose their white coat and grow a darker-colored coat.

(C) Competition for mates in the spring causes each fox to increase its camouflage with the environment by producing a darker-appearing coat.

(D) The lower temperatures in winter denature the pigment molecules in the arctic fox coat, causing the coat to become lighter in color.

Answer A

This option is correct. It demonstrates the ability to explain the influence of environmental factors on the phenotype of an organism. When the genes of the arctic fox that produce the dark coat are blocked by cold temperatures, a white coat results. In the summer these genes are not blocked, and a darker coat results.

6 Gene Expression and Regulation MCQ

150. The table below describes the action of two genes involved in the regulation of nervous system development in the nematode *C. elegans*.

	Gene *A*	Gene *B*	Observation
Pattern 1	Inactive	Inactive	No neurons develop.
Pattern 2	Inactive	Active	No neurons develop.
Pattern 3	Active	Inactive	Greater-than-normal number of neurons develop.
Pattern 4	Active	Active	Normal number of neurons develop.

Which of the following claims is best supported by the data?

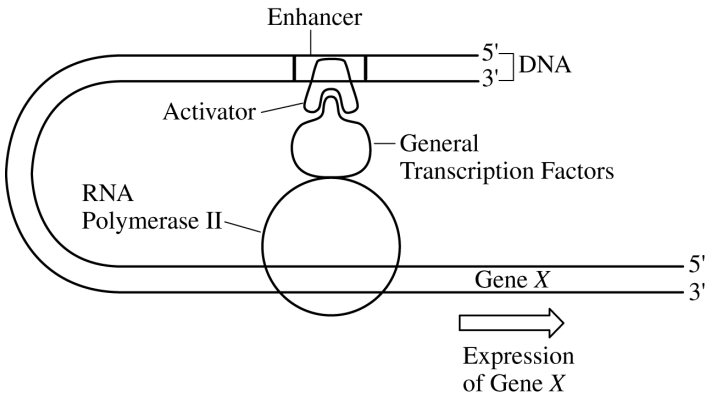
(A) Gene *A* promotes neuron development; gene *B* promotes programmed cell death in neuronal precursors. ✓

(B) Gene *A* promotes programmed cell death in neuronal precursors; gene *B* promotes neuron development.

(C) Gene *B* must be active before gene *A* can function.

(D) Gene *B* must be inactive before gene *A* can function.

151.



The figure above depicts the DNA-protein complex that is assembled at the transcriptional start site of gene *X* when the expression of gene *X* is activated in liver cells. Previous studies have shown that gene *X* is never expressed in nerve cells. Based on the diagram, which of the following most likely contributes to the specific expression pattern of gene *X*?

(A) Expression of gene *X* produces large amounts of tRNA but undetectable amounts of mRNA.

(B) The general transcription factors inhibit the activation of gene *X* in liver cells by blocking the activator from binding to RNA polymerase II.

(C) The activator is a sequence-specific DNA-binding protein that is present in some tissues but not in other tissues. ✓

(D) The enhancer is a unique DNA segment that is added to the nuclear DNA of some cells of an organism during the process of mitotic cell division but not other cells.

6 Gene Expression and Regulation MCQ

152. Histone methyltransferases are a class of enzymes that methylate certain amino acid sequences in histone proteins. A research team found that transcription of gene R decreases when histone methyltransferase activity is inhibited.

Which scientific claim is most consistent with these findings?

(A) DNA methylation inhibits transcription of gene R.

(B) Histone modifications of genes are usually not reversible.

(C) Histone methylation condenses the chromatin at gene R so transcription factors cannot bind to DNA.

(D) Histone methylation opens up chromatin at gene R so transcription factors can bind to DNA more easily. ✓

Answer D

Correct. This claim is consistent with the findings presented. Histone methylation in this case increases transcription of gene R, so blocking histone methyltransferase activity would inhibit transcription.

153. Pit vipers are snakes that have deep sensory openings (pits) between their nostrils and eyes. These pits enable the snakes to sense heat emitted by endotherms, such as mammals, allowing the snakes to find their prey. In an experiment, researchers discovered that nerve cells in the pits transcribe the gene $TRPA1$ 400 times more often than do other nerve cells in the snakes. The researchers claim that the $TRPA1$ gene is involved in the snakes' ability to sense heat.

Which of the following would be the best control to add to the researchers' experiment?

(A) Measuring translation of $TRPA1$ mRNA in pit nerve cells

(B) Measuring translation of $TRPA1$ mRNA in nerve cells of mammals

(C) Measuring $TRPA1$ transcription in nerve cells of snakes lacking pits ✓

(D) Measuring $TRPA1$ transcription in nerve cells of mammals

Answer C

Correct. Comparing the transcription rate of the $TRPA1$ gene in the pit nerve cells of snakes with pits relative to nerve cells in snakes lacking pits would serve as a control to indicate if the high level of $TRPA1$ gene transcription is exclusive to pit nerve cells.

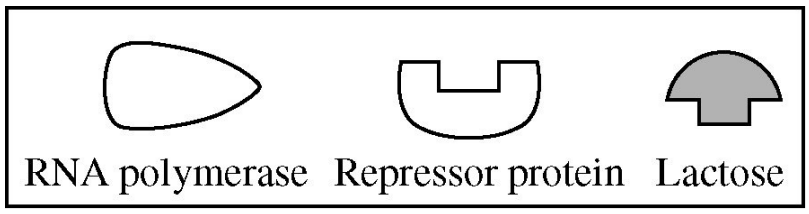
154. Maximal transcription of the *lac* operon requires

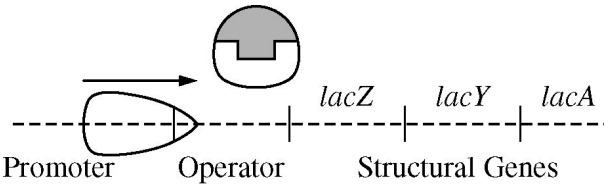
6 Gene Expression and Regulation MCQ

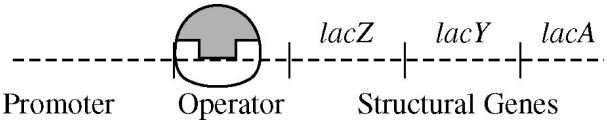
(A) the presence of high levels of glucose
(B) the removal of the operator region of the operon
(C) the presence of cyclic AMP and lactose ✓
(D) the activation of the repressor protein
(E) the presence of cyclin and cyclin-dependent kinase

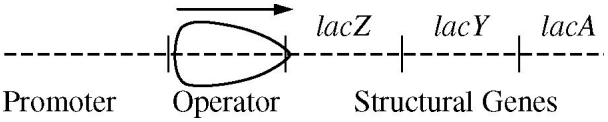
155. Lactose digestion in *E. coli* begins with its hydrolysis by the enzyme *β*-galactosidase. The gene encoding *β*-galactosidase, *lacZ*, is part of a coordinately regulated operon containing other genes required for lactose utilization.

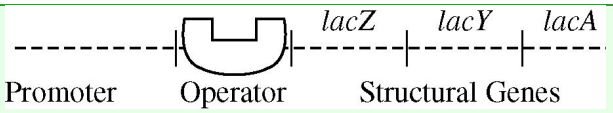
Which of the following figures correctly depicts the interactions at the *lac* operon when lactose is NOT being utilized? (The legend below defines the shapes of the molecules illustrated in the options.)



(A)


(B)


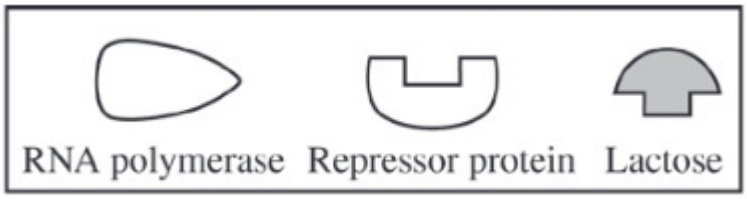
(C)


(D) ✓


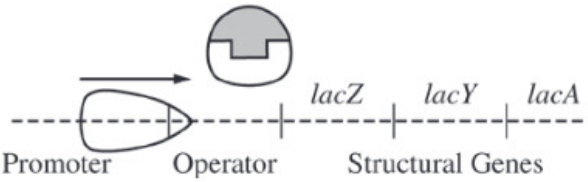
6 Gene Expression and Regulation MCQ

156. Lactose digestion in *E. coli* begins with its hydrolysis by the enzyme *β*-galactosidase. The gene encoding *β*-galactosidase, *lacZ*, is part of a coordinately regulated operon containing other genes required for lactose utilization.

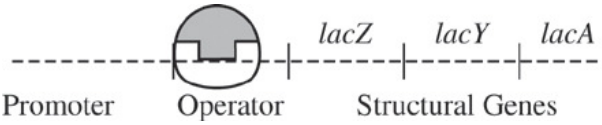
Which of the following figures correctly depicts the interactions at the *lac* operon when lactose is NOT being utilized? (The legend below defines the shapes of the molecules illustrated in the options.)



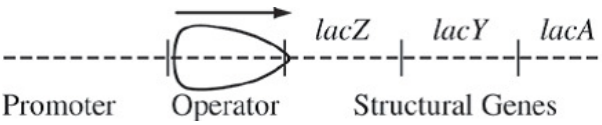
(A)



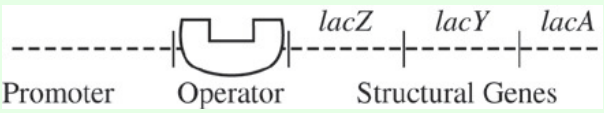
(B)



(C)



(D)



6 Gene Expression and Regulation MCQ

157. Figure 1 represents a metabolic process involving the regulation of lactose metabolism by *E. coli* bacteria. Lactose is utilized for energy by *E. coli* when glucose is not present. Allolactose is an isomer of lactose that is in the environment of these bacteria when lactose is present. The CAP site prevents the binding of RNA polymerase when glucose is present in the environment. The $lacZ$, $lacY$, and $lacA$ genes code for proteins needed for lactose metabolism.

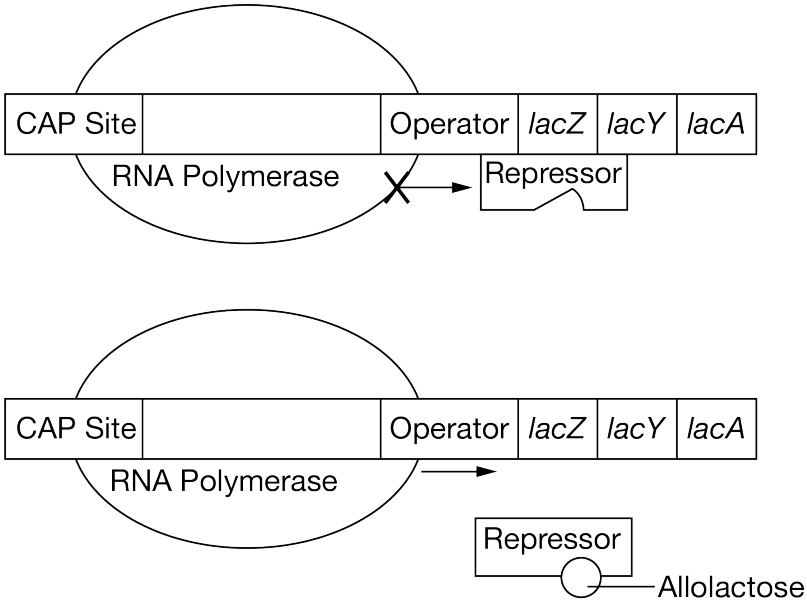


Figure 1. Model of *lac* operon, comparing repressed and active states

Which is a scientific claim that is consistent with the information provided and Figure 1 ?

(A) The presence of excess lactose blocks the functioning of RNA polymerase in this operon.

(B) When bound to the operator, the repressor protein prevents lactose metabolism in *E. coli*. ✓

(C) The binding of the repressor protein to the operator enables *E. coli* to metabolize lactose.

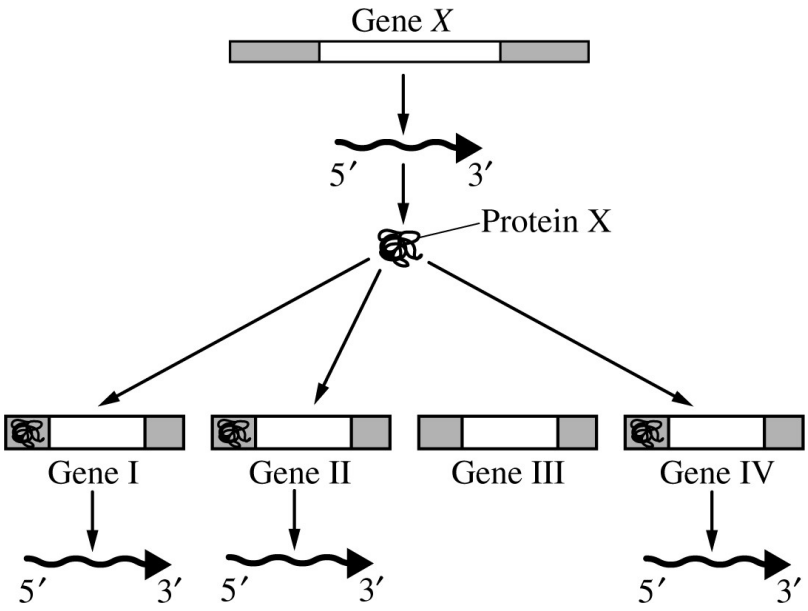
(D) Allolactose acts as an inducer that binds to the operator, allowing *E. coli* to metabolize lactose.

Answer B

Correct. When the repressor protein is bound to the operator, the *lac* operon is turned off and the genes for lactose metabolism cannot be transcribed.

6 Gene Expression and Regulation MCQ

158. Phenotype is determined, in part, by which genes are expressed. The diagram below illustrates how the product of gene *X* regulates the expression of other genes.



Which of the following statements best explains how protein X regulates gene expression?

(A) Protein X is responsible for processing pre-mRNA.

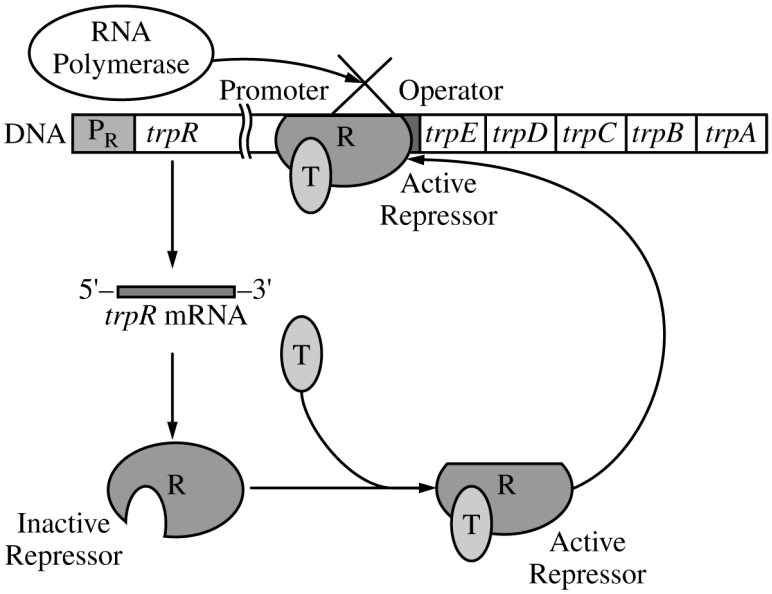
(B) Protein X is responsible for activating transcription of some genes but not others. ✓

(C) Protein X is a member of some cytoplasmic protein complexes but not others.

(D) Protein X causes specific base-pair changes to produce new alleles.

6 Gene Expression and Regulation MCQ

159.



R	= Repressor molecule
T	= Tryptophan molecule
P_R	= Promoter for the *trpR* gene
trpR	= Repressor gene
Operator	= Binding site for repressor molecule

The *Trp* operon is a coordinately regulated group of genes (*trpA–trpE*) that are required for tryptophan biosynthesis in *E. coli*. Based on the figure above, which of the following correctly explains the regulation of the *Trp* operon?

(A) In the absence of tryptophan, the repressor is active and binds to the *Trp* operator, preventing RNA polymerase from transcribing the operon.

(B) In the presence of tryptophan, the repressor is active and binds to the *Trp* operator, preventing RNA polymerase from transcribing the operon. ✓

(C) In the absence of tryptophan, the *trpR* gene is inactive, preventing the production of the repressor that blocks expression of the operon.

(D) In the presence of tryptophan, the *trpR* gene is inactive, preventing the production of the repressor that blocks expression of the operon.

6 Gene Expression and Regulation MCQ

160. Figure 1 illustrates processes related to control of transcription and translation in a cell.

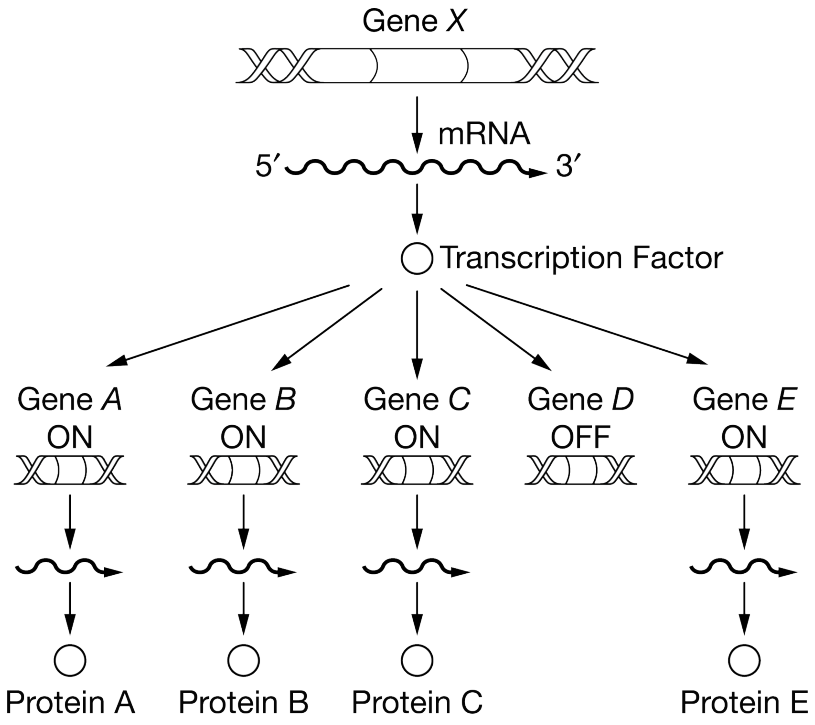


Figure 1. Model of a relationship between a transcription factor and selected genes

Which of the following scientific claims is most consistent with the information provided in Figure 1 ?

(A) Gene X codes for a transcription factor required for transcription of gene D.

(B) A single transcription factor regulates transcription similarly, regardless of the specific gene.

(C) Transcription of genes A, B, and C is necessary to transcribe gene E.

(D) Different genes may be regulated by the same transcription factor. ✓

Answer D

Correct. The same transcription factor may regulate the transcription of multiple genes and exhibit differential regulatory effects on different genes.

6 Gene Expression and Regulation MCQ

161. Figure 1 represents a portion of a process that occurs during protein synthesis.

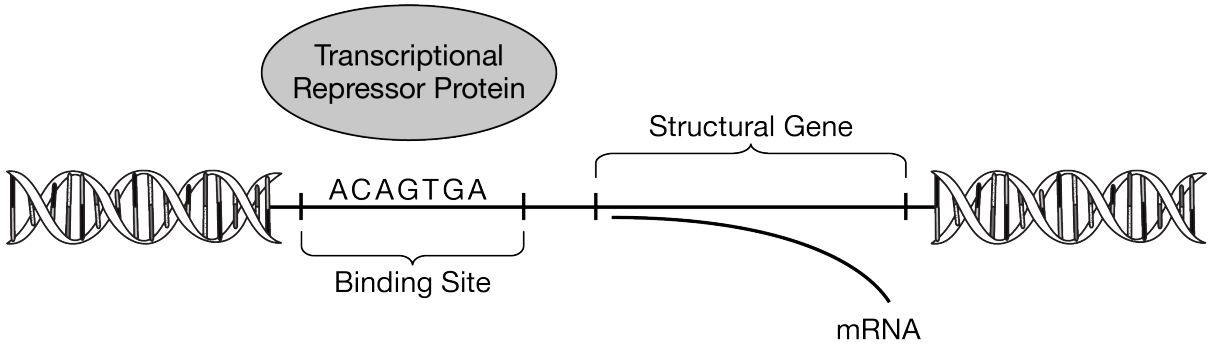


Figure 1. Model of selected features of DNA transcription

Which claim is most consistent with the information provided by the diagram and current scientific understanding of gene regulation and expression?

(A) Reversible changes in the DNA sequence may influence how a gene is expressed in a cell.

(B) Some sequences of DNA can interact with regulatory proteins that control transcription. ✓

(C) This is an inducible operon controlled by several regulatory factors.

(D) The transcription factor may produce mutations in the binding site at the promoter sequence inhibiting the synthesis of the protein.

Answer B

Correct. The regulatory base sequence (called a silencer) interacts with regulatory proteins such as this transcriptional repressor, which inhibits transcription.

6 Gene Expression and Regulation MCQ

The three-spined stickleback (Gasterosteus aculeatus) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mostly of individuals with pronounced pelvic spines, as shown in Figure 1. Individuals in freshwater stickleback populations, on the other hand, typically have reduced pelvic spines, as shown in Figure 2.

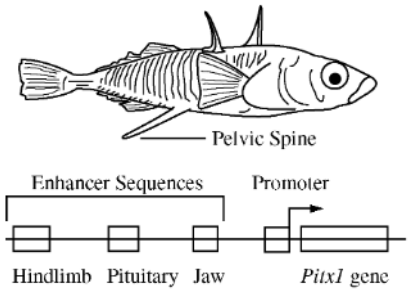


Figure 1. Marine stickleback

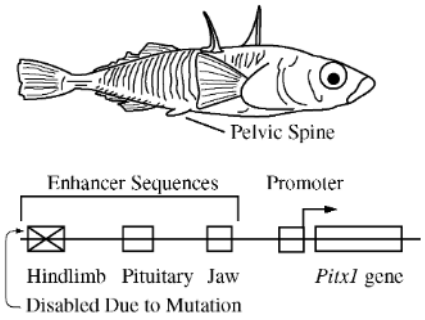


Figure 2. Freshwater stickleback

As represented in Figure 1 and Figure 2, the phenotypic difference between marine and freshwater sticklebacks involves Pitx1, a gene that influences the formation of the jaw, pituitary gland, and pelvic spine. Enhancer sequences upstream of the Pitx1 genetic locus regulate expression of the Pitx1 gene at the appropriate times and in the appropriate tissues during development. Previous studies have found that a mutation in the hindlimb enhancer interferes with the formation of a pronounced pelvic spine.

162. Which of the following best explains how sticklebacks in the same population with identical copies of the Pitx1 gene can still show phenotypic variation in the pelvic spine character?

(A) The Pitx1 gene is carried on different chromosomes in different individuals.

(B) Expression of the Pitx1 gene is affected by mutations at other genetic loci. ✓

(C) The genetic code of the Pitx1 gene is translated differently in males and females.

(D) The subcellular location of the Pitx1 gene changes when individuals move to a new environment.

163. A mutation that affects Pitx1 gene function in all tissue types is most likely to be at which of the following genetic loci?

(A) Hindlimb enhancer

(B) Pituitary enhancer

(C) Jaw enhancer

(D) Promoter ✓

6 Gene Expression and Regulation MCQ

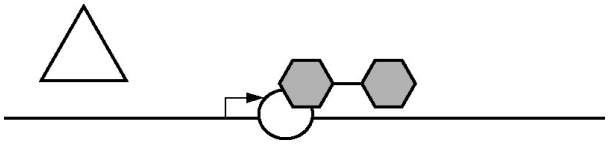
164. 26. The *lac* operon in *E. coli* consists of genes that code for enzymes necessary for the breakdown of lactose. When lactose is absent, the operon is inactive because a repressor protein binds to a specific site in the *lac* operon. When lactose is present, lactose molecules bind to the repressor protein, causing the repressor protein to dissociate from the binding site. In the absence of glucose (a preferred energy source for bacteria), the protein CAP binds to a regulatory site near the *lac* promoter to activate transcription of the *lac* operon.

The following symbols represent the macromolecules involved in regulation of the *lac* operon.

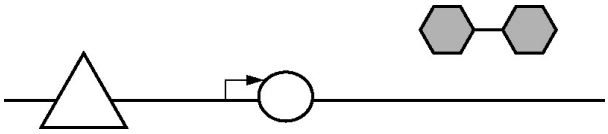


In the diagrams below, the horizontal line represents the *lac* operon and the bent arrow represents the transcription start site of the *lac* operon. Which of the following diagrams best represents the scenario in which lactose is available to the cell and glucose is absent?

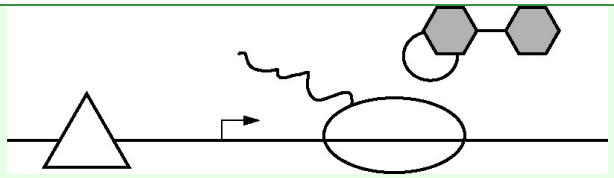
(A)



(B)



(C) ✓



(D)

